

数学物理方程



2 分离变量法

张 丹 副教授/博导

能源与动力工程学院

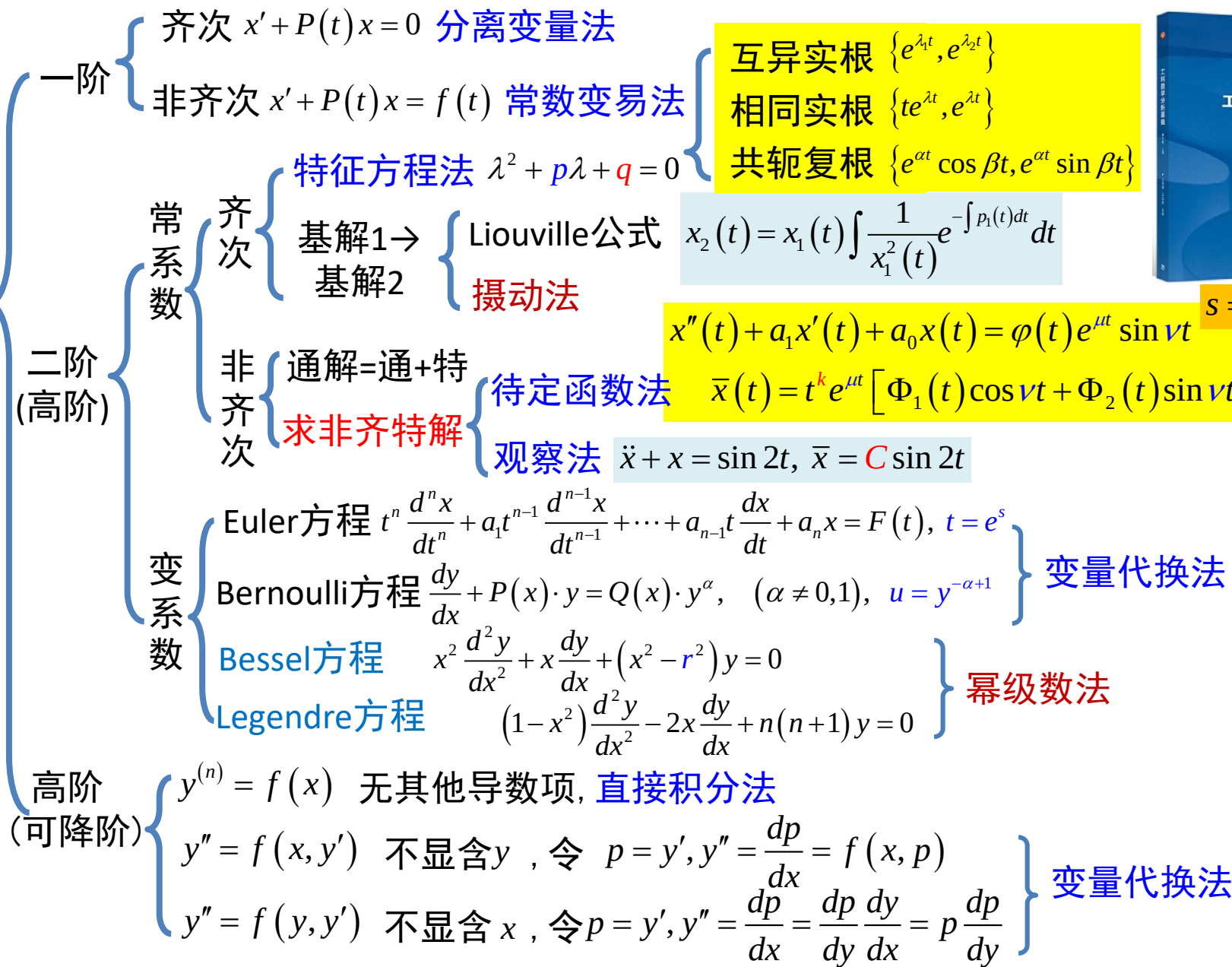
数学与统计学院(兼)

动力工程多相流国家重点实验室

线性ODE的解法

$$x^{<n>} + P_1 x^{<n-1>} + P_2 x^{<n-2>} + \cdots + P_{n-1} x^{<1>} + P_n x^{<0>} = F$$

线性常微分方程解法



$$s = \mu \pm \nu i$$

线性ODE的解法

$$x^{<n>} + P_1 x^{<n-1>} + P_2 x^{<n-2>} + \cdots + P_{n-1} x^{<1>} + P_n x^{<0>} = F$$



例1 求下列齐次ODE的通解

(1) $y'' - 3y' + 2y = 0$

解 二阶线性常系数ODE

特征方程 $\lambda^2 - 3\lambda + 2 = 0$

特征根 $\lambda_1 = 1, \lambda_2 = 2$

互异实根, 基解组 $\{e^x, e^{2x}\}$

通解 $y = C_1 e^x + C_2 e^{2x}$

(2) $y'' + 4y' + 13y = 0$

解 二阶线性常系数ODE

特征方程 $\lambda^2 + 4\lambda + 13 = 0$

特征根 $\lambda_{1,2} = -2 \pm 3i$

共轭复根, 基解组 $\{e^{-2x} \cos 3x, e^{-2x} \sin 3x\}$

通解 $y = C_1 e^{-2x} \cos 3x + C_2 e^{-2x} \sin 3x$

二阶常系数ODE $x''(t) + px'(t) + qx(t) = 0$

特征方程

$$\lambda^2 + p\lambda + q = 0$$

互异实根 $\{e^{\lambda_1 t}, e^{\lambda_2 t}\}$

相同实根 $\{te^{\lambda t}, e^{\lambda t}\}$

共轭复根 $\{e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t\}$

(3) $y'' + 4y' + 4y = 0$

解 二阶线性常系数ODE

特征方程 $\lambda^2 + 4\lambda + 4 = 0$

特征根 $\lambda_{1,2} = -2$

相同实根, 基解组 $\{e^{-2x}, xe^{-2x}\}$

通解 $y = C_1 e^{-2x} + C_2 x e^{-2x}$

二阶齐次ODE求解的核心是寻找2个基解

线性ODE的解法

$$x^{<n>} + P_1 x^{<n-1>} + P_2 x^{<n-2>} + \cdots + P_{n-1} x^{<1>} + P_n x^{<0>} = F$$

例2 求下列ODE的通解

(1) $y' - 2y = 0$

解法1 一阶线性常系数ODE

观察法 易知 $y = 0$ 是原方程的解

当 $y \neq 0$ 时, 分离变量法

$$\frac{dy}{dx} = 2y \rightarrow \frac{dy}{y} = 2dx \rightarrow \ln y = 2x + C$$

原ODE通解 $y = C \exp(2x)$

$C = 0$ 时
兼容 $y = 0$

解法2 特征方程法

原ODE对应特征方程 $s - 2 = 0$

特征根 $s = 2$, 对应基解 $\exp(2x)$

即得原ODE通解 $y = C \exp(2x)$

1. 验证0/常数是否为解? 能否被通解包含?

2. 但数学物理方程更加关心的是非0解

(2) $y' - 2y = x + 2$

解 一阶线性非齐次ODE, 常数变易法

变易对应齐次方程通解系数

对应齐次方程的通解左例已求出 $y = C \exp(2x)$

设非齐次方程的通解为 $y = C(x) \exp(2x)$

则 $y' = C'(x) \exp(2x) + 2C(x) \exp(2x)$ 代入原方程

$$C'(x) \exp(2x) + C(x) \cdot 2 \exp(2x) - 2 \cdot C(x) \exp(2x) = x + 2$$

化简得 $C'(x) = (x + 2) \exp(-2x)$ 积分有

$$C(x) = \int (x + 2) \exp(-2x) dx = -\frac{1}{2} \left(x + \frac{5}{2} \right) \exp(-2x) + C$$

原ODE通解为

$$\begin{aligned} y &= C(x) \exp(2x) = \left[-\frac{1}{2} \left(x + \frac{5}{2} \right) e^{-2x} + C \right] \exp(2x) \\ &= C \exp(2x) - \frac{x}{2} - \frac{5}{4} \end{aligned}$$

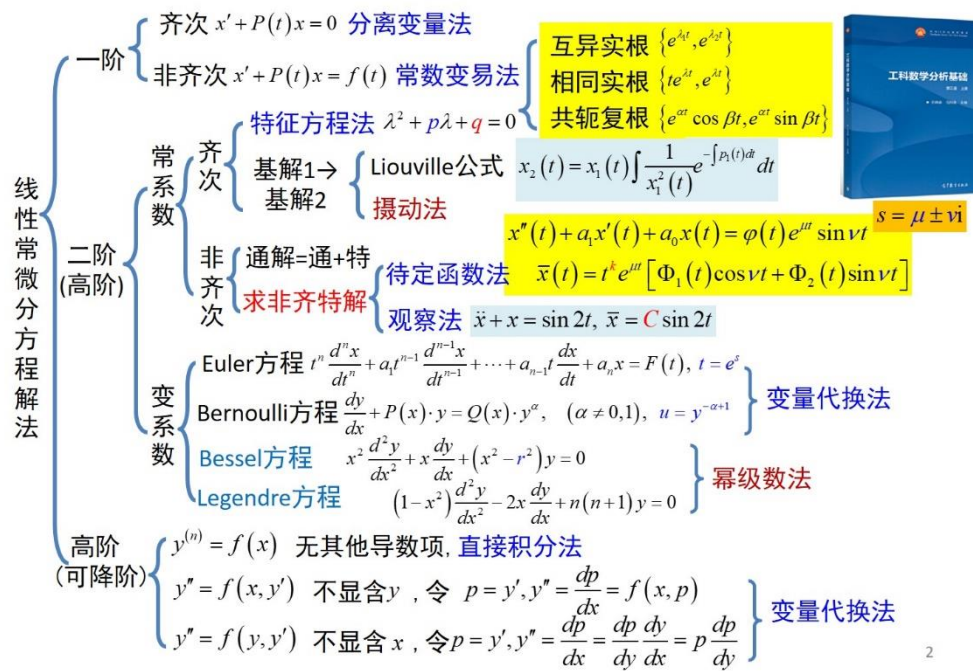
快速计算手册

边界条件齐次化常用辅助函数 $u(x,t) = v(x,t) + w(x,t)$

边界类型	边界条件	辅助函数
[1,1]	$u(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{l}x + g_1(t)$
[2,1]	$u_x(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = g_1(t)(x-l) + g_2(t)$
[1,2]	$u(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = g_2(t)x + g_1(t)$
[2,2]	$u_x(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{2l}x^2 + g_1(t)x$

线性ODE的解法

$$x^{<n>} + P_1 x^{<n-1>} + P_2 x^{<n-2>} + \dots + P_{n-1} x^{<1>} + P_n x^{<0>} = F$$



1. 数学建模和基本原理介绍

2. 分离变量法

2.1 特征值问题

矩阵的特征值

一个二阶微分算子的特征值

2.2 分离变量法

2.3 基于MATLAB的定解 问题求解方法

3. Bessel函数

4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

2.1 特征值问题

为什么要有特征值

1. 数学本质: 物理/社会现象变量关系→抽象提炼定义运算→研究运算→简化运算
2. 运算本质=定义在给定{集合|元素}上的操作
3. 运算多样/对象复杂/结果繁复→如何简化运算?→前提是如何衡量运算/操作效果?

$$\text{操作}(\text{元素}_i) = \lambda \cdot \text{元素}_i \quad \{\text{集合}|\text{元素}_i \text{属于操作对象集合}\}$$

反向评价法 若定义的 操作 对某非零元素无可奈何→即上式存在非零解
→反过来该元素(非零解)可作为特征→衡量操作的作用范围/效果→进而
探析简化运算的途径

$$Ax = \lambda x, \quad x \in R^n$$

特征向量 特征值

2.1 特征值问题

利用熟悉的矩阵特征值/特征向量→正交分解→简化运算

矩阵特征值

n 阶实矩阵 A 左乘列向量是针对 n 维列向量的线性变换

矩阵 A 的特征值是如下方程的非零解

$$Ax = \lambda x, \quad x \in \mathbb{R}^n$$

特征(列)向量 特征值

条件

- 表述1 矩阵 A 有 n 个线性无关的特征向量
- 表述2 特征值的代数重数=特征向量的几何重数

$$T_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{in} \end{bmatrix}$$

n 个特征向量组成的方阵

对角阵右乘方阵=对角元素乘以对应列

$$A[T_1, T_2, \dots, T_n] = [T_1, T_2, \dots, T_n] \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} = [\lambda_1 T_1, \lambda_2 T_2, \dots, \lambda_n T_n]$$



$$AT_i = \lambda_i T_i$$

$$T = [T_1, T_2, \dots, T_n]$$

左乘运算对特征向量无可奈何

- 正交性←特征向量线性无关
- 完备性 $T = [T_1, T_2, \dots, T_n]$ 可作为 $\mathbb{R}^n$ 的一组基, $\mathbb{R}^n$ 中任意向量均可由该组基表示

对向量的左乘(线性变换)→特征值/特征向量→特征方向上的标量计算(系数计算)

2.1 特征值问题

特征值如何简化运算-方阵左乘

例1 矩阵左乘列向量的计算. 二维矩阵左乘(向量)=旋转+放缩

$$A\mathbf{x} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

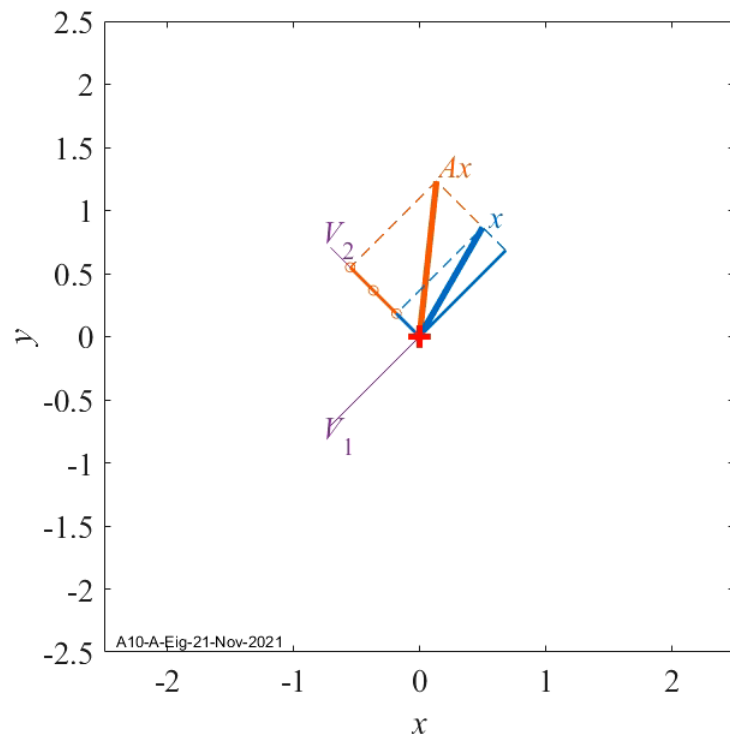
直接法: 按左乘定义计算

间接法: 按特征向量分解

$$A\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \in \mathbf{R}^n$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

特征向量 特征值



将 $\mathbf{x}$ 按特征方向 $\mathbf{v}_1, \mathbf{v}_2$ 分解 $\rightarrow$ 每个方向按特征值放缩 $\rightarrow$ 合成得到左乘结果 $A\mathbf{x}$

1. 运算对象 $\rightarrow$ 用特征向量(函数)的线性组合表示 $\rightarrow$ 对象规范化的通用手段
2. 按特征方向分解 $\rightarrow$ 方阵左乘运算 $\rightarrow$ 各特征方向的系数计算 $\rightarrow$ 运算大大简化

2.1 特征值问题

例2 若方阵 A 可对角化, 利用正交分解求解线性方程组 $A\mathbf{x} = \mathbf{b}$, $\mathbf{b} \in \mathbb{R}^n$

解 将 A 的 n 个线性无关的特征向量作为一组基 $\{\mathbf{T}_1, \mathbf{T}_2, \dots, \mathbf{T}_n\}$

将 $\mathbf{x}, \mathbf{b}$ 按照此基展开 $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{T}_i$, $\mathbf{b} = \sum_{i=1}^n b_i \mathbf{T}_i$

正交/完备

原方程可化为 $A\mathbf{x} = \mathbf{b}$ 求得即得解

$$A \sum_{i=1}^n x_i \mathbf{T}_i = \sum_{i=1}^n b_i \mathbf{T}_i$$

$$A\mathbf{T}_i = \lambda_i \mathbf{T}_i \quad \hookrightarrow \quad \sum_{i=1}^n x_i A\mathbf{T}_i = \sum_{i=1}^n b_i \mathbf{T}_i$$

$$\sum_{i=1}^n x_i \lambda_i \mathbf{T}_i = \sum_{i=1}^n b_i \mathbf{T}_i$$

$$x_i \lambda_i = b_i$$

基线性无关, 比较系数可得 $x_i = \frac{b_i}{\lambda_i}$

原方程的解可表示为 $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{T}_i = \sum_{i=1}^n \frac{b_i}{\lambda_i} \mathbf{T}_i$

2.1 特征值问题

例3 若矩阵 A 可对角化, 利用正交分解求解非齐次ODE方程组

$$\frac{dx(t)}{dt} = Ax + f(t), \quad x(t), x(0) = x^0 \in \mathbb{R}^n$$

正交/完备

解 将 A 的 n 个线性无关的特征向量作为一组基 $\{T_1, T_2, \dots, T_n\}$

将 x, x^0, f 分别按照此基展开 $x(t) = \sum_{i=1}^n x_i(t) T_i, \quad x^0 = \sum_{i=1}^n x_i^0 T_i, \quad f(t) = \sum_{i=1}^n f_i(t) T_i$

原方程可化为 $\frac{dx(t)}{dt} = Ax + f(t)$ 求得即得解

$$\sum_{i=1}^n \frac{dx_i(t)}{dt} T_i = A \sum_{i=1}^n x_i(t) T_i + \sum_{i=1}^n f_i(t) T_i$$

$$\sum_{i=1}^n \frac{dx_i(t)}{dt} T_i = \sum_{i=1}^n x_i(t) AT_i + \sum_{i=1}^n f_i(t) T_i$$

$AT_i = \lambda_i T_i$

$$\sum_{i=1}^n \frac{dx_i(t)}{dt} T_i = \sum_{i=1}^n x_i(t) \cdot \lambda_i T_i + \sum_{i=1}^n f_i(t) T_i$$

$$\sum_{i=1}^n \frac{dx_i(t)}{dt} T_i = \sum_{i=1}^n [x_i(t) \cdot \lambda_i + f_i(t)] T_i$$

每个特征方向上的ODE具有相同形式

$$\frac{dx_i(t)}{dt} = \lambda_i \cdot x_i(t) + f_i(t)$$

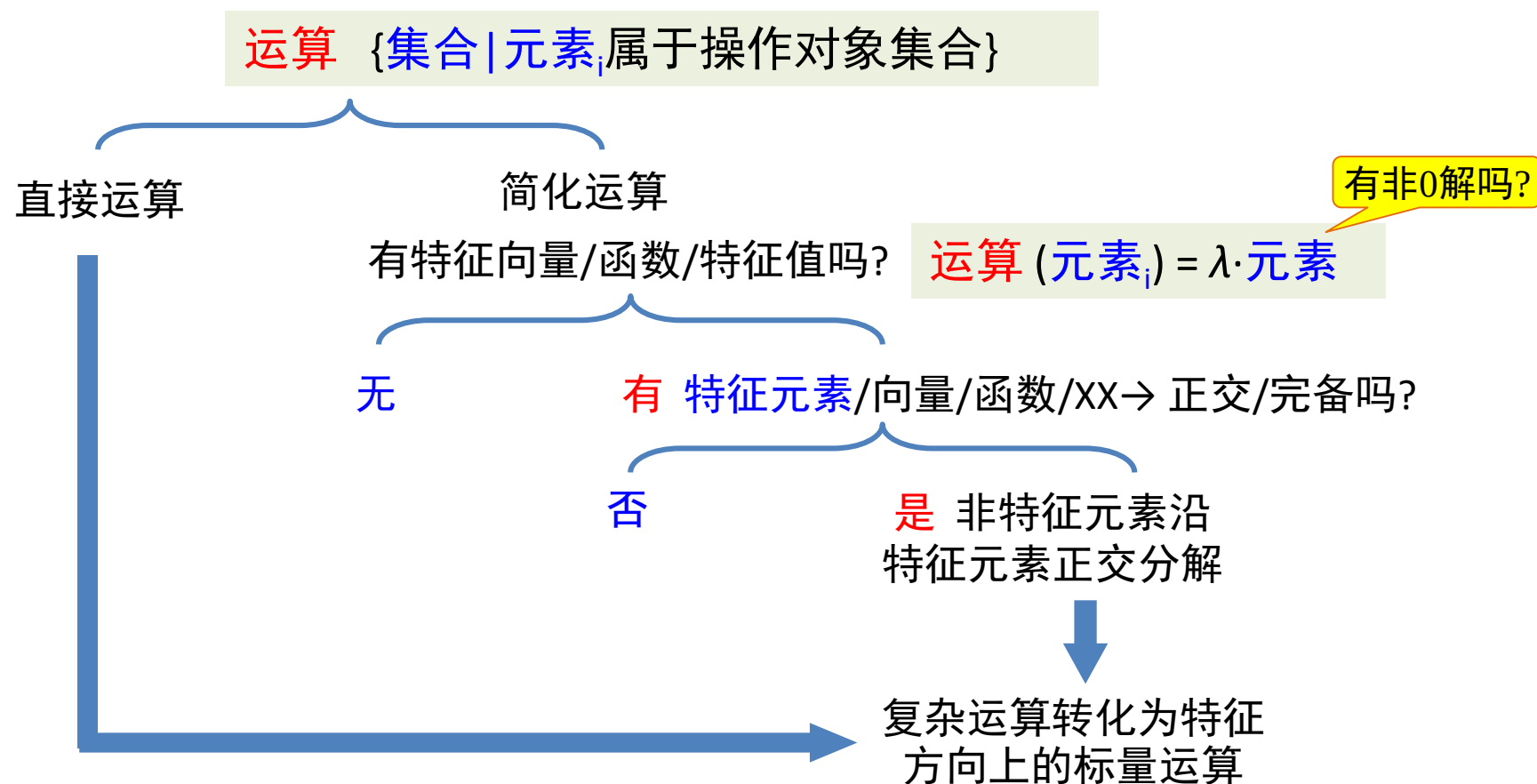
常数变易法

原方程的解为 $x(t) = \sum_{i=1}^n x_i(t) T_i, \quad x_i(t) = x_i^0 e^{\lambda_i t} + \int_0^t f_i(\tau) \cdot e^{\lambda_i(t-\tau)} d\tau$

2.1 特征值问题

小结

1. 特征值/特征向量(函数/方向)是针对运算效能的**反向衡量法**
2. 利用运算无可奈何的**特征元素 (向量/函数/方向)**可简化运算
3. 该简化运算的 思想/方法 不限于矩阵运算



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$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值

波动/导热/Poisson 未知函数空间变化率一侧相同

$$\left. \begin{aligned} \frac{\partial^2 u}{\partial \tau^2} - a^2 \frac{\partial^2 u}{\partial x^2} &= f \\ \frac{\partial u}{\partial \tau} - a^2 \frac{\partial^2 u}{\partial x^2} &= f \\ \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= f \end{aligned} \right\}$$

定义二阶微分算子

$$A = -\frac{d^2}{dx^2}$$

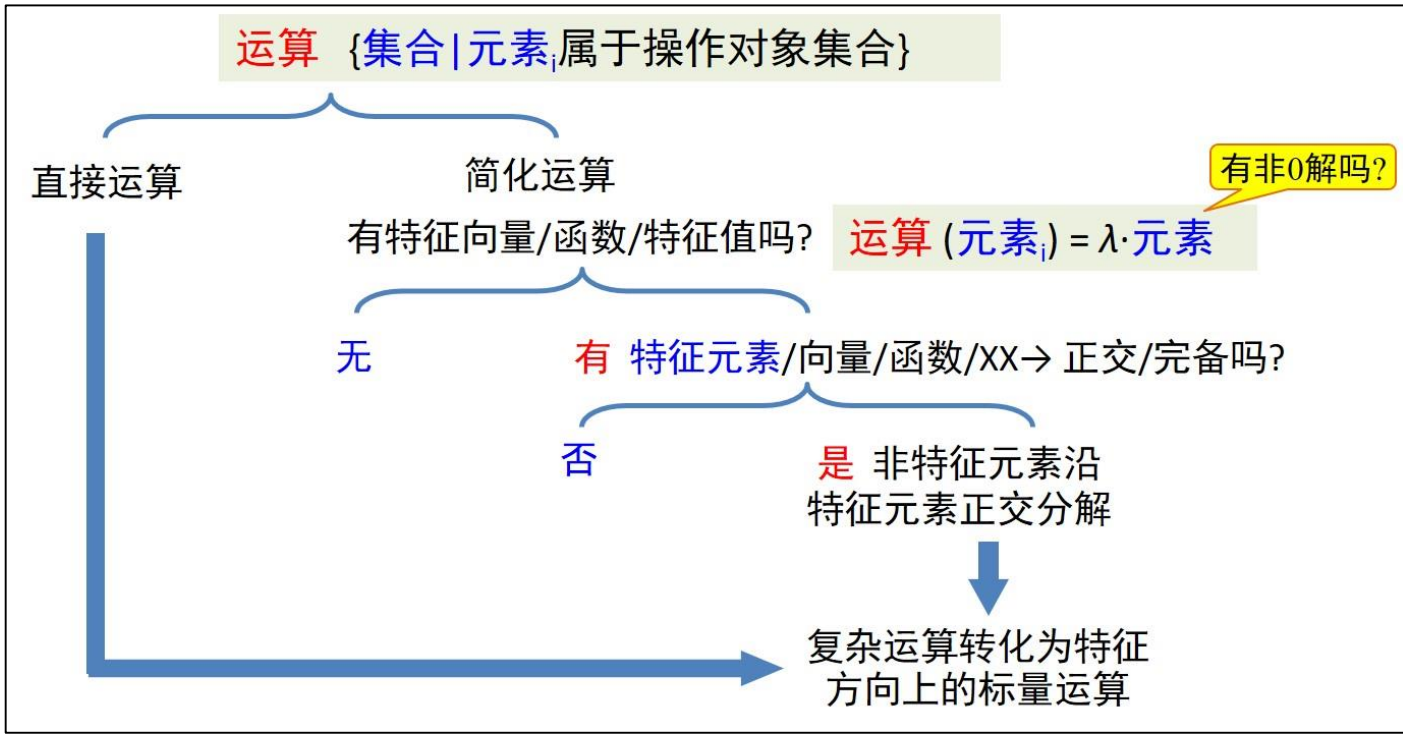
-Δ运算的一维形式

对象集合

- ① 第1/2/3类均可
- ②
- ③

$$H = \{X(x) \in C^2[0, l] \mid X(x) \text{ 齐次 } x=0, l\}$$

数学意义 作用在函数空间 H 上的线性变换



2.2 分离变量法

预备知识1 函数正交性/正交函数系

函数正交 若函数 $f_1(x), f_2(x)$ 满足 $\int_a^b f_1(x)f_2(x)dx=0$, 称函数 $f_1(x), f_2(x)$ 在区间 $[a, b]$ 上正交

① 什么区间

② 什么函数

正交函数系 若定义在区间 $[a, b]$ 上的函数列 $\{f_1(x), f_2(x), f_2(x), \cdots, f_n(x), \cdots\}$ 中,

任意两个函数满足 $\int_a^b f_i(x)f_j(x)dx \begin{cases} =0, & (i \neq j) \\ \neq 0, & (i = j) \end{cases}$, 则称该函数系为正交

函数系

③ 平方模是多少

没0/函数名不同/系数不同

三角函数系 $\{1, \cos x, \cos 2x, \cos nx, \cdots, \sin x, \sin 2x, \sin nx, \cdots\}$ 在区间 $[-\pi, \pi]$ 上为正交函数系

平方模 { 不同函数定积分为0 { $\int_{-\pi}^{\pi} 1 \cdot dx = 2\pi$

相同函数平方模不同 { $\int_{-\pi}^{\pi} \sin^2 nx \cdot dx = \left[\frac{1}{2n} (nx - \sin nx \cos nx) \right]_{-\pi}^{\pi} = \left[\frac{x}{2} - \frac{1}{4n} \sin 2nx \right]_{-\pi}^{\pi} = \pi$

$\int_{-\pi}^{\pi} \cos^2 nx \cdot dx = \left[\frac{1}{2n} (nx + \sin nx \cos nx) \right]_{-\pi}^{\pi} = \left[\frac{x}{2} + \frac{1}{4n} \sin 2nx \right]_{-\pi}^{\pi} = \pi$

正交区间可推广至[整数 π , 整数 $\pi+2\pi$]

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值

$$A = -\frac{d^2}{dx^2}$$

$$AX(x) = \lambda X(x), \quad X(x) \in H$$

$$H = \{X(x) \in C^2[0, l] \mid X(x) \text{ 齐次 } x=0, l\}$$

特征值/特征函数 → 是否存在? 是正是负? 具体是多少? → 依赖于边界条件

算子 A 的特征值问题

= 带有齐次边界条件 ODE 定解问题

[1, 1] 算子在给定区间两端都是第一类齐次边界

$$AX(x) = \lambda X(x) \rightarrow -\frac{d^2 X(x)}{dx^2} = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X(0) = X(l) = 0 \end{cases}$$

ODE 边值问题

设 $X(x)$ 是相应于 λ 的非零解

同乘方程两端 $X(x)X''(x) + \lambda X^2(x) = 0$

积分得 $\int_0^l X(x)X''(x)dx + \lambda \int_0^l X^2(x)dx = 0$

分部积分

$$X(x)X'(x)|_0^l - \int_0^l [X'(x)]^2 dx + \lambda \int_0^l X^2(x)dx = 0$$

= 0

移项有

$$\lambda = \frac{\int_0^l [X'(x)]^2 dx}{\int_0^l X^2(x) \cdot dx} \geq 0$$

算子 $A = -\frac{d^2}{dx^2}$ 的特征值非负 → 能否求出

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值 [1, 1]

$$AX(x) = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X(0) = X(l) = 0 \end{cases}$$

算子 **A** 的特征值问题
= 带有**齐次**边界条件ODE定解问题

算子 **A** 的特征值非负

$$\lambda = \frac{\int_0^l [X'(x)]^2 dx}{\int_0^l X^2(x) \cdot dx} \geq 0$$

可否解出具体值?

当 $\lambda = 0$ 时, 方程的通解为 $X(x) = C_1x + C_2$
根据边界条件得 $C_1 = C_2 = 0, X(x) = 0$
方程为**零解**, 所以 $\lambda = 0$ **不是特征值**

特征函数要
求非零解

当 $\lambda > 0$ 时, 方程的通解为 $X(x) = C_1 \cos(\sqrt{\lambda}x) + C_2 \sin(\sqrt{\lambda}x)$
根据边界条件得 $C_1 = 0, C_2 \sin(\sqrt{\lambda}l) = 0 \quad \sqrt{\lambda} \cdot l = n\pi, n \geq 1$
特征值/特征函数

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), n \geq 1$$

1. 特征值/特征函数系的表示包含**3**要素
2. 0是否为特征值需要单独讨论
3. 特征值 λ 是否包含 0 与 特征值**序号** n 是否从0开始 没有必然联系

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值 [1, 1]

特征值/特征函数系性质

$$AX(x) = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X(0) = X(l) = 0 \end{cases}$$

$$A = -\frac{d^2}{dx^2}$$

$$\left\{ \begin{array}{l} \text{特征值 非负/递增 } \lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad n \geq 1 \\ \text{特征函数系 } \left\{ X_n(x) = \sin\left(\frac{n\pi}{l}x\right) \mid n \geq 1 \right\} \end{array} \right.$$

正交

$$\int_{-l}^l \sin\left(\frac{n\pi}{l}x\right) \sin\left(\frac{m\pi}{l}x\right) dx = \frac{l}{\pi} \int_{-\pi}^{\pi} \sin(n\theta) \sin(m\theta) d\theta = \begin{cases} 0, & n \neq m \\ l, & n = m \end{cases}$$

$$\theta = \frac{\pi}{l}x, x = \frac{l}{\pi}\theta$$

完备 任意分段光滑的函数可展开为Fourier正弦级数

$$f(x) = \sum_{n=1}^{\infty} \left(b_n \sin \frac{n\pi x}{l} \right)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi}{l} x \cdot dx, \quad (n = 1, 2, 3 \dots)$$



J. Fourier
1768-1830
French Math.
Physical
Scientist

1811, Fourier级数

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} \cdot dx, \quad (n = 0, 1, 2, 3 \dots)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n = 1, 2, 3 \dots)$$

$$A = -\frac{d^2 f(x)}{dx^2}$$

对 $f(x)$ 的二阶求导运算 $\rightarrow$ 将函数 $f(x)$ 按特征函数系展开 $\rightarrow$ 转化为对其展开后各项的比例放缩 $\rightarrow$ 算子 A 代表的线性变换可正交分解 $\rightarrow$ 运算大大简化

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

特征值如何简化计算 [1, 1]

$$AX(x) = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X(0) = X(l) = 0 \end{cases}$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\left(\frac{n\pi}{l}x\right), \quad n \geq 1$$

例1

$$H = \{X(x) \in C^2[0, l] \mid X(x) \text{ 齐次 } x=0, l\}$$

$$A[f(x)] = -\frac{d^2 f(x)}{dx^2}$$



按特征函数系展开

$$f(x) = \sum_{n=1}^{\infty} f_n X_n(x)$$

$$= -\frac{d^2 \left[\sum_{n=1}^{\infty} f_n X_n(x) \right]}{dx^2}$$



f_n 与 x 无关, 求导/求和可交换次序

$$= \sum_{n=1}^{\infty} \left\{ f_n \frac{-d^2 [X_n(x)]}{dx^2} \right\}$$



$$-\frac{d^2 X(x)}{dx^2} = \lambda X(x)$$

利用特征值定义化简

$$= \sum_{n=1}^{\infty} f_n \cdot \lambda_n \cdot X_n(x)$$

求导运算 → 转化为简单的系数计算

以上为[1,1]型 → 改变边界条件 → 相应特征函数系也会改变 → Sturm-Liouville定理

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值 Sturm-Liouville定理

向不同类齐次边界组合下的推广 $H = \{X(x) \in C^2[0,l] \mid X(x) \text{ 齐次 } x=0,l\}$

$$AX(x) = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X^{<k>}(0) = X^{<m>}(l) = 0, & 0 \leq k, m \leq 1, k, m \in \mathbb{Z}^+ \end{cases}$$



J. Sturm
1803-1855
French Math.



J. Liouville
1809-1882
French Math.

- [1, 1]
- [1, 2]
- [2, 1]
- [2, 2]
- [1, 3]
- [3, 1]
- [2, 3]
- [3, 2]
- [3, 3]

① 存在

② 特征值 非负/递增 $0 \leq \lambda_1 < \lambda_2 < \dots < \lambda_n < \dots \rightarrow \infty$

③ 正交

③ 完备

特征函数系 $\{X_n(x) \mid n \geq 1\}$

在区间 $[0,l]$ 上, 任意分段光滑的函数可按特征函数系展开为广义Fourier级数

$$\int_0^l X_n(x) X_m(x) dx = \begin{cases} 0, & n \neq m \\ \int_0^l X_n^2(x) dx, & n = m \end{cases}$$

平方模

$$f(x) = \sum_{n=1}^{\infty} [f_n \cdot X_n(x)]$$

$$f_n = \frac{\int_0^l f(x) \cdot X_n(x) \cdot dx}{\int_0^l X_n^2(x) \cdot dx}, \quad (n=1, 2, 3 \dots)$$

任意齐次边界 $\rightarrow H$ 内任意函数可用特征函数系表示 $\rightarrow$ 均可简化为特征方向标量运算

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值 [1, 2]

$$AX(x) = \lambda X(x) \rightarrow -\frac{d^2 X(x)}{dx^2} = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X(0) = X'(l) = 0 \end{cases}$$

算子 A 的特征值非负

$$\lambda = \frac{\int_0^l [X'(x)]^2 dx}{\int_0^l X^2(x) \cdot dx} \geq 0$$

当 $\lambda = 0$ 时, 方程的通解为 $X(x) = C_1 x + C_2, X'(x) = C_1$

根据边界条件得 $C_1 = C_2 = 0, X(x) = 0$

方程为零解, 所以 $\lambda = 0$ 不是特征值

特征函数要求非零解

当 $\lambda > 0$ 时, 方程的通解为 $X(x) = C_1 \cos(\sqrt{\lambda} x) + C_2 \sin(\sqrt{\lambda} x)$

$$X'(x) = -C_1 \sqrt{\lambda} \sin(\sqrt{\lambda} x) + C_2 \sqrt{\lambda} \cos(\sqrt{\lambda} x)$$

由边界条件 $C_1 = 0, C_2 \sqrt{\lambda} \cos(\sqrt{\lambda} l) = 0,$

$$\sqrt{\lambda} \cdot l = \left(n + \frac{1}{2}\right) \pi = \frac{2n+1}{2} \pi, n \geq 0$$

特征值/特征函数 $\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l} x\right]$
 $(n = 0, 1, 2, 3 \dots)$

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值 [2, 1]

$$AX(x) = \lambda X(x) \rightarrow -\frac{d^2 X(x)}{dx^2} = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X'(0) = X(l) = 0 \end{cases}$$

算子 A 的特征值非负

$$\lambda = \frac{\int_0^l [X'(x)]^2 dx}{\int_0^l X^2(x) \cdot dx} \geq 0$$

当 $\lambda = 0$ 时, 方程的通解为 $X(x) = C_1 x + C_2$, $X'(x) = C_1$

根据边界条件得 $C_1 = C_2 = 0$, $X(x) = 0$

方程为零解, 所以 $\lambda = 0$ 不是特征值

特征函数要求非零解

当 $\lambda > 0$ 时, 方程的通解为 $X(x) = C_1 \cos(\sqrt{\lambda}x) + C_2 \sin(\sqrt{\lambda}x)$

$$X'(x) = -C_1 \sqrt{\lambda} \sin(\sqrt{\lambda}x) + C_2 \sqrt{\lambda} \cos(\sqrt{\lambda}x)$$

由边界条件 $C_2 = 0$, $C_1 \cos(\sqrt{\lambda}l) = 0$,

$$\sqrt{\lambda} \cdot l = \left(n + \frac{1}{2}\right)\pi = \frac{2n+1}{2}\pi, \quad n \geq 0$$

特征值/特征函数 $\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2$, $X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$
($n = 0, 1, 2, 3, \dots$)

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值 [2, 2]

$$AX(x) = \lambda X(x) \rightarrow -\frac{d^2 X(x)}{dx^2} = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X'(0) = X'(l) = 0 \end{cases}$$

算子 A 的特征值非负

$$\lambda = \frac{\int_0^l [X'(x)]^2 dx}{\int_0^l X^2(x) \cdot dx} \geq 0$$

当 $\lambda = 0$ 时, 方程的通解为 $X(x) = C_1 x + C_2$, $X'(x) = C_1$

根据边界条件得 $C_1 = 0$, $X(x) = C_2 \neq 0$

方程为 **非零解**, 所以 $\lambda = 0$ 是特征值, 通常取 $X(x) = 1$

当 $\lambda > 0$ 时, 方程的通解为 $X(x) = C_1 \cos(\sqrt{\lambda}x) + C_2 \sin(\sqrt{\lambda}x)$

$$X'(x) = -C_1 \sqrt{\lambda} \sin(\sqrt{\lambda}x) + C_2 \sqrt{\lambda} \cos(\sqrt{\lambda}x)$$

由边界条件 $C_2 = 0$, $C_1 \sqrt{\lambda} \sin(\sqrt{\lambda}l) = 0$, $\sqrt{\lambda} \cdot l = n\pi$, $n > 0$

特征值/特征函数

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), \\ (n = 0, 1, 2, 3 \dots)$$

兼容 $\lambda=0$

$$A = -\frac{d^2}{dx^2}$$

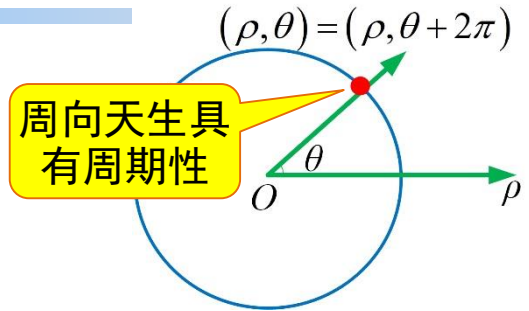
2.1 特征值问题

极坐标定解问题→周向/径向分离变量→周向存在特征值

二阶微分算子的特征值 周期性边界

$$A = -\frac{d^2}{dx^2} \rightarrow \begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0, & 0 \leq \theta \leq 2\pi \\ \Phi(0) = \Phi(2\pi), \Phi'(0) = \Phi'(2\pi) \end{cases}$$

$$A\Phi(x) = \lambda\Phi(x)$$



当 $\lambda = 0$ 时, 通解为 $\Phi(\theta) = C_1\theta + C_2, \Phi'(\theta) = C_1$ 由边界条件 $C_1 = 0, \Phi(\theta) = C_2 \neq 0$
 方程为**非零解** 所以 $\lambda = 0$ **是**特征值, 特征函数可取 $\Phi(\theta) = 1$

当 $\lambda > 0$ 时, 方程的通解为 $\Phi(\theta) = C_1 \cos(\sqrt{\lambda}\theta) + C_2 \sin(\sqrt{\lambda}\theta)$
 $\Phi'(x) = -C_1\sqrt{\lambda} \sin(\sqrt{\lambda}\theta) + C_2\sqrt{\lambda} \cos(\sqrt{\lambda}\theta)$ 由边界条件

$$\begin{cases} C_1 = C_1 \cos(2\pi\sqrt{\lambda}) + C_2 \sin(2\pi\sqrt{\lambda}) \\ C_2\sqrt{\lambda} = -C_1\sqrt{\lambda} \sin(2\pi\sqrt{\lambda}) + C_2\sqrt{\lambda} \cos(2\pi\sqrt{\lambda}) \end{cases}$$

→

$$\begin{bmatrix} 1 - \cos(2\pi\sqrt{\lambda}) & -\sin(2\pi\sqrt{\lambda}) \\ \sin(2\pi\sqrt{\lambda}) & 1 - \cos(2\pi\sqrt{\lambda}) \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

不能同时为0

系数行列式=0, $1 - \cos(2\pi\sqrt{\lambda}) = 0$, 解得 $\lambda_n = n^2, \Phi_n(\theta) = \{\cos n\theta, \sin n\theta\}, n = 1, 2, 3 \dots$

特征值/特征函数

$$\lambda_0 = 0, \Phi_0(\theta) = 1, n = 0$$

$$\lambda_n = n^2, \Phi_n(\theta) = \{\cos n\theta, \sin n\theta\}, n = 1, 2, 3 \dots$$

2个特征函数

若视 $\Phi_0(\theta) = 1 = \cos(0 \cdot \theta)$, 则 $n=0$ 与 $n = 1, 2, 3 \dots$ 可合并→但三角函数系中**1**与**sin/cos**平方模不同 →展开极易出错→最好单列



2.1 特征值问题

小结

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3\cdots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3\cdots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3\cdots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0, 1, 2, 3\cdots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = \{\cos n\theta, \sin n\theta\}, (n=1, 2, 3\cdots)$

2.1 特征值问题

例1 特征值问题 $\begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X(0) = X'(l) = 0 \end{cases}$

的特征值和特征函数是

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3\cdots)$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0, 1, 2, 3\cdots)$$

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3\cdots)$$

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3\cdots)$$

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数		
齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3\cdots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3\cdots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3\cdots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0, 1, 2, 3\cdots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1, 2, 3\cdots)$

2.1 特征值问题

例2 求特征值问题 $\begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < \pi \\ X'(0) = X(\pi) = 0 \end{cases}$

$$\lambda_n = \frac{(2n+1)^2}{4}, \quad X_n(x) = \cos\left[\frac{(2n+1)}{2}x\right], \quad (n = 0, 1, 2, 3 \cdots)$$

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数		
齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n = 1, 2, 3 \cdots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n = 0, 1, 2, 3 \cdots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n = 0, 1, 2, 3 \cdots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n = 0, 1, 2, 3 \cdots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n = 0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n = 1, 2, 3 \cdots)$

2.1 特征值问题

研究特征值问题的目的→将给定函数按特征函数系正交分解

特征函数系 平方模

将函数按 A 算子特征函数正交分解
→Fourier展开的变体→平方模是关键

平方模

$$\int_{-l}^l 1^2 \cdot dx = 2l$$
$$\theta = \frac{\pi}{l} x, x = \frac{l}{\pi} \theta$$
$$\int_{-l}^l \sin^2\left(\frac{n\pi}{l} x\right) dx = \frac{l}{\pi} \int_{-\pi}^{\pi} \sin^2(n\theta) d\theta = l$$
$$\int_{-l}^l \cos^2\left(\frac{n\pi}{l} x\right) dx = \frac{l}{\pi} \int_{-\pi}^{\pi} \cos^2(n\theta) d\theta = l$$
$$\int_{-l}^l \sin^2\left[\frac{(2n+1)\pi}{2l} x\right] dx = \frac{l}{\pi} \int_{-\pi}^{\pi} \sin^2\left[\frac{(2n+1)}{2} \cdot \theta\right] d\theta = \frac{l}{\pi} \left[\frac{\theta}{2} - \frac{1}{4 \cdot \frac{(2n+1)}{2}} \sin\left(2 \frac{(2n+1)}{2} \theta\right) \right]_{-\pi}^{\pi} = l$$
$$\int_{-l}^l \cos^2\left[\frac{(2n+1)\pi}{2l} x\right] dx = \frac{l}{\pi} \int_{-\pi}^{\pi} \cos^2\left[\frac{(2n+1)}{2} \cdot \theta\right] d\theta = \frac{l}{\pi} \left[\frac{\theta}{2} + \frac{1}{4 \cdot \frac{(2n+1)}{2}} \sin\left(2 \frac{(2n+1)}{2} \theta\right) \right]_{-\pi}^{\pi} = l$$

A 算子特征函数正交区间是[-l, l]
平方模仅有1函数是 2l, 其余特征函数均为 l

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数		
齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x\right), (n=1, 2, 3 \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l} x\right], (n=0, 1, 2, 3 \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l} x\right], (n=0, 1, 2, 3 \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l} x\right), (n=0, 1, 2, 3 \dots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1, 2, 3 \dots)$

2.1 特征值问题

按特征函数系正交分解



间接阐明了三角函数的完备性

函数在 $[-\pi, \pi]$ 上:
1829, Dirichlet条件
1. 分段单调;
2. 除有限个第一类间断点外都连续,
则该函数的Fourier级数在 $[-\pi, \pi]$ 上收敛

1822, 定义在 $[0, l]$ 上函数Fourier展开

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$
$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} \cdot dx, \quad (n = 0, 1, 2, 3 \dots)$$
$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n = 1, 2, 3 \dots)$$

利用正交函数系将函数展开→正交分解

$$f(x) = \sum_{n=0}^{\infty} f_n X_n(x) = f_0 \cdot 1 + f_1 \cdot X_1(x) + \dots + f_n \cdot X_n(x) + \dots$$

*n*是特征值序号起点

展开系数计算通式

$$f_n = \frac{1}{\int_a^b X_n^2(x) dx} \int_a^b f(x) \cdot X_n(x) dx$$

$$f(x) = \sum_{n=1}^{\infty} f_n \cdot \sin\left(\frac{n\pi}{l} x\right) \Rightarrow f_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi}{l} x\right) dx = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi}{l} x\right) dx$$

$$f(x) = \sum_{n=0}^{\infty} f_n \cdot \cos\left(\frac{n\pi}{l} x\right) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n \cdot \cos\left(\frac{n\pi}{l} x\right)$$

积分区间对折

$$f_0 = \frac{1}{2l} \int_{-l}^l f(x) \cdot 1 \cdot dx = \frac{2}{2l} \int_0^l f(x) dx = \frac{1}{l} \int_0^l f(x) dx$$
$$f_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi}{l} x\right) dx = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi}{l} x\right) dx$$

快速计算手册

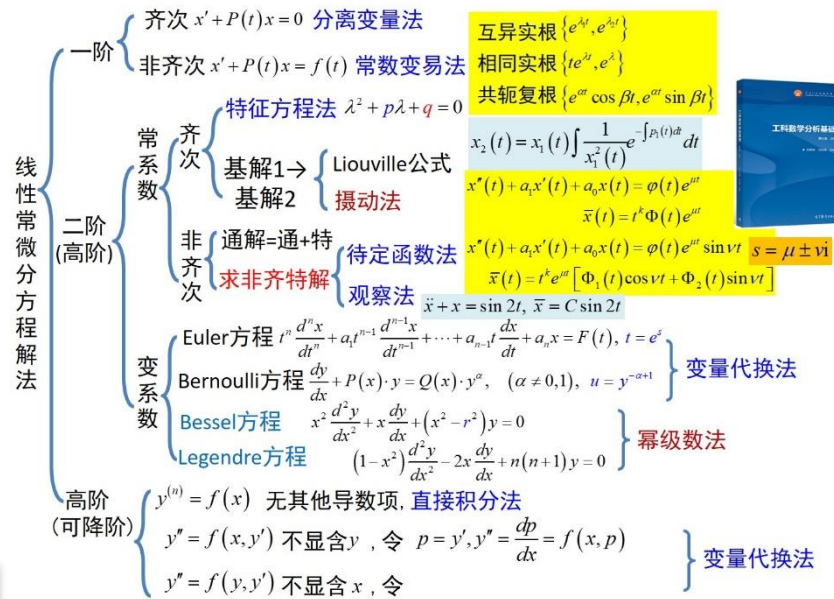
边界条件齐次化常用辅助函数 $u(x,t) = v(x,t) + w(x,t)$

边界类型	边界条件	辅助函数
[1,1]	$u(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{l}x + g_1(t)$
[2,1]	$u_x(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = g_1(t)(x-l) + g_2(t)$
[1,2]	$u(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = g_2(t)x + g_1(t)$
[2,2]	$u_x(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{2l}x^2 + g_1(t)x$

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1,2,3\cdots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0,1,2,3\cdots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0,1,2,3\cdots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0,1,2,3\cdots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1,2,3\cdots)$

线性ODE的解法 $x^{<n>} + P_1 x^{<n-1>} + P_2 x^{<n-2>} + \cdots + P_{n-1} x^{<1>} + P_n x^{<0>} = F$



1. 数学建模和基本原理介绍

2. 分离变量法

3. Bessel函数

4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

2.1 特征值问题

2.2 分离变量法

2.3 基于MATLAB的定解问题求解方法

弦振动方程定解问题

热传导方程定解问题

平面位势方程边值问题

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例1

求解一维波动定解问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

1齐

1齐

解 第1步 分离变量 根据齐次方程/齐次边界, 假设 $u(x, t) = X(x)T(t)$

代入齐次波动方程有 $X(x)T''(t) - a^2 X''(x)T(t) = 0$

$$\frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$$

分离常数

左右只可能同等于某个常数 $-\lambda$

边值问题

初值问题

化为两个ODE $X''(x) + \lambda X(x) = 0$ $T''(t) + \lambda a^2 T(t) = 0$

由边界条件 $X(0)T(t) = X(l)T(t) = 0$ $T(t)$ 不恒为0

$$\begin{cases} X(0) = X(l) = 0 \\ X''(x) + \lambda X(x) = 0 \end{cases}$$

特征值/特征函数

$$A = -\frac{d^2}{dx^2}$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right) (n=1, 2, 3, \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right] (n=0, 1, 2, 3, \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right] (n=0, 1, 2, 3, \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right) (n=0, 1, 2, 3, \dots)$

找出特征值/特征函数→边值问题得解→获得分离常数

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例1 求解一维波动定解问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

1齐

1齐

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3\cdots)$$

第2步 求解初值问题 将特征值 λ_n 代入初值ODE $T''(t) + \lambda a^2 T(t) = 0$

得到初值问题通解 $T_n(t) = C_{1n} \sin(a\sqrt{\lambda_n}t) + C_{2n} \cos(a\sqrt{\lambda_n}t)$

每个 λ_n 对应的解 $u_n(x, t) = T_n(t) X_n(x) = \left[C_{1n} \sin(a\sqrt{\lambda_n}t) + C_{2n} \cos(a\sqrt{\lambda_n}t) \right] \sin\left(\frac{n\pi}{l}x\right)$

所有 λ_n 对应的解
叠加→形式解

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} \left[C_{1n} \sin\left(\frac{n\pi a}{l}t\right) + C_{2n} \cos\left(\frac{n\pi a}{l}t\right) \right] \sin\left(\frac{n\pi}{l}x\right)$$

应满足初始条件

$$u(x, 0) = \sum_{n=1}^{\infty} C_{2n} \sin\left(\frac{n\pi}{l}x\right) = \varphi(x)$$

$$u_t(x, 0) = \sum_{n=1}^{\infty} C_{1n} \frac{n\pi a}{l} \cdot \sin\left(\frac{n\pi}{l}x\right) = \psi(x)$$

系数不能吸收

初始条件按
特征函数系
正交分解

$$C_{2n} = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi x}{l} \cdot dx, (n=1, 2, 3\cdots)$$

$$C_{1n} = \frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi x}{l} \cdot dx, (n=1, 2, 3\cdots)$$

此 x 仅是
积分变量

累加包括

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

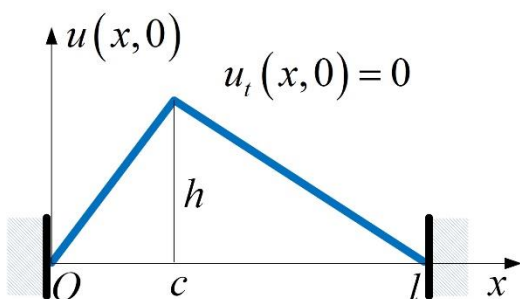
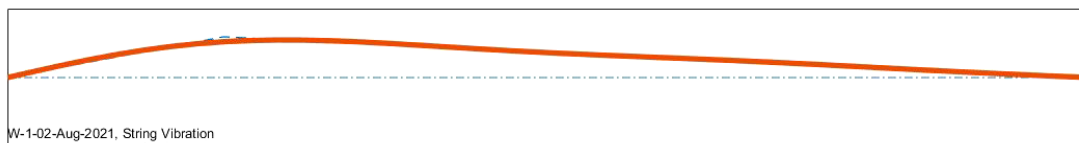
例1 求解一维波动定解问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 & \text{波动/一维/齐次} \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 & \text{1齐} \quad \text{1齐} \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} \left[C_{1n} \sin\left(\frac{n\pi a}{l} t\right) + C_{2n} \cos\left(\frac{n\pi a}{l} t\right) \right] \sin\left(\frac{n\pi}{l} x\right)$$

$$C_{1n} = \frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi x}{l} \cdot dx, \quad C_{2n} = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n=1, 2, 3, \dots)$$

讨论 两端固定/初速度为0的弦振动



根据初始速度为0

$$\psi(x) = 0, \quad C_{1n} = \frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi x}{l} \cdot dx = 0$$

初始位移可表示为

$$u(x, 0) = \varphi(x) = \begin{cases} \frac{h}{c} x, & 0 < x \leq c \\ \frac{-hx + hl}{l - c}, & c \leq x < l \end{cases}$$

$$\begin{aligned} C_{2n} &= \frac{2}{l} \left[\int_0^c \varphi(x) \sin \frac{n\pi x}{l} \cdot dx + \int_c^l \varphi(x) \sin \frac{n\pi x}{l} \cdot dx \right] \\ &= \frac{2}{l} \left[\int_0^c \frac{h}{c} x \sin \frac{n\pi x}{l} \cdot dx + \int_c^l \frac{-hx + hl}{l - c} \sin \frac{n\pi x}{l} \cdot dx \right] \\ &= \frac{2hl^2}{\pi^2 c(l - c)} \frac{1}{n^2} \sin \frac{n\pi}{l} c \end{aligned}$$

该定解问题的解为

$$u(x, \tau) = \frac{2hl^2}{\pi^2 c(l - c)} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi c}{l} \cos\left(\frac{n\pi a}{l} \tau\right) \sin\left(\frac{n\pi}{l} x\right)$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

分离变量的目标与推广

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f$$

$f = 0$

齐次PDE定解问题→可直接分离变量求解

分离变量法: PDE/ODE有质的不同

变量分离目标

1. 分离成独立函数 $u(x, t) = X(x)T(t)$
2. 找出各函数独立满足的ODE
3. 得出各ODE独立的定解条件

时/空解耦
的多个ODE
定解问题

若 $u(x, t) = X(x) + T(t)$ { 不同量纲不能相加
无法找出各自ODE独立定解条件

齐次PDE的求解并未体现特征值问题简化计算的功能

$f \neq 0$

无法直接分离变量→但特征函数系 $\{X_n(x)\}$ 正交/完备→正交分解
 $u(x, t) = \sum_{n=0,1}^{\infty} X_n(x)T_n(t) \rightarrow T_n(t)$ 待定函数法利用初始条件确定→特征函数法/正交分解法/广义分离变量法

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

例2 求解一维波动定解问题, 其中自由项 A 为常数

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐1齐

解 边界已经齐次

第1步 特征值问题, 设 $u(x, t) = X(x)T(t)$

代入对应的齐次方程

$$X(x)T''(t) - a^2 X''(x)T(t) = 0$$

$$\frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$$

$$T''(t) + \lambda a^2 T(t) = 0, X''(x) + \lambda X(x) = 0$$

齐次边界 $T(t)$ 不恒为0 $X(0) = X(l) = 0$

边界条件由边值问题实现

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$ ($n=1, 2, 3 \dots$)
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3 \dots$)
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3 \dots$)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right)$ ($n=0, 1, 2, 3 \dots$)

特征值/特征函数

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3 \dots)$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例2 求解一维波动定解问题, 其中自由项 A 为常数

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐	
1齐	1齐

特征值/特征函数

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right) \\ (n=1, 2, 3, \dots)$$

第2步 正交分解

待定函数

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l}x\right)$$

自由项/初始条件按特征函数系展开

定义在 $[0, l]$ 上函数 $f(x)$ 的Fourier展开/正交分解

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x) \\ f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx, \\ f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

$$A = \sum_{n=1}^{\infty} f_n X_n(x) = \sum_{n=1}^{\infty} f_n \sin\left(\frac{n\pi}{l}x\right)$$

$$f_n = \frac{2}{l} \int_0^l A \sin\left(\frac{n\pi}{l}x\right) dx = \frac{2A}{n\pi} [1 - \cos n\pi]$$

$$\varphi(x) = \sum_{n=1}^{\infty} \varphi_n X_n(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi}{l}x\right)$$

$$\rightarrow \varphi_n = \frac{2}{l} \int_0^l \varphi(x) \sin\left(\frac{n\pi}{l}x\right) dx$$

$$\psi(x) = \sum_{n=1}^{\infty} \psi_n X_n(x) = \sum_{n=1}^{\infty} \psi_n \sin\left(\frac{n\pi}{l}x\right)$$

$$\psi_n = \frac{2}{l} \int_0^l \psi(x) \sin\left(\frac{n\pi}{l}x\right) dx$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

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波动/一维/非齐

1齐

1齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right) \quad (n=1, 2, 3\cdots)$$

$$A = \sum_{n=1}^{\infty} f_n X_n(x),$$

$$\varphi(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi}{l}x\right), \psi(x) = \sum_{n=1}^{\infty} \psi_n \sin\left(\frac{n\pi}{l}x\right)$$

第3步 建立初值问题ODE $T_n(t)$

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l}x\right)$$

假设可逐项求导

$$\sum_{n=1}^{\infty} T_n''(t) X_n(x) + a^2 \sum_{n=1}^{\infty} T_n(t) (-X_n''(x)) = \sum_{n=1}^{\infty} f_n X_n(x)$$

$X_n(x)$ 是 A 运算特征函数 $X_n''(x) + \lambda X_n(x) = 0$

特征值简化运算

$$\sum_{n=1}^{\infty} T_n''(t) X_n(x) + a^2 \sum_{n=1}^{\infty} T_n(t) [\lambda_n X_n(x)] = \sum_{n=1}^{\infty} f_n X_n(x)$$

$$\sum_{n=1}^{\infty} [T_n''(t) + a^2 \lambda_n T_n(t)] X_n(x) = \sum_{n=1}^{\infty} f_n X_n(x)$$

比较系数 $T_n''(t) + a^2 \lambda_n T_n(t) = f_n$

用原PDE初始条件→初值问题ODE初始条件

$$u(x, 0) = \sum_{n=1}^{\infty} T_n(0) X_n(x) = \varphi(x) = \sum_{n=1}^{\infty} \varphi_n X_n(x)$$

$$u_t(x, 0) = \sum_{n=1}^{\infty} T_n'(0) X_n(x) = \psi(x) = \sum_{n=1}^{\infty} \psi_n X_n(x)$$

建立了初值问题ODE $T_n(t)$

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = \varphi_n, T_n'(0) = \psi_n \end{cases}$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

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波动/一维/非齐

1齐1齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$$

$(n=1, 2, 3\cdots)$

$A = \sum_{n=1}^{\infty} f_n X_n(x),$

$$\varphi(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi}{l}x\right), \psi(x) = \sum_{n=1}^{\infty} \psi_n \sin\left(\frac{n\pi}{l}x\right)$$

第4步 求初值问题

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = \varphi_n, T_n'(0) = \psi_n \end{cases}$$

二阶线性非齐次ODE定解问题

通解=齐次通解+非齐特解

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$$

f_n 为常数, 故 $\bar{T}_n$ 也为常数

$$T_n'(t) = -C_{1n} a\sqrt{\lambda_n} \sin(a\sqrt{\lambda_n}t) + C_{2n} a\sqrt{\lambda_n} \cos(a\sqrt{\lambda_n}t)$$

$$T_n''(t) = -C_{1n} a^2 \lambda_n \cos(a\sqrt{\lambda_n}t) - C_{2n} a^2 \lambda_n \sin(a\sqrt{\lambda_n}t)$$

$$\begin{aligned} &T_n''(t) + a^2 \lambda_n T_n(t) \\ &= -C_{1n} a^2 \lambda_n \cos(a\sqrt{\lambda_n}t) - C_{2n} a^2 \lambda_n \sin(a\sqrt{\lambda_n}t) \\ &+ C_{1n} a^2 \lambda_n \cos(a\sqrt{\lambda_n}t) + C_{2n} a^2 \lambda_n \sin(a\sqrt{\lambda_n}t) \\ &+ a^2 \lambda_n \bar{T}_n = f_n \end{aligned}$$

$\Rightarrow \bar{T}_n = \frac{f_n}{a^2 \lambda_n}$

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \frac{f_n}{a^2 \lambda_n}$$

根据初始条件确定系数

$$\begin{cases} T_n(0) = C_{1n} + \frac{f_n}{a^2 \lambda_n} = \varphi_n \\ T_n'(0) = C_{2n} a\sqrt{\lambda_n} = \psi_n \end{cases}$$

$$\Rightarrow \begin{cases} C_{1n} = \varphi_n - \frac{f_n}{a^2 \lambda_n} \\ C_{2n} = \frac{l}{n\pi a} \psi_n \end{cases}$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

例2 求解一维波动定解问题, 其中自由项 A 为常数

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐	
1齐	1齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right) \\ (n=1, 2, 3, \dots)$$

初值解代入得出原定解问题的解

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x)$$

$$C_{1n} = \varphi_n - \frac{f_n}{a^2 \lambda_n}, C_{2n} = \frac{l}{n\pi a} \psi_n$$

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \frac{f_n}{a^2 \lambda_n}$$

$$= \sum_{n=1}^{\infty} \left[\left(\varphi_n - \frac{f_n}{a^2 \lambda_n} \right) \cos(a\sqrt{\lambda_n}t) + \frac{l \cdot \psi_n}{n\pi a} \sin(a\sqrt{\lambda_n}t) + \frac{f_n}{a^2 \lambda_n} \right] \sin\left(\frac{n\pi}{l}x\right)$$

$$= \sum_{n=1}^{\infty} \left[\left(\varphi_n - \frac{l^2}{n^2 \pi^2 a^2} f_n \right) \cos \frac{n\pi a}{l} t + \frac{l}{n\pi a} \psi_n \cdot \sin \frac{n\pi a}{l} t + \frac{l^2}{n^2 \pi^2 a^2} f_n \right] \sin \frac{n\pi}{l} x$$

其中

$$f_n = \frac{2}{l} \int_0^l A \sin\left(\frac{n\pi}{l}x\right) dx = \frac{2A}{n\pi} (1 - \cos n\pi)$$

$$\varphi_n = \frac{2}{l} \int_0^l \varphi(x) \sin\left(\frac{n\pi}{l}x\right) dx, \psi_n = \frac{2}{l} \int_0^l \psi(x) \sin\left(\frac{n\pi}{l}x\right) dx$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

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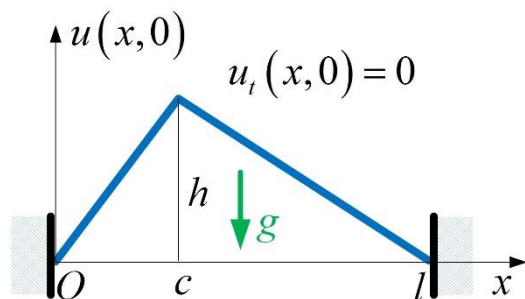
波动/一维/非齐

1齐

1齐

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right) \\ (n=1, 2, 3, \dots)$$

讨论 考虑重力的弦振动



根据初始速度为0

$$\psi(x) = 0, \psi_n = \frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi x}{l} \cdot dx = 0$$

初始位移可表示为

$$u(x, 0) = \varphi(x) = \begin{cases} \frac{h}{c} x, & 0 < x \leq c \\ \frac{-hx + hl}{l - c}, & c \leq x < l \end{cases}$$

$$u(x, t) = \sum_{n=1}^{\infty} \left[\left(\varphi_n - \frac{l^2}{n^2 \pi^2 a^2} f_n \right) \cos \frac{n\pi a}{l} t + \frac{l}{n\pi a} \psi_n \cdot \sin \frac{n\pi a}{l} t + \frac{l^2}{n^2 \pi^2 a^2} f_n \right] \sin \frac{n\pi}{l} x$$

$$f_n = \frac{2A}{n\pi} [1 - \cos n\pi], \varphi_n = \frac{2}{l} \int_0^l \varphi(x) \sin\left(\frac{n\pi}{l} x \right) dx, \psi_n = \frac{2}{l} \int_0^l \psi(x) \sin\left(\frac{n\pi}{l} x \right) dx$$

$$\varphi_n = \frac{2}{l} \left[\int_0^c \varphi(x) \sin \frac{n\pi x}{l} \cdot dx + \int_c^l \varphi(x) \sin \frac{n\pi x}{l} \cdot dx \right]$$

$$= \frac{2hl^2}{\pi^2 c(l-c)} \frac{1}{n^2} \sin \frac{n\pi}{l} c$$

$$u(x, t) = \sum_{n=1}^{\infty} \left[\varphi_n \cos \frac{n\pi a}{l} t + \frac{l^2}{n^2 \pi^2 a^2} \left(1 - \cos \frac{n\pi a}{l} t \right) f_n \right] \sin \frac{n\pi}{l} x$$

$$= \sum_{n=1}^{\infty} \left[\frac{2hl^2}{\pi^2 c(l-c)} \frac{1}{n^2} \sin \frac{n\pi c}{l} \cdot \cos \frac{n\pi a}{l} t + \frac{2Al^2}{\pi^3 a^2} \frac{1}{n^3} \left(1 - \cos \frac{n\pi a}{l} t \right) \left(1 - (-1)^n \right) \right] \sin \frac{n\pi}{l} x$$

$$\cos n\pi = (-1)^n$$

2.2 分离变量法

一维波动定解问题

例2 求解一维波动定解问题, 其中自由项 A 为常数

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

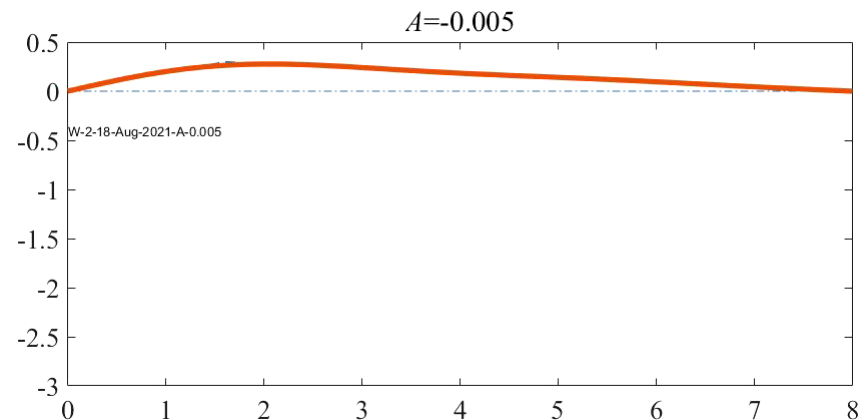
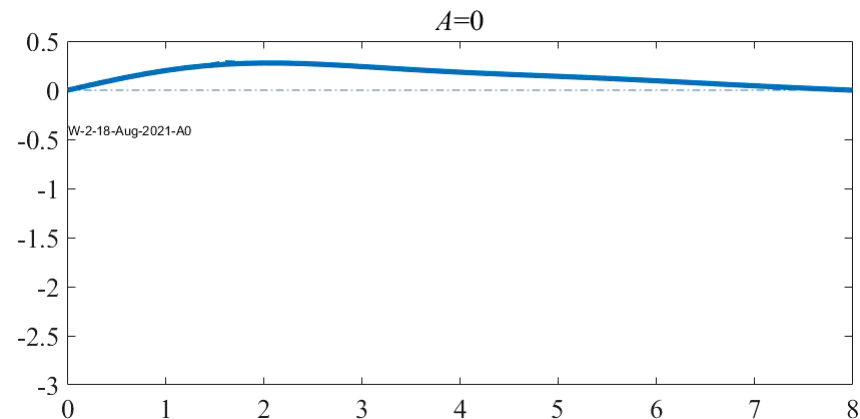
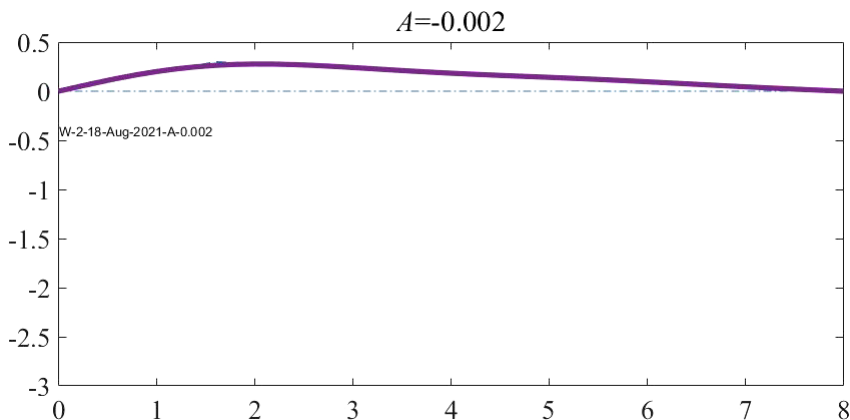
1齐 1齐

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right) \\ (n=1, 2, 3, \dots)$$

讨论

$$u(x, t) = \sum_{n=1}^{\infty} \left[\frac{2hl^2}{\pi^2 c(l-c)} \frac{1}{n^2} \sin \frac{n\pi c}{l} \cdot \cos \frac{n\pi a}{l} t + \frac{2Al^2}{\pi^3 a^2} \frac{1}{n^3} \left(1 - \cos \frac{n\pi a}{l} t \right) \left(1 - (-1)^n \right) \right] \sin \frac{n\pi}{l} x$$



$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

驻波法的由来

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

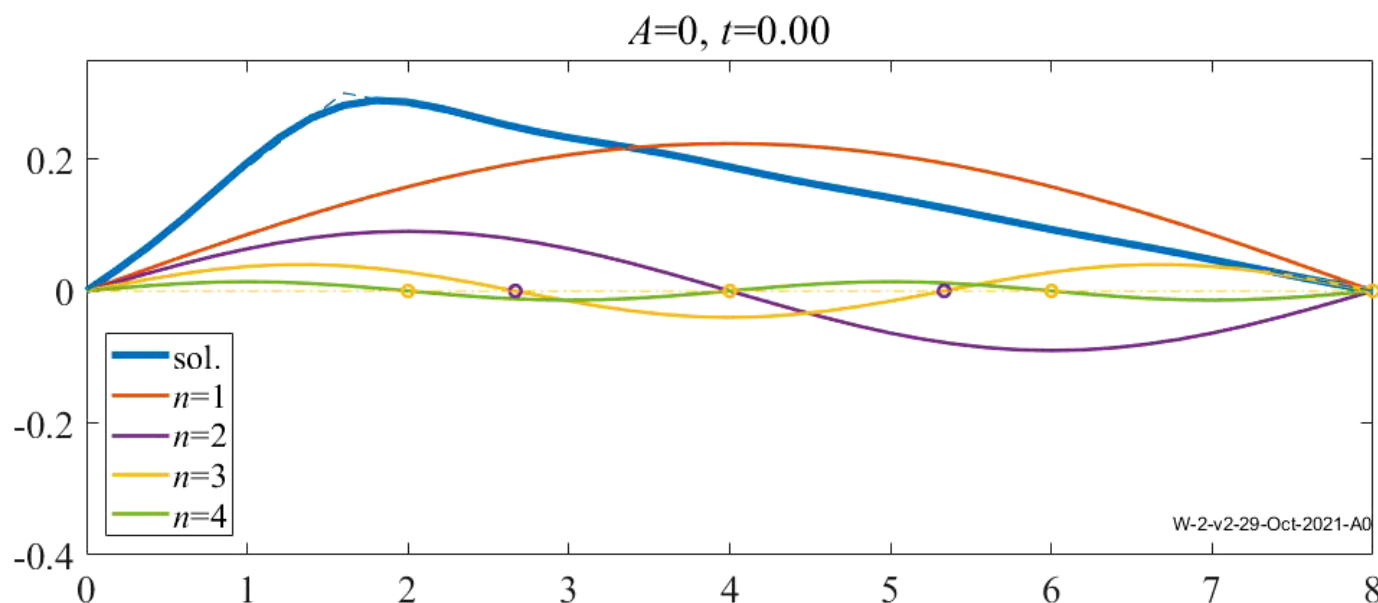
1齐

$$u(x, t) = \sum_{n=1}^{\infty} \left[\frac{2hl^2}{\pi^2 c(l-c)} \frac{1}{n^2} \sin \frac{n\pi c}{l} \cdot \cos \frac{n\pi a}{l} t + \frac{2Al^2}{\pi^3 a^2} \frac{1}{n^3} \left(1 - \cos \frac{n\pi a}{l} t \right) (1 - \cos n\pi) \right] \sin \frac{n\pi}{l} x$$

1. 解由无数简谐振动叠加

2. 各振动都有振幅始终为0的点 → 该振动称为驻波 → 分离变量法 = 驻波法 = 特征函数法

3. 驻波中振幅始终为0的点称为节点 $\sin \frac{n\pi}{l} x = 0, \frac{n\pi}{l} x = k\pi, x = k \frac{l}{n}, (1 \leq k \leq n-1)$



驻波的运用

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

预备知识 固有频率/共振 固有频率→与时间有关的固有属性→齐次PDE分离→ $T(t)$ 体现

$u_{tt} - a^2 u_{xx} = 0$ 分离变量 $u(x, t) = X(x)T(t)$

边界条件→特征值/特征函数系 $\frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$

$X''(x) + \lambda X(x) = 0 \quad T''(t) + \lambda a^2 T(t) = 0$

无论何种齐次边界→初值通解形式一致→弦上任一质点都做周期性运动

$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t)$ 以[1,1]型边界为例

固有频率←特征值←边界类型

$\omega_n = a\sqrt{\lambda_n} = a \frac{n\pi}{l}, (n=1, 2, 3\cdots)$

$-\frac{d^2 X(x)}{dx^2} = \lambda X(x)$

$[a\sqrt{\lambda_n}] = \left[\frac{m}{s} \sqrt{\frac{1}{m^2}} \right] = \left[\frac{1}{s} \right] = [Hz]$

$\lambda = -\frac{1}{X(x)} \cdot \frac{d^2 X(x)}{dx^2} \Rightarrow [\lambda_n] = \left[\frac{1}{m^2} \right]$

最低阶固有频率→固有主频

$$\left\{ \begin{aligned} \omega_1 &= a\sqrt{\lambda_1} = \frac{1 \cdot \pi a}{l} \\ \omega_2 &= a\sqrt{\lambda_2} = \frac{2 \cdot \pi a}{l} \\ &\vdots \\ \omega_k &= a\sqrt{\lambda_k} = \frac{k \cdot \pi a}{l} \\ &\vdots \\ \omega_n &= a\sqrt{\lambda_n} = \frac{n \cdot \pi a}{l} \end{aligned} \right.$$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$ ($n=1, 2, 3\cdots$)
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3\cdots$)
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3\cdots$)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right)$ ($n=0, 1, 2, 3\cdots$)

- 1.共振:周期性外力频率等于任一阶固有频率
- 2.外力一次只能等于某一个固有频率
- 3. k 阶共振本质是该阶初值问题特解不同

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

例3 周期性外力作用下的弦振动, $x=0$ 端为自由端, $x=l$ 端固定, 外力密度 $\sin \omega t$

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

2齐

1齐

解 第1步 特征值问题, 设 $u(x, t) = X(x)T(t)$

代入对应的齐次PDE

$$X(x)T''(t) - a^2 X''(x)T(t) = 0$$
$$\frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$$

分离变量法精髓

$$T''(t) + \lambda a^2 T(t) = 0, X''(x) + \lambda X(x) = 0$$

齐次边界 $T(t)$ 不恒为0 $X'(0) = X(l) = 0$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$ ($n=1, 2, 3 \dots$)
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3 \dots$)
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3 \dots$)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right)$ ($n=0, 1, 2, 3 \dots$)

特征值/特征函数

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3 \dots)$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例3 周期性外力作用下的弦振动, $x=0$ 端为自由端, $x=l$ 端固定, 外力密度 $\sin \omega t$

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐	
2齐	1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n = 0, 1, 2, 3 \dots)$$

第2步 正交分解

分离变量的形式解

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \cos \left[\frac{(2n+1)\pi}{2l} x \right]$$

自由项/初始条件 按特征函数系展开

$$\rho^{-1} \sin \omega t = \sum_{n=0}^{\infty} f_n X_n(x) = \sum_{n=0}^{\infty} f_n \cos \left[\frac{(2n+1)\pi}{2l} x \right]$$

$$f_n = \frac{2}{l} \int_0^l \rho^{-1} \sin \omega t \cdot \cos \left[\frac{(2n+1)\pi}{2l} x \right] dx = \frac{(-1)^n}{2n+1} \cdot \frac{4}{\pi \rho} \sin \omega t = k_n \sin \omega t$$

简化后续
表达式

定义在 $[0, l]$ 上函数 $f(x)$
的Fourier展开/正交分解

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x)$$

$$f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx,$$

$$f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

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波动/一维/非齐

2齐

1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n = 0, 1, 2, 3, \dots)$$

第3步 建立初值ODE

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \cos \left[\frac{(2n+1)\pi}{2l} x \right]$$

假设可逐项求导

$$\sum_{n=1}^{\infty} T_n''(t) X_n(x) + a^2 \sum_{n=1}^{\infty} T_n(t) (-X_n''(x)) = \sum_{n=1}^{\infty} f_n X_n(x)$$

$X_n(x)$ 是特征函数 $X_n''(x) + \lambda_n X_n(x) = 0$

$$\sum_{n=1}^{\infty} T_n''(t) X_n(x) + a^2 \sum_{n=1}^{\infty} T_n(t) [\lambda_n X_n(x)] = \sum_{n=1}^{\infty} f_n X_n(x)$$

$$\sum_{n=1}^{\infty} [T_n''(t) + a^2 \lambda_n T_n(t)] X_n(x) = \sum_{n=1}^{\infty} f_n X_n(x)$$

比较系数 $T_n''(t) + a^2 \lambda_n T_n(t) = f_n$

用原PDE初始条件 $\rightarrow$ 初值ODE初始条件

$$u(x, 0) = \sum_{n=1}^{\infty} T_n(0) X_n(x) = 0$$

特征函数非0

$$u_t(x, 0) = \sum_{n=1}^{\infty} T_n'(0) X_n(x) = 0$$

建立了初值问题ODE

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = 0, T_n'(0) = 0 \end{cases}$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

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波动/一维/非齐
2齐 1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n = 0, 1, 2, 3 \dots)$$

第4步 求初值问题

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n = k_n \sin \omega t \\ T_n(0) = 0, T_n'(0) = 0 \end{cases}$$

二阶线性非齐次ODE定解问题
通解=齐次通解+非齐特解

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$$

特解 $\bar{T}_n$ 的取法需分类讨论
 $\pm \omega i$ 是否为 $s^2 + a^2 \lambda_n = 0$ 的根

固有频率
的平方

否 $\left\{ \begin{aligned} \omega^2 &\neq a^2 \lambda_n, \bar{T}_n = \bar{C}_{1n} \cos \omega t + \bar{C}_{2n} \sin \omega t \\ \text{是} \quad \omega^2 &= a^2 \lambda_n = \omega_n^2, \bar{T}_n = t(\bar{C}_{1n} \cos \omega t + \bar{C}_{2n} \sin \omega t) \end{aligned} \right.$

辅助等式可用于后续化简

二阶ODE非齐次项含三角函数时特解取法

$$\ddot{x} + a_1 \dot{x} + a_2 x = e^{\mu t} \varphi(t) \sin vt$$

$\downarrow$

$$\bar{x} = t^k e^{\mu t} [Z_1(t) \cos vt + Z_2(t) \sin vt]$$

$s^2 + a_1 s + a_2 = 0, \mu \pm iv \begin{cases} \text{非特征根 } k=0 \\ \text{是特征根 } k=1 \end{cases}$

$Z_1(t), Z_2(t)$ 是与 $\varphi(t)$ 同次幂实多项式



λ_n 不止一个 $\rightarrow \pm \omega i$ 是不是根? 是哪个根 $\rightarrow$
结合物理意义 $\rightarrow$ 初值问题分类计算框架

2.2 分离变量法

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题 周期性外力作用下的弦振动, 左端为自由端, 右端固定, 外力密度 $\sin \omega t$

例3
$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐
2齐 1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n=0, 1, 2, 3, \dots)$$

$$\begin{cases} T_n(0) = 0, T'_n(0) = 0 \\ T''_n(t) + a^2 \lambda_n T_n(t) = f_n = e^{0t} k_n \sin \omega t \end{cases}$$

二阶ODE非齐次项含三角函数时特解取法

$$\ddot{x} + a_1 \dot{x} + a_2 x = e^{\mu t} \varphi(t) \sin vt$$

$\downarrow$

$$\bar{x} = t^k e^{\mu t} [Z_1(t) \cos vt + Z_2(t) \sin vt]$$
$$s^2 + a_1 s + a_2 = 0, \mu \pm iv \begin{cases} \text{非特征根 } k=0 \\ \text{是特征根 } k=1 \end{cases}$$

$Z_1(t), Z_2(t)$ 是与 $\varphi(t)$ 同次幂实多项式

$$-\frac{d^2 X(x)}{dx^2} = \lambda X(x)$$
$$\lambda = -\frac{1}{X(x)} \cdot \frac{d^2 X(x)}{dx^2}$$

特征方程 $s^2 + a^2 \lambda_n = 0$

特征根 $s = \pm a\sqrt{\lambda_n}i = \pm \omega_n i$

$\omega_n = a\sqrt{\lambda_n}$

$\mu + vi = 0 \pm \omega i$

$$[a\sqrt{\lambda_n}] = \left[\frac{m}{s} \sqrt{\frac{1}{m^2}} \right] = \left[\frac{1}{s} \right] = [Hz]$$

数学 $\omega = a\sqrt{\lambda_n}$

物理 $\omega = \omega_n = a\sqrt{\lambda_n}$

是不是特征根?

是不是固有频率?

固有频率/不止一个/n阶固有频率

辅助等式→后续共振情形解的化简

ω 不等于任意固有频率 $\omega^2 \neq \omega_n^2 = a^2 \lambda_n$
无共振, 所有初值问题特解相同
 $0 \pm \omega i$ 非特征根 $\bar{T}_n = \bar{C}_{1n} \cos \omega t + \bar{C}_{2n} \sin \omega t$

ω 等于 k 次固有频率 $\omega^2 = \omega_k^2 = a^2 \lambda_k$
有共振, 但仅 k 次谐波对应的特解与众不同
 $\bar{T}_n = \begin{cases} \bar{C}_{1n} \cos \omega t + \bar{C}_{2n} \sin \omega t & (n \neq k) \text{ 非特征根} \\ t(\bar{C}_{1k} \cos \omega t + \bar{C}_{2k} \sin \omega t) & (n = k, \omega = a\sqrt{\lambda_k}) \text{ 是特征根} \end{cases}$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例3 周期性外力作用下的弦振动, $x=0$ 端为自由端, $x=l$ 端固定, 外力密度 $\sin \omega t$

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

2齐

1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n = 0, 1, 2, 3 \dots)$$

第4步 求初值问题

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = k_n \sin \omega t \\ T_n(0) = 0, T_n'(0) = 0, n \geq 0 \end{cases}$$

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \frac{k_n}{a^2 \lambda_n - \omega^2} \sin \omega t$$

$$T_n'(t) = -C_{1n} a\sqrt{\lambda_n} \sin(a\sqrt{\lambda_n}t) + C_{2n} a\sqrt{\lambda_n} \cos(a\sqrt{\lambda_n}t) - \frac{k_n \omega}{a^2 \lambda_n - \omega^2} \cos \omega t$$

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$$

由初始条件 $C_{1n} = 0, C_{2n} = \frac{k_n \omega}{a\sqrt{\lambda_n}(\omega^2 - a^2 \lambda_n)}$

1 非共振 $\omega^2 \neq a^2 \lambda_n, \bar{T}_n = \bar{C}_{1n} \cos \omega t + \bar{C}_{2n} \sin \omega t$

$$\bar{T}_n' = -\bar{C}_{1n} \omega \sin \omega t + \bar{C}_{2n} \omega \cos \omega t$$

$$\bar{T}_n'' = -\bar{C}_{1n} \omega^2 \cos \omega t - \bar{C}_{2n} \omega^2 \sin \omega t$$

代入比较系数 $\bar{C}_{1n} = 0, \bar{C}_{2n} = \frac{k_n}{a^2 \lambda_n - \omega^2}$

$$\bar{T}_n = \frac{k_n}{a^2 \lambda_n - \omega^2} \sin \omega t$$

$$T_n(t) = \frac{k_n}{\omega^2 - a^2 \lambda_n} \left(\frac{\omega}{a\sqrt{\lambda_n}} \sin(a\sqrt{\lambda_n}t) - \sin \omega t \right)$$

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x)$$

原定解问题非共振解

$$= \sum_{n=0}^{\infty} \left[\frac{k_n}{\omega^2 - a^2 \lambda_n} \left(\frac{\omega}{a\sqrt{\lambda_n}} \sin(a\sqrt{\lambda_n}t) - \sin \omega t \right) \right] \cos \frac{(2n+1)\pi}{2l} x$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例3 周期性外力作用下的弦振动, $x=0$ 端为自由端, $x=l$ 端固定, 外力密度 $\sin \omega t$

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

2齐	1齐
----	----

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n = 0, 1, 2, 3 \dots)$$

第4步 求初值问题 $T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$

2 共振

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = k_n \sin \omega t \\ T_n(0) = 0, T_n'(0) = 0, n \geq 0 \end{cases}$$

系统固有频率 ω_n 有多个, 外力频率 ω 一次只能等于一个谐波频率
不妨设 $\omega^2 = \omega_k^2 = a^2 \lambda_k, (n=k)$

{

非共振谐波

$n \neq k$

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = k_n \sin \omega t \\ T_n(0) = 0, T_n'(0) = 0, n \neq k \end{cases}$$

与非共振相同

叠加原理

$$u(x, t) = T_k(t) X_k(x) + \sum_{n=0, n \neq k}^{\infty} T_n(t) X_n(x)$$

共振谐波

$n = k$

$$\begin{cases} T_k''(t) + a^2 \lambda_k T_k(t) = k_k \sin \omega t \\ T_k(0) = 0, T_k'(0) = 0, n = k \end{cases}$$

本例取 $n=k, \omega = a\sqrt{\lambda_k} = \frac{(2n+1)\pi a}{2l}$

特解形式 $\bar{T}_k = t(\bar{C}_1 \cos \omega t + \bar{C}_2 \sin \omega t)$ 下面先确定特解系数

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例3 周期性外力作用下的弦振动, $x=0$ 端为自由端, $x=l$ 端固定, 外力密度 $\sin \omega t$

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐	
2齐	1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n = 0, 1, 2, 3 \dots)$$

第4步 求初值问题 共振

$$\begin{cases} T_k''(t) + a^2 \lambda_k T_k(t) = k_k \sin \omega t \\ T_k(0) = 0, T_k'(0) = 0, n = k \end{cases}$$

$$\omega^2 = a^2 \lambda_k$$

$$T_k(t) = C_1 \cos(a\sqrt{\lambda_k}t) + C_2 \sin(a\sqrt{\lambda_k}t) + \bar{T}_k$$

$$\bar{T}_k = t(\bar{C}_1 \cos \omega t + \bar{C}_2 \sin \omega t)$$

$$\begin{aligned} \bar{T}_k' &= \bar{C}_1 \cos \omega t + \bar{C}_2 \sin \omega t \\ &\quad + t(-\bar{C}_1 \omega \sin \omega t + \bar{C}_2 \omega \cos \omega t) \end{aligned}$$

$$\begin{aligned} \bar{T}_k'' &= (-\bar{C}_1 \omega^2 t + 2\bar{C}_2 \omega) \cos \omega t \\ &\quad - (\bar{C}_2 \omega^2 t + 2\bar{C}_1 \omega) \sin \omega t \end{aligned}$$

$$\begin{aligned} T_k''(t) + a^2 \lambda_k T_k(t) &= (-\bar{C}_1 \omega^2 t + 2\bar{C}_2 \omega + a^2 \lambda_k \cdot t \bar{C}_1) \cos \omega t \\ &\quad - (\bar{C}_2 \omega^2 t + 2\bar{C}_1 \omega + a^2 \lambda_k \cdot t \bar{C}_2) \sin \omega t \\ &= (-\bar{C}_1 \omega^2 t + 2\bar{C}_2 \omega + \omega^2 t \bar{C}_1) \cos \omega t \\ &\quad - (\bar{C}_2 \omega^2 t + 2\bar{C}_1 \omega + \omega^2 t \bar{C}_2) \sin \omega t \\ &= (+2\bar{C}_2 \omega) \cos \omega t - (2\bar{C}_1 \omega + 2\omega^2 t \bar{C}_2) \sin \omega t = k_k \sin \omega t \end{aligned}$$

代入比较系数 $\bar{C}_1 = -\frac{k_k}{2\omega}, \bar{C}_2 = 0, \bar{T}_k = -\frac{k_k t}{2\omega} \cos \omega t$

该初值问题的通解为

$$T_k(t) = C_1 \cos(a\sqrt{\lambda_k}t) + C_2 \sin(a\sqrt{\lambda_k}t) - \frac{k_k t}{2\omega} \cos \omega t$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例3 周期性外力作用下的弦振动, $x=0$ 端为自由端, $x=l$ 端固定, 外力密度 $\sin \omega t$

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

2齐

1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n=0, 1, 2, 3, \dots)$$

第4步 求初值问题 共振

$$\begin{cases} T_k''(t) + a^2 \lambda_k T_k(t) = k_k \sin \omega t \\ T_k(0) = 0, T_k'(0) = 0, n = k \end{cases}$$

$$T_k(t) = C_1 \cos(a\sqrt{\lambda_k}t) + C_2 \sin(a\sqrt{\lambda_k}t) - \frac{k_k t}{2\omega} \cos \omega t$$

$$T_k'(t) = -C_1 a\sqrt{\lambda_k} \sin(a\sqrt{\lambda_k}t) + C_2 a\sqrt{\lambda_k} \cos(a\sqrt{\lambda_k}t) + \frac{k_k t}{2} \sin \omega t - \frac{k_k}{2\omega} \cos \omega t$$

由初始条件 $C_1 = 0, C_2 = \frac{k_k}{2\omega^2}$

$$T_k(t) = \frac{k_k}{2\omega^2} \sin \omega t - \frac{k_k t}{2\omega} \cos \omega t \quad \text{初值问题解毕}$$

共振发生时原定解问题的解

$$\begin{aligned} u(x, t) &= \sum_{n=0}^{\infty} T_n(t) X_n(x) \\ &= T_k(t) X_k(x) + \sum_{n=0, n \neq k}^{\infty} T_n(t) X_n(x) \\ &= \left(\frac{k_k}{2\omega^2} \sin \omega t - \frac{k_k t}{2\omega} \cos \omega t \right) \cos \left[\frac{(2k+1)\pi}{2l} x \right] \\ &\quad + \sum_{n=0, n \neq k}^{\infty} \left[\frac{k_n}{\omega^2 - a^2 \lambda_n} \cdot \left(\frac{\omega}{a\sqrt{\lambda_n}} \sin(a\sqrt{\lambda_n}t) - \sin \omega t \right) \right] \cos \frac{(2n+1)\pi}{2l} x \end{aligned}$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

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$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

2齐

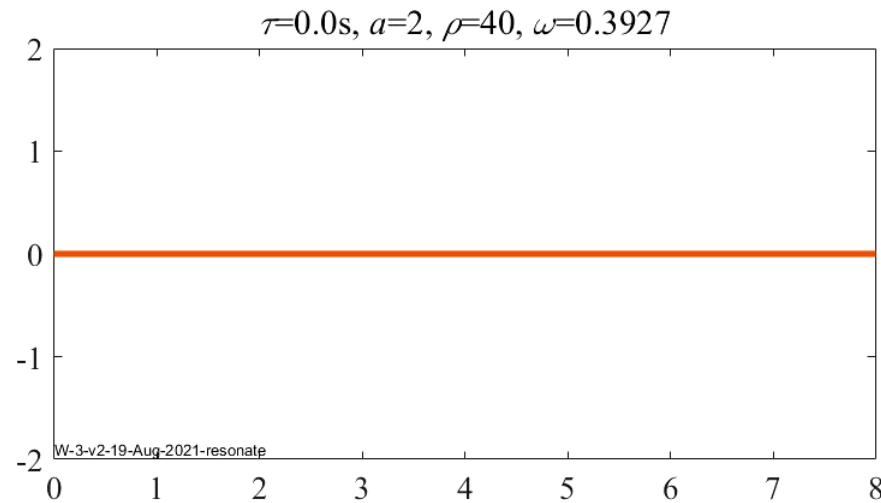
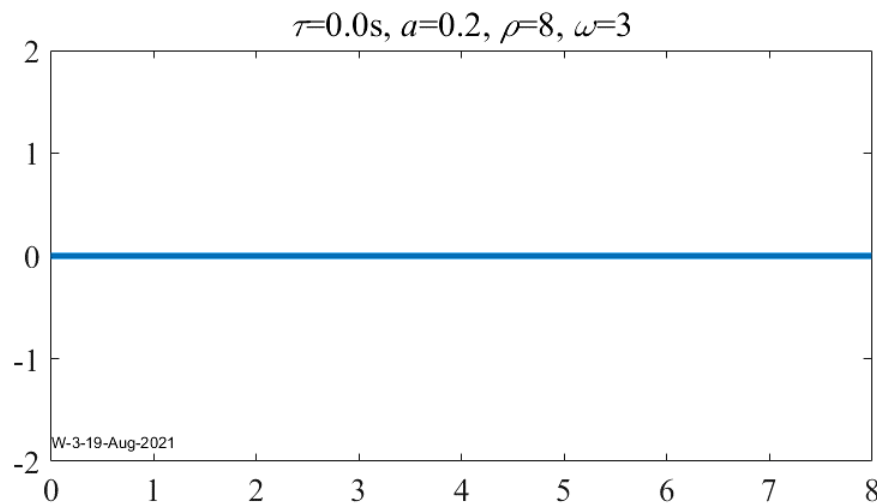
1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right]$$

$$k_n = \frac{(-1)^n}{2n+1} \cdot \frac{4}{\pi \rho}, (n=0, 1, 2, 3, \dots)$$

讨论

$$u(x, t) = \sum_{n=0}^{\infty} \left[\frac{k_n}{\omega^2 - a^2 \lambda_n} \left(\frac{\omega}{a \sqrt{\lambda_n}} \sin(a \sqrt{\lambda_n} t) - \sin \omega t \right) \right] \cos \frac{(2n+1)\pi}{2l} x$$



$$u(x, t) = \left(\frac{k_k}{2\omega^2} \sin \omega t - \frac{k_k t}{2\omega} \cos \omega t \right) \cos \frac{\pi}{2l} x + \sum_{n=0, n \neq k}^{\infty} \left[\frac{k_n}{\omega^2 - a^2 \lambda_n} \left(\frac{\omega}{a \sqrt{\lambda_n}} \sin(a \sqrt{\lambda_n} t) - \sin \omega t \right) \right] \cos \frac{(2n+1)\pi}{2l} x$$

2.2 分离变量法

一维波动定解问题

例3

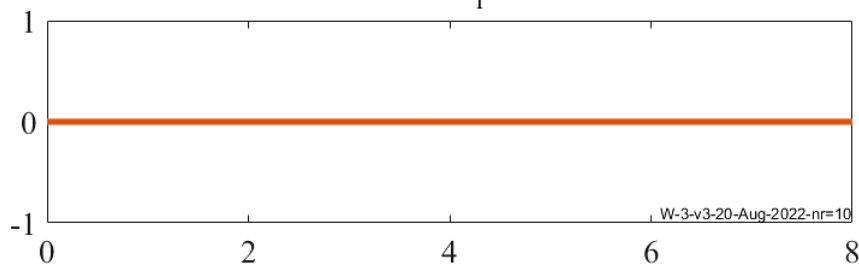
$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

讨论

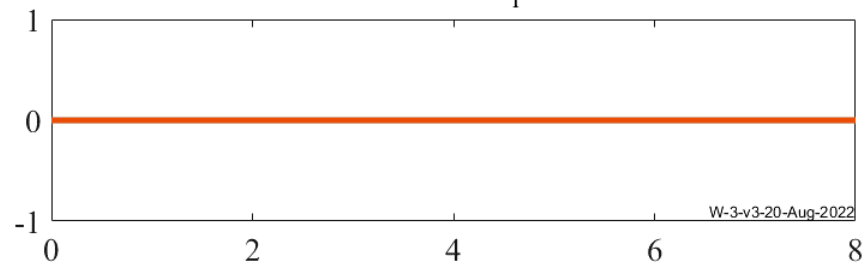
$$u(x, t) = \sum_{n=0}^{\infty} \left[\frac{k_n}{\omega^2 - a^2 \lambda_n} \begin{pmatrix} \frac{\omega}{a\sqrt{\lambda_n}} \sin(a\sqrt{\lambda_n}t) \\ -\sin \omega t \end{pmatrix} \right] \cos \frac{(2n+1)\pi}{2l} x$$

$$u(x, t) = \left(\frac{k_k}{2\omega^2} \sin \omega t - \frac{k_k t}{2\omega} \cos \omega t \right) \cos \frac{\pi}{2l} x + \sum_{n=0, n \neq k}^{\infty} \left[\frac{k_n}{\omega^2 - a^2 \lambda_n} \begin{pmatrix} \frac{\omega}{a\sqrt{\lambda_n}} \sin(a\sqrt{\lambda_n}t) \\ -\sin \omega t \end{pmatrix} \right] \cos \frac{(2n+1)\pi}{2l} x$$

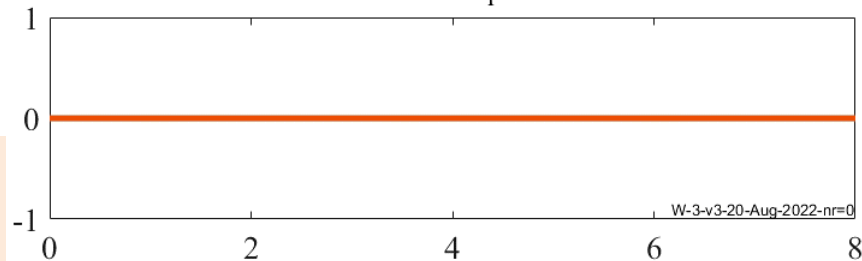
$$\tau=0.0s, a=2, \rho=5, n_r=10, \omega=8.2467$$



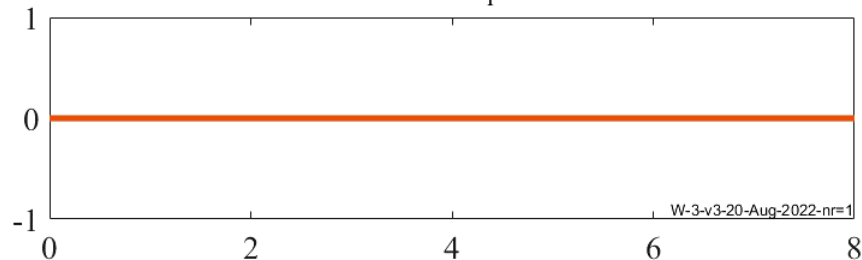
$$\tau=0.0s, a=2, \rho=10, n_r=\text{NaN}, \omega=3$$



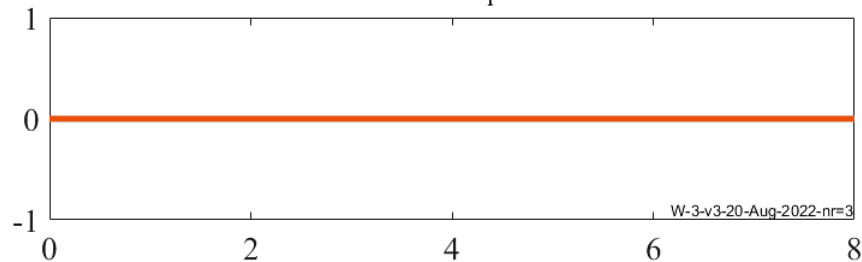
$$\tau=0.0s, a=2, \rho=5, n_r=0, \omega=0.3927$$



$$\tau=0.0s, a=2, \rho=5, n_r=1, \omega=1.1781$$



$$\tau=0.0s, a=2, \rho=5, n_r=3, \omega=2.7489$$



2.2 分离变量法

话说共振

不同振幅以相同的相位叠加→能量向空间某点集中

抑制

- 改变固有频率←改变质量
- 改变强迫外力的频率
- 改变谐振振动相位←加阻尼

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

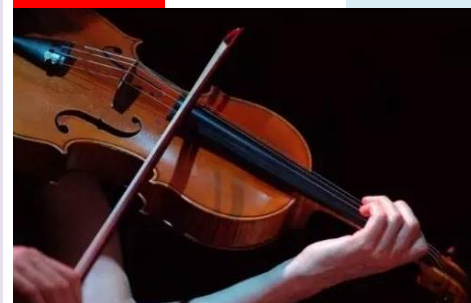
$$\frac{\partial^2 u}{\partial t^2} + b \frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$



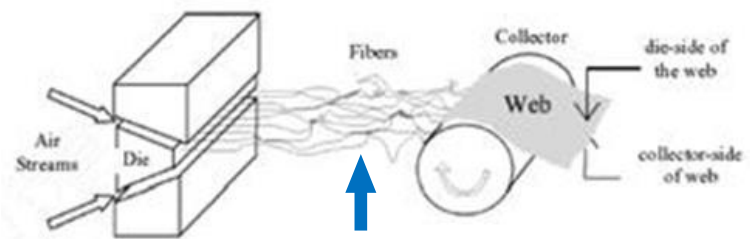
$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right) \quad (n=1, 2, 3, \dots)$$

强化



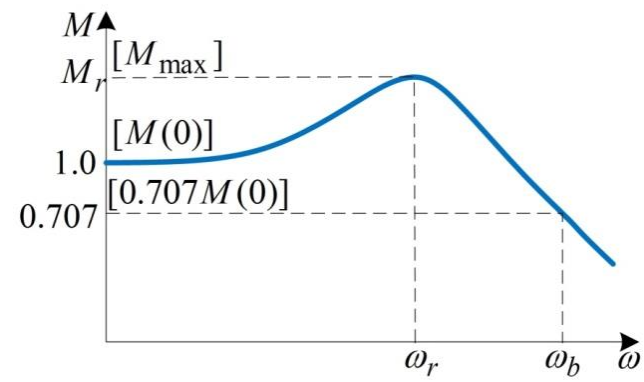
按弦→产生特有频率声波
琴腔共振→放大特有频率振幅



熔喷短纤快速生产工艺无接触超声切割

测试

复杂系统固有主频的测量



2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

例4

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1非齐



解 第1步 边界条件齐次化

设 $u(x, t) = v(x, t) + w(x, t)$

可取 $w(x, t) = \frac{F \sin \omega t}{l} x$ 代入原方程

$$w_t = \frac{F \omega \cos \omega t}{l} x$$
$$w_{tt} = \frac{-F \omega^2 \sin \omega t}{l} x$$

$$w_x = \frac{F \sin \omega t}{l}$$
$$w_{xx} = 0$$

改造PDE

$$u_{tt} - a^2 u_{xx} = A$$
$$v_{tt} + w_{tt} - a^2 v_{xx} - a^2 w_{xx} = A$$
$$v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x$$

改造初始条件

$$u(x, 0) = f(x)$$
$$v(x, 0) + w(x, 0) = f(x)$$
$$v(x, 0) = f(x)$$

边界条件齐次化常用辅助函数 $u(x, t) = v(x, t) + w(x, t)$

边界类型	边界条件	辅助函数
[1,1]	$u(0, t) = g_1(t), u(l, t) = g_2(t)$	$w(x, t) = \frac{g_2(t) - g_1(t)}{l} x + g_1(t)$
[2,1]	$u_x(0, t) = g_1(t), u(l, t) = g_2(t)$	$w(x, t) = g_1(t)(x - l) + g_2(t)$
[1,2]	$u(0, t) = g_1(t), u_x(l, t) = g_2(t)$	$w(x, t) = g_2(t)x + g_1(t)$
[2,2]	$u_x(0, t) = g_1(t), u_x(l, t) = g_2(t)$	$w(x, t) = \frac{g_2(t) - g_1(t)}{2l} x^2 + g_1(t)x$

$$u_t(x, 0) = 0$$
$$v_t(x, 0) + w_t(x, 0) = 0$$
$$v_t(x, 0) + \frac{F \omega x}{l} = 0$$
$$v_t(x, 0) = -\frac{F \omega x}{l}$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

例4

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐1非齐

$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐1齐

第2步 特征值问题 $v(x, t) = X(x)T(t)$

代入对应的齐次PDE

$$X(x)T''(t) - a^2 X''(x)T(t) = 0$$

$$\frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$$

$$T''(t) + \lambda a^2 T(t) = 0, X''(x) + \lambda X(x) = 0$$

齐次边界 $T(t)$ 不恒为0 $X(0) = X(l) = 0$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$ ($n=1, 2, 3 \dots$)
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3 \dots$)
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3 \dots$)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right)$ ($n=0, 1, 2, 3 \dots$)

特征值/特征函数

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3 \dots)$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例4

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐	1非齐
----	-----

$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐	1齐
----	----

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

第3步 正交分解

$$v(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l}x\right)$$

自由项/初始条件 按特征函数系展开

$$A + \frac{F \omega^2 \sin \omega t}{l} x = \sum_{n=1}^{\infty} f_n X_n(x) = \sum_{n=1}^{\infty} f_n \sin\left(\frac{n\pi}{l}x\right)$$

$$f(x) = \sum_{n=1}^{\infty} \varphi_n X_n(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi}{l}x\right)$$

$$-\frac{F \omega x}{l} = \sum_{n=1}^{\infty} \psi_n X_n(x) = \sum_{n=1}^{\infty} \psi_n \sin\left(\frac{n\pi}{l}x\right)$$

定义在[0, l]上函数 f(x) 的Fourier展开/正交分解

$$\begin{aligned} f(x) &= f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x) \\ f_0 &= \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx, \\ f_n &= \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx \end{aligned}$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

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波动/一维/非齐

1齐

1非齐

$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1齐

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right), (n=1, 2, 3, \dots)$$

第3步 正交分解 自由项 按特征函数系展开

$$\textcircled{1} A + \frac{F \omega^2 \sin \omega t}{l} x = \sum_{n=1}^{\infty} f_n X_n(x) = \sum_{n=1}^{\infty} f_n \sin\left(\frac{n\pi}{l} x \right)$$

$$f_{n1} = \frac{2}{l} \int_0^l A \sin \frac{n\pi x}{l} \cdot dx \quad f_{n2} = \frac{2}{l} \int_0^l \frac{F \omega^2 \sin \omega t}{l} x \cdot \sin \frac{n\pi x}{l} \cdot dx$$

$$= \frac{2A}{l} \left[-\frac{l}{n\pi} \cos \frac{n\pi x}{l} \right]_0^l = \frac{2F \omega^2 \sin \omega t}{l^2} \int_0^l x \cdot \sin \frac{n\pi x}{l} \cdot dx$$

$$= -\frac{2A}{n\pi} \left[\cos \frac{n\pi x}{l} \right]_0^l = \frac{2F \omega^2 \sin \omega t}{l^2} \left[-\frac{l}{n\pi} x \cos \frac{n\pi x}{l} + \frac{l^2}{n^2 \pi^2} \sin \frac{n\pi x}{l} \right]_0^l = (-1)^{n+1} \frac{2F \omega^2 \sin \omega t}{n\pi}$$

$$= \frac{2A}{n\pi} [1 - (-1)^n] \quad \cos n\pi = (-1)^n$$

$$\begin{aligned} f_n &= \frac{2A}{n\pi} [1 - (-1)^n] + (-1)^{n+1} \frac{2F \omega^2}{n\pi} \sin \omega t \\ &= f_A + f_B \sin \omega t \end{aligned}$$

定义在 $[0, l]$ 上函数 $f(x)$ 的Fourier展开/正交分解

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x)$$

$$f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx,$$

$$f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例4

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1非齐

$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1齐

第3步 正交分解 初始条件 按特征函数系展开

$$f(x) = \sum_{n=1}^{\infty} \varphi_n X_n(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi}{l} x\right)$$

➡

$$\varphi_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx$$

$$-\frac{F \omega x}{l} = \sum_{n=1}^{\infty} \psi_n X_n(x) = \sum_{n=1}^{\infty} \psi_n \sin\left(\frac{n\pi}{l} x\right)$$

$$\begin{aligned} \psi_n &= \frac{2}{l} \int_0^l -\frac{F \omega x}{l} \sin \frac{n\pi x}{l} \cdot dx = -\frac{2F \omega}{l^2} \int_0^l x \sin \frac{n\pi x}{l} \cdot dx = -\frac{2F \omega}{l^2} \left[-\frac{l}{n\pi} x \cos \frac{n\pi}{l} x + \frac{l^2}{n^2 \pi^2} \sin \frac{n\pi}{l} x \right] \Big|_0^l \\ &= -\frac{2F \omega}{l^2} \left[-\frac{l}{n\pi} x \cos \frac{n\pi}{l} x \right] \Big|_0^l = (-1)^n \frac{2F \omega}{n\pi} \end{aligned}$$

$$\psi_n = (-1)^n \frac{2F \omega}{n\pi}$$

定义在[0, l]上函数 f(x) 的Fourier展开/正交分解

$$\begin{aligned} f(x) &= f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x) \\ f_0 &= \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx, \\ f_n &= \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx \end{aligned}$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例4

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐	
1齐	1非齐

$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐	
1齐	1齐

第4步 建立初值问题ODE

$$v(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l} x\right)$$

假设可逐项求导, 代入原方程

$$\begin{aligned} \sum_{n=1}^{\infty} T_n''(t) X_n(x) - a^2 \sum_{n=1}^{\infty} T_n(t) X_n''(x) &= \sum_{n=1}^{\infty} f_n X_n(x) \\ \sum_{n=1}^{\infty} T_n''(t) X_n(x) - a^2 \sum_{n=1}^{\infty} T_n(t) [-\lambda_n X_n(x)] &= \sum_{n=1}^{\infty} f_n X_n(x) \\ \sum_{n=1}^{\infty} [T_n''(t) + a^2 \lambda_n T_n(t)] X_n(x) &= \sum_{n=1}^{\infty} f_n X_n(x) \end{aligned}$$

初值问题ODE $T_n''(t) + a^2 \lambda_n T_n(t) = f_n$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x\right), (n=1, 2, 3, \dots)$$

由PDE初始条件

$$v(x, 0) = \sum_{n=1}^{\infty} T_n(0) X_n(x) = \sum_{n=1}^{\infty} \phi_n X_n(x)$$

$$v_t(x, 0) = \sum_{n=1}^{\infty} T_n'(0) X_n(x) = \sum_{n=1}^{\infty} \psi_n X_n(x)$$

初值问题ODE定解问题为

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = \phi_n, T_n'(0) = \psi_n \end{cases}$$

2.2 分离变量法

一维波动定解问题

例4
$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0,t) = 0, u(l,t) = F \sin \omega t, & t \geq 0 \\ u(x,0) = f(x), u_t(x,0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐1非齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1,2,3,\dots)$$
$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x,t)$$
$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F\omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0,t) = 0, v(l,t) = 0, & t \geq 0 \\ v(x,0) = f(x), v_t(x,0) = -\frac{F\omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐1齐

第5步 求初值问题, 关键是确定 $\bar{T}_n$ 形式

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$$

二阶ODE非齐次项含三角函数时特解取法

$$\ddot{x} + a_1 \dot{x} + a_2 x = e^{\mu t} \varphi(t) \sin vt$$

$\downarrow$

$$\bar{x} = t^k e^{\mu t} [Z_1(t) \cos vt + Z_2(t) \sin vt]$$

$\downarrow$

$$s^2 + a_1 s + a_2 = 0, \mu \pm i v \begin{cases} \text{非特征根 } k=0 \\ \text{是特征根 } k=1 \end{cases}$$

$Z_1(t), Z_2(t)$ 是与 $\varphi(t)$ 同次幂实多项式

$$\begin{cases} T_n(0) = \varphi_n, T'_n(0) = \psi_n \\ T''_n(t) + a^2 \lambda_n T_n(t) = f_n = f_A + e^{0t} f_B \sin \omega t \end{cases}$$

特征方程 $s^2 + a^2 \lambda_n = 0$

特征根 $s = \pm a\sqrt{\lambda_n} i = \pm \omega_n i$

数学 $\omega = a\sqrt{\lambda_n}$ 物理 $\omega = \omega_n = a\sqrt{\lambda_n}$ **n次固有频率**

是不是特征根? $\omega^2 = \omega_n^2 = a^2 \lambda_n$ 是不是固有频率?

辅助等式 $\rightarrow$ 后续共振情形解的化简

ω 不等于任意频率 $\omega^2 \neq \omega_n^2 = a^2 \lambda_n$
无共振, 所有谐波对应的特解相同

ω 等于k次固有频率 $\omega^2 = \omega_k^2 = a^2 \lambda_k$
有共振, k次谐波对应的特解与众不同

$$\bar{T}_n = \begin{cases} \bar{C}_{1n} + \bar{C}_{2n} \cos \omega t + \bar{C}_{3n} \sin \omega t & (n \neq k) \\ \bar{C}_{1k} + t(\bar{C}_{2k} \cos \omega t + \bar{C}_{3k} \sin \omega t) & (n = k) \end{cases}$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例4

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1非齐

$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

第5步 求初值问题 情形1 非共振

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n = f_A + f_B \sin \omega t \\ T_n(0) = \varphi_n, T_n'(0) = \psi_n \end{cases}$$

通解=齐次通解+非齐特解

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$$

$$\bar{T}_n = \bar{C}_{1n} + \bar{C}_{2n} \cos \omega t + \bar{C}_{3n} \sin \omega t$$

根据自由项形式, 观察法特解可化简

$$\bar{T}_n = \bar{C}_{1n} + \bar{C}_{2n} \sin \omega t$$

各阶导数

$$\bar{T}_n' = \omega \bar{C}_{2n} \cos \omega t$$

$$\bar{T}_n'' = -\omega^2 \bar{C}_{2n} \sin \omega t$$

$$\begin{aligned} f_n &= \frac{2A}{n\pi} [1 - (-1)^n] + (-1)^{n+1} \frac{2F \omega^2}{n\pi} \sin \omega t \\ &= f_A + f_B \sin \omega t \end{aligned}$$

代入ODE

$$\begin{aligned} T_n''(t) + a^2 \lambda_n T_n(t) &= f_n \\ -\omega^2 \bar{C}_{2n} \sin \omega t + a^2 \lambda_n (\bar{C}_{1n} + \bar{C}_{2n} \sin \omega t) &= f_A + f_B \sin \omega t \\ a^2 \lambda_n \bar{C}_{1n} + (a^2 \lambda_n - \omega^2) \bar{C}_{2n} \sin \omega t &= f_A + f_B \sin \omega t \end{aligned}$$

特解系数

$$\bar{C}_{1n} = \frac{f_A}{a^2 \lambda_n}, \quad \bar{C}_{2n} = \frac{f_B}{a^2 \lambda_n - \omega^2}$$

特解

$$\bar{T}_n = \frac{f_A}{a^2 \lambda_n} + \frac{f_B}{a^2 \lambda_n - \omega^2} \sin \omega t$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例4

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1非齐

$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1齐

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right), (n=1, 2, 3, \dots)$$

第5步 求初值问题

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = \varphi_n, T_n'(0) = \psi_n \end{cases}$$

ODE定解问题 通解

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$$

$$= C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \frac{f_A}{a^2 \lambda_n} + \frac{f_B}{a^2 \lambda_n - \omega^2} \sin \omega t$$

$$T_n'(t) = -a\sqrt{\lambda_n} C_{1n} \sin(a\sqrt{\lambda_n}t) + a\sqrt{\lambda_n} C_{2n} \cos(a\sqrt{\lambda_n}t) + \frac{\omega f_B}{a^2 \lambda_n - \omega^2} \cos \omega t$$

$$T_n(0) = C_{1n} + \frac{f_A}{a^2 \lambda_n} = \varphi_n$$

$$C_{1n} = \varphi_n - \frac{f_A}{a^2 \lambda_n}$$

$$T_n'(0) = a\sqrt{\lambda_n} C_{2n} + \frac{\omega f_B}{a^2 \lambda_n - \omega^2} = \psi_n$$

$$C_{2n} = \frac{1}{a\sqrt{\lambda_n}} \left(\psi_n - \frac{\omega f_B}{a^2 \lambda_n - \omega^2} \right)$$

$$\begin{aligned} f_n &= \frac{2A}{n\pi} \left[1 - (-1)^n \right] \\ &\quad + (-1)^{n+1} \frac{2F\omega^2}{n\pi} \sin \omega t \\ &= f_A + f_B \sin \omega t \end{aligned}$$

非共振时初值ODE的解

$$T_n(t) = \left(\varphi_n - \frac{f_A}{a^2 \lambda_n} \right) \cos(a\sqrt{\lambda_n}t) + \left[\frac{1}{a\sqrt{\lambda_n}} \left(\psi_n - \frac{\omega f_B}{a^2 \lambda_n - \omega^2} \right) \right] \sin(a\sqrt{\lambda_n}t) + \frac{f_A}{a^2 \lambda_n} + \frac{f_B}{a^2 \lambda_n - \omega^2} \sin \omega t$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例4

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐	1非齐
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$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐	1齐
----	----

第6步 回代 情形1 非共振原定解问题的解

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

$$u(x, t) = w(x, t) + v(x, t)$$

$$= \frac{F \sin \omega t}{l} x + \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l}x\right)$$

加齐次化辅助函数

$$= \frac{F \sin \omega t}{l} x + \sum_{n=1}^{\infty} \left[\left(\varphi_n - \frac{f_A}{a^2 \lambda_n} \right) \cos(a\sqrt{\lambda_n}t) + \frac{1}{a\sqrt{\lambda_n}} \left(\psi_n - \frac{\omega f_B}{a^2 \lambda_n - \omega^2} \right) \sin(a\sqrt{\lambda_n}t) + \frac{f_A}{a^2 \lambda_n} + \frac{f_B}{a^2 \lambda_n - \omega^2} \sin \omega t \right] \sin\left(\frac{n\pi}{l}x\right)$$

$$\begin{aligned} f_n &= \frac{2A}{n\pi} [1 - (-1)^n] + (-1)^{n+1} \frac{2F\omega^2}{n\pi} \sin \omega t \\ &= f_A + f_B \sin \omega t \\ \varphi_n &= \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx \\ \psi_n &= (-1)^n \frac{2F\omega}{n\pi} \end{aligned}$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

例4
$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐	
1齐	1非齐

$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐	
1齐	1齐

第5步 求初值问题 情形2 共振

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = \varphi_n, T_n'(0) = \psi_n \end{cases}$$

通解=齐次通解+非齐特解

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$$

$$\bar{T}_n = \begin{cases} \bar{C}_{1n} + \bar{C}_{2n} \cos \omega t + \bar{C}_{3n} \sin \omega t & (n \neq k) \\ \bar{C}_1 + t(\bar{C}_2 \cos \omega t + \bar{C}_3 \sin \omega t) & (n = k) \end{cases}$$

若自由项频率=任一固有频率 $\omega^2 = \omega_k^2 = a^2 \lambda_k$

即可发生共振, 不妨取 $\omega^2 = a^2 \lambda_k$

特解形式应为 $\bar{T}_k = \bar{C}_1 + t(\bar{C}_2 \cos \omega t + \bar{C}_3 \sin \omega t)$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

$$f_n = \frac{2A}{n\pi} [1 - (-1)^n] + (-1)^{n+1} \frac{2F \omega^2}{n\pi} \sin \omega t = f_A + f_B \sin \omega t$$

各阶导数 $\bar{T}_k' = (\bar{C}_2 \cos \omega t + \bar{C}_3 \sin \omega t) + t(-\omega \bar{C}_2 \sin \omega t + \omega \bar{C}_3 \cos \omega t)$

$$\begin{aligned} \bar{T}_k'' &= (-\omega \bar{C}_2 \sin \omega t + \omega \bar{C}_3 \cos \omega t) \\ &\quad + (-\omega \bar{C}_2 \sin \omega t + \omega \bar{C}_3 \cos \omega t) \\ &\quad + t(-\omega^2 \bar{C}_2 \cos \omega t - \omega^2 \bar{C}_3 \sin \omega t) \\ &= -2\omega \bar{C}_2 \sin \omega t + 2\omega \bar{C}_3 \cos \omega t \\ &\quad - \omega^2 t \bar{C}_2 \cos \omega t - \omega^2 t \bar{C}_3 \sin \omega t \\ &= (2\omega \bar{C}_3 - \omega^2 t \bar{C}_2) \cos \omega t - (\omega^2 t \bar{C}_3 + 2\omega \bar{C}_2) \sin \omega t \end{aligned}$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例4
$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐
1齐 1非齐

$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐
1齐 1齐

第5步 求初值问题 情形2 共振

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = \varphi_n, T_n'(0) = \psi_n \end{cases}$$

$$f_n = \frac{2A}{n\pi} [1 - (-1)^n] + (-1)^{n+1} \frac{2F \omega^2}{n\pi} \sin \omega t = f_A + f_B \sin \omega t$$

$$\bar{T}_k = \bar{C}_1 + t(\bar{C}_2 \cos \omega t + \bar{C}_3 \sin \omega t)$$

$$\bar{T}_k'' = (2\omega \bar{C}_3 - \omega^2 t \bar{C}_2) \cos \omega t - (\omega^2 t \bar{C}_3 + 2\omega \bar{C}_2) \sin \omega t$$

代入初值ODE

$$(2\omega \bar{C}_3 - \omega^2 t \bar{C}_2) \cos \omega t - (\omega^2 t \bar{C}_3 + 2\omega \bar{C}_2) \sin \omega t + a^2 \lambda_k [\bar{C}_1 + t(\bar{C}_2 \cos \omega t + \bar{C}_3 \sin \omega t)] = f_A + f_B \sin \omega t$$

$$(2\omega \bar{C}_3 - \omega^2 t \bar{C}_2 + a^2 \lambda_k t \bar{C}_2) \cos \omega t + a^2 \lambda_k \bar{C}_1 + (a^2 \lambda_k t \bar{C}_3 - \omega^2 t \bar{C}_3 - 2\omega \bar{C}_2) \sin \omega t = f_A + f_B \sin \omega t$$

$$\omega^2 = a^2 \lambda_k$$

$$2\omega \bar{C}_3 \cos \omega t + a^2 \lambda_k \bar{C}_1 - 2\omega \bar{C}_2 \sin \omega t = f_A + f_B \sin \omega t$$

↓ 比较系数

特解
$$\bar{T}_k = \frac{f_A}{a^2 \lambda_k} - \frac{f_B}{2\omega} t \cos \omega t$$

←
$$\bar{C}_1 = \frac{f_A}{a^2 \lambda_k}, \bar{C}_2 = -\frac{f_B}{2\omega}, \bar{C}_3 = 0$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例4

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1非齐

$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1齐

第5步 求初值问题

情形2 共振

$$\begin{cases} T_k''(t) + a^2 \lambda_k T_k(t) = f_k \\ T_k(0) = \varphi_k, T_k'(0) = \psi_k \end{cases}$$

$$f_n = \frac{2A}{n\pi} [1 - (-1)^n] + (-1)^{n+1} \frac{2F \omega^2}{n\pi} \sin \omega t = f_A + f_B \sin \omega t$$

共振时的通解

$$T_k(t) = C_{11} \cos(a \sqrt{\lambda_k} t) + C_{21} \sin(a \sqrt{\lambda_k} t) + \frac{f_A}{a^2 \lambda_k} - \frac{f_B}{2\omega} t \cos \omega t$$

$$T_k'(t) = -a \sqrt{\lambda_k} C_{11} \sin(a \sqrt{\lambda_k} t) + a \sqrt{\lambda_k} C_{21} \cos(a \sqrt{\lambda_k} t) + \frac{f_B}{2\omega} \cos \omega t + \frac{f_B}{2} t \sin \omega t$$

代入初始条件

$$T_k(0) = C_{11} + \frac{f_A}{a^2 \lambda_k} = \varphi_k \quad C_{11} = \varphi_k - \frac{f_A}{a^2 \lambda_k}$$

$$T_k'(0) = a \sqrt{\lambda_k} C_{21} + \frac{f_B}{2\omega} = \psi_k \quad C_{21} = \frac{1}{a \sqrt{\lambda_k}} \left(\psi_k - \frac{f_B}{2\omega} \right)$$

除 $n = k$ 外其他阶次谐波未共振, $T_n(t)$ 的解与情形1非共振完全相同

$$T_k(t) = \left(\varphi_k - \frac{f_A}{a^2 \lambda_k} \right) \cos(a \sqrt{\lambda_k} t) + \frac{1}{a \sqrt{\lambda_k}} \left(\psi_k - \frac{f_B}{2\omega} \right) \sin(a \sqrt{\lambda_k} t) + \frac{f_A}{a^2 \lambda_k} - \frac{f_B}{2\omega} t \cos \omega t$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例4

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1非齐

$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1齐

第6步 回代 情形2 共振原定解问题的解

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right), (n=1, 2, 3, \dots)$$

$$\begin{aligned} u(x, t) &= w(x, t) + v(x, t) \\ &= \frac{F \sin \omega t}{l} x + T_k(t) \sin\left(\frac{k\pi}{l} x \right) + \sum_{n=1, n \neq k}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l} x \right) \end{aligned}$$

$n=k$ 单列

$$\begin{aligned} &= \frac{F \sin \omega t}{l} x + \left[\left(\varphi_k - \frac{f_A}{a^2 \lambda_k} \right) \cos(a \sqrt{\lambda_k} t) + \frac{1}{a \sqrt{\lambda_k}} \left(\psi_k - \frac{f_B}{2\omega} \right) \sin(a \sqrt{\lambda_k} t) + \frac{f_A}{a^2 \lambda_k} - \frac{f_B}{2\omega} t \cos \omega t \right] \sin\left(\frac{k\pi}{l} x \right) \\ &\quad + \sum_{n=1, n \neq k}^{\infty} \left[\left(\varphi_n - \frac{f_A}{a^2 \lambda_n} \right) \cos(a \sqrt{\lambda_n} t) + \frac{1}{a \sqrt{\lambda_n}} \left(\psi_n - \frac{\omega f_B}{a^2 \lambda_n - \omega^2} \right) \sin(a \sqrt{\lambda_n} t) \right. \\ &\quad \left. + \frac{f_A}{a^2 \lambda_n} + \frac{f_B}{a^2 \lambda_n - \omega^2} \sin \omega t \right] \sin\left(\frac{n\pi}{l} x \right) \end{aligned}$$

$$\begin{aligned} f_n &= \frac{2A}{n\pi} \left[1 - (-1)^n \right] + (-1)^{n+1} \frac{2F \omega^2}{n\pi} \sin \omega t \\ &= f_A + f_B \sin \omega t \end{aligned}$$

$$\varphi_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \psi_n = (-1)^n \frac{2F \omega}{n\pi}$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例4

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1非齐

$$\begin{cases} v_{tt} - a^2 v_{xx} = A + \frac{F \omega^2 \sin \omega t}{l} x, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x), v_t(x, 0) = -\frac{F \omega x}{l}, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1齐

第6步 回代 情形2 共振原定解问题的解

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

利用 $\omega^2 = a^2 \lambda_k$
 $\omega_n^2 = a^2 \lambda_n$ 可进一步化简

$$\begin{aligned} f_n &= \frac{2A}{n\pi} [1 - (-1)^n] + (-1)^{n+1} \frac{2F \omega^2}{n\pi} \sin \omega t \\ &= f_A + f_B \sin \omega t \\ \varphi_n &= \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \psi_n = (-1)^n \frac{2F \omega}{n\pi} \end{aligned}$$

$$\begin{aligned} u(x, t) &= w(x, t) + v(x, t) \\ &= \frac{F \sin \omega t}{l} x + \left[\left(\varphi_k - \frac{f_A}{\omega^2} \right) \cos \omega t + \frac{1}{\omega} \left(\psi_k - \frac{f_B}{2\omega} \right) \sin \omega t + \frac{f_A}{\omega^2} - \frac{f_B}{2\omega} t \cos \omega t \right] \sin \left(\frac{k\pi}{l} x \right) \\ &\quad + \sum_{n=1, n \neq k}^{\infty} \left[\left(\varphi_n - \frac{f_A}{\omega_n^2} \right) \cos \omega_n t + \frac{1}{\omega_n} \left(\psi_n - \frac{\omega f_B}{\omega_n^2 - \omega^2} \right) \sin \omega_n t \right. \\ &\quad \left. + \frac{f_A}{\omega_n^2} + \frac{f_B}{\omega_n^2 - \omega^2} \sin \omega t \right] \sin \left(\frac{n\pi}{l} x \right) \end{aligned}$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$



2.2 分离变量法

一维波动定解问题

波动/一维/非齐

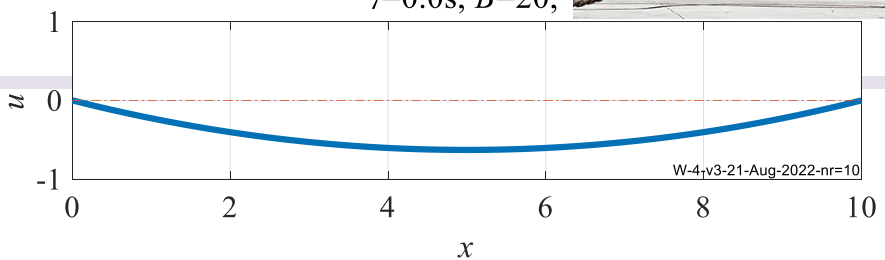
1齐	1非齐
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例4

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

形状因子

$\tau=0.0s, B=20,$



讨论 假设位移初始分布为悬链线

$$f(x) = B \cosh\left[\frac{1}{B}\left(x - \frac{l}{2}\right)\right] - B \cosh\left(\frac{l}{2B}\right)$$

按特征函数系展开

$$f(x) = \sum_{n=1}^{\infty} \varphi_n X_n(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi}{l}x\right) \quad \varphi_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx$$

$$\int e^{ax} \sin bxdx = \frac{e^{ax} (a \sin bx - b \cos bx)}{a^2 + b^2} + C$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$f(x) = \frac{B}{2} \overset{\textcircled{1}}{\exp\left(-\frac{l}{2B}\right)} \exp\left(\frac{x}{B}\right) + \frac{B}{2} \overset{\textcircled{2}}{\exp\left(\frac{l}{2B}\right)} \exp\left(-\frac{x}{B}\right) - B \overset{\textcircled{3}}{\cosh\left(\frac{l}{2B}\right)}$$

$$\left\{ \begin{aligned} \varphi_{n1} &= \left[(-1)^{n+1} \exp\left(\frac{l}{B}\right) + 1 \right] \frac{B}{\frac{1}{B^2} + \left(\frac{n\pi}{l}\right)^2} \frac{n\pi}{l^2} \exp\left(-\frac{l}{2B}\right) \\ \varphi_{n2} &= \left[(-1)^{n+1} \exp\left(-\frac{l}{B}\right) + 1 \right] \frac{B}{\frac{1}{B^2} + \left(\frac{n\pi}{l}\right)^2} \frac{n\pi}{l^2} \exp\left(\frac{l}{2B}\right) \\ \varphi_{n3} &= \left[(-1)^n - 1 \right] \frac{2}{n\pi} B \cosh\left(\frac{l}{2B}\right) \end{aligned} \right.$$

$$\varphi_n = \frac{2B}{\frac{1}{B^2} + \left(\frac{n\pi}{l}\right)^2} \frac{n\pi}{l^2} \left[(-1)^{n+1} \cosh\left(\frac{l}{2B}\right) + \cosh\left(\frac{l}{2B}\right) \right] + \left[(-1)^n - 1 \right] \frac{2}{n\pi} B \cosh\left(\frac{l}{2B}\right)$$

2.2 分离变量法

一维波动定解问题

波动/一维/非齐

1齐

1非齐

例4

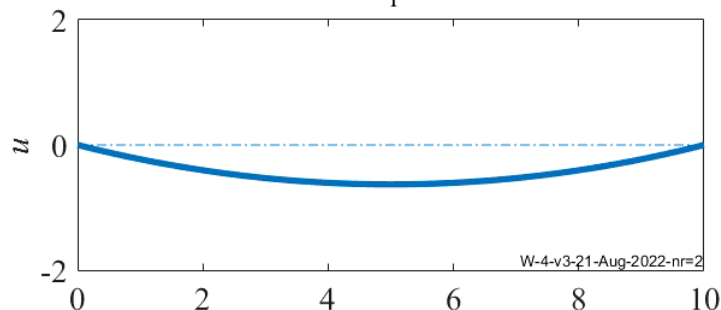
$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = F \sin \omega t, & t \geq 0 \\ u(x, 0) = f(x), u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

讨论

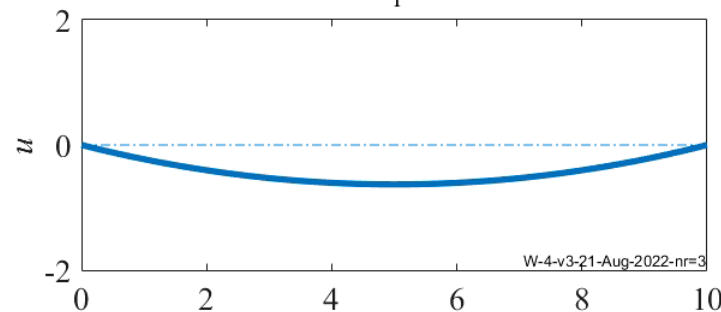
$$f(x) = B \cosh \left[\frac{1}{B} \left(x - \frac{l}{2} \right) \right] - B \cosh \left(\frac{l}{2B} \right)$$



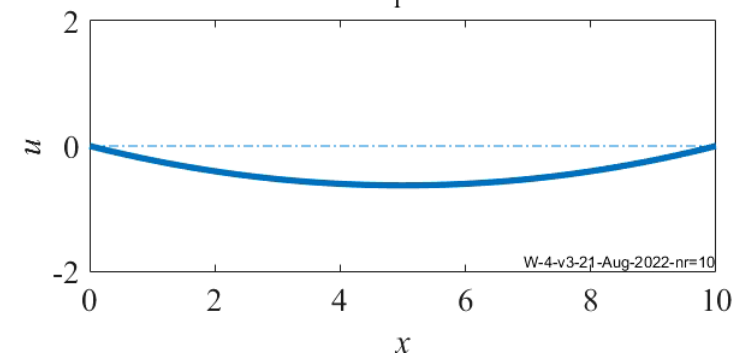
$\tau=0.0s, a=5, n_r=2, \omega=3.1416$



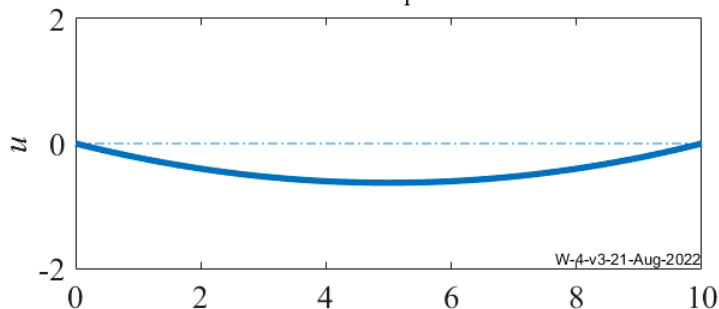
$\tau=0.0s, a=5, n_r=3, \omega=4.7124$



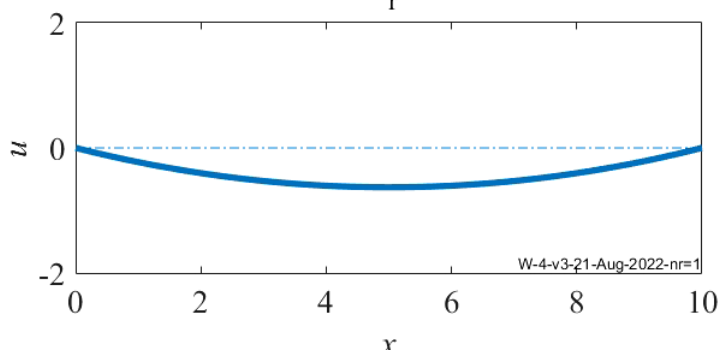
$\tau=0.0s, a=5, n_r=10, \omega=15.708$



$\tau=0.0s, a=5, n_r=\text{NaN}, \omega=3$



$\tau=0.0s, a=5, n_r=1, \omega=1.5708$





2.2 分离变量法

技巧小结

1. 复杂的级数解尽量用中间变量表示→减少笔误/易于程序化

$$\begin{aligned}
 u(x, t) &= w(x, t) + v(x, t) \\
 &= \frac{F \sin \omega t}{l} x + T_k(t) \sin\left(\frac{k\pi}{l} x\right) + \sum_{n=1, n \neq k}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l} x\right) \\
 &= \frac{F \sin \omega t}{l} x + \left[\left(\varphi_k - \frac{f_A}{a^2 \lambda_k} \right) \cos(a\sqrt{\lambda_k} t) + \frac{1}{a\sqrt{\lambda_k}} \left(\psi_k - \frac{f_B}{2\omega} \right) \sin(a\sqrt{\lambda_k} t) + \frac{f_A}{a^2 \lambda_k} + \frac{f_B}{2\omega} t \cos \omega t \right] \sin\left(\frac{k\pi}{l} x\right) \\
 &\quad + \sum_{n=1, n \neq k}^{\infty} \left\{ \left(\varphi_n - \frac{f_A}{a^2 \lambda_n} \right) \cos(a\sqrt{\lambda_n} t) + \frac{1}{a\sqrt{\lambda_n}} \left(\psi_n - \frac{\omega f_B}{a^2 \lambda_n - \omega^2} \right) \sin(a\sqrt{\lambda_n} t) + \frac{f_A}{a^2 \lambda_n} + \frac{f_B}{a^2 \lambda_n - \omega^2} \sin \omega t \right\} \sin\left(\frac{n\pi}{l} x\right)
 \end{aligned}$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x\right), (n=1, 2, 3, \dots)$$

$$f_n = \frac{2A}{n\pi} [1 - (-1)^n] + (-1)^{n+1} \frac{2F\omega^2}{n\pi} \sin \omega t$$

$$= f_A + f_B \sin \omega t$$

$$\varphi_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \psi_n = (-1)^n \frac{2F\omega}{n\pi}$$

2. 几个常用的简化表达式 $\cos(n\pi) = (-1)^n, \sin \frac{(2n+1)\pi}{2} = (-1)^n, n = 0, 1, 2, \dots$

3. 特征函数包含1时→其平方模与sin/cos不同→计算小心/级数结果单列

4. 若初始分布恰好为某个特征函数→可仅针对该 n 值对应的特征函数展开

5. 波动方程自由项/边界条件为周期性函数→考虑共振→本质是初值ODE特解取法

6. 量纲正确是解正确的必要条件 $\left[\frac{f_A}{a^2 \lambda_k} \right] = \left[\frac{\text{N}}{\text{kg} \cdot \text{Hz}^2} \right] = \left[\frac{\text{kg} \cdot \text{m} \cdot \text{s}^2}{\text{kg} \cdot \text{s}^2} \right] = [\text{m}] = [u]$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例5 一根长为 l 的细弦两端固定, 该弦在周期性外力作用下发生横振动, 则可使该弦发生共振的**最小频率**为

$$\omega_1 = a\sqrt{\lambda_1} = a\sqrt{\left(\frac{1 \cdot \pi}{l}\right)^2} = \frac{a\pi}{l}$$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right) \\ (n=1, 2, 3 \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right] \\ (n=0, 1, 2, 3 \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right] \\ (n=0, 1, 2, 3 \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right) \\ (n=0, 1, 2, 3 \dots)$

1. 数学建模和基本原理介绍

2. 分离变量法

3. Bessel函数

4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

2.1 特征值问题

2.2 分离变量法

2.3 基于MATLAB的定解问题求解方法

弦振动方程定解问题

热传导方程定解问题

平面位势方程边值问题

2.2 分离变量法

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维导热定解问题

例1 齐次恒温边界下的热传导

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = x(l - x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1齐1齐

解 第1步 特征值问题, 设 $u(x, t) = X(x)T(t)$

代入对应的齐次方程

$$X(x)T'(t) - a^2 X''(x)T(t) = 0$$
$$\frac{X''(x)}{X(x)} = \frac{T'(t)}{a^2 T(t)} = -\lambda$$

1阶导数

$$T'(t) + \lambda a^2 T(t) = 0, X''(x) + \lambda X(x) = 0$$

齐次边界 $T(t)$ 不恒为0 $X(0) = X(l) = 0$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$ ($n = 1, 2, 3 \dots$)
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n = 0, 1, 2, 3 \dots$)
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n = 0, 1, 2, 3 \dots$)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right)$ ($n = 0, 1, 2, 3 \dots$)

特征值/特征函数

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n = 1, 2, 3 \dots)$$

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维导热定解问题

例1 齐次恒温边界下的热传导

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = x(l - x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次	
1齐	1齐

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right), \quad (n=1, 2, 3, \dots)$$

第2步 正交分解, 设定解问题的解可表示为

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l} x \right)$$

从1开始

自由项/初始条件按特征函数系展开

$$x(l - x) = \sum_{n=1}^{\infty} \varphi_n X_n(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi}{l} x \right)$$

$$\begin{aligned} \varphi_n &= \frac{2}{l} \int_0^l x(l - x) \sin\left(\frac{n\pi}{l} x \right) dx = \frac{2}{l} \left[l \int_0^l x \sin \frac{n\pi x}{l} \cdot dx - \int_0^l x^2 \sin \frac{n\pi x}{l} \cdot dx \right] \\ &= \frac{4l^2}{\pi^3} \cdot \frac{1 - \cos n\pi}{n^3} = \frac{4l^2}{\pi^3} \cdot \frac{1 - (-1)^n}{n^3} \end{aligned}$$

定义在 $[0, l]$ 上函数 $f(x)$ 的Fourier展开/正交分解

$$\begin{aligned} f(x) &= f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x) \\ f_0 &= \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx, \\ f_n &= \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx \end{aligned}$$

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维导热定解问题

例1 齐次恒温边界下的热传导

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = x(l - x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次	
1齐	1齐

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, \quad X_n(x) = \sin\left(\frac{n\pi}{l} x \right), \quad (n=1, 2, 3, \dots)$$

第3步 建立初值ODE

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l} x \right)$$

假设可逐项求导, 代入原方程

$$\sum_{n=1}^{\infty} T_n'(t) X_n(x) - a^2 \sum_{n=1}^{\infty} T_n(t) X_n''(x) = 0$$

$$\sum_{n=1}^{\infty} T_n'(t) X_n(x) - a^2 \sum_{n=1}^{\infty} T_n(t) [-\lambda_n X_n(x)] = 0$$

$$\sum_{n=1}^{\infty} [T_n'(t) + a^2 \lambda_n T_n(t)] X_n(x) = 0$$

所以 $T_n'(t) + a^2 \lambda_n T_n(t) = 0$

根据初始条件

$$u(x, 0) = \sum_{n=1}^{\infty} T_n(0) X_n(x) = \sum_{n=1}^{\infty} \varphi_n X_n(x)$$

初值问题可表示为

$$\begin{cases} T_n'(t) + a^2 \lambda_n T_n(t) = 0 \\ T_n(0) = \varphi_n \end{cases}$$

$$X_n''(x) + \lambda_n X_n(x) = 0$$

2.2 分离变量法

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维导热定解问题

例1 齐次恒温边界下的热传导

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = x(l - x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1齐

1齐

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right), \quad (n=1, 2, 3, \dots)$$

第4步 求初值问题

$$\begin{cases} T_n'(t) + a^2 \lambda_n T_n(t) = 0 \\ T_n(0) = \varphi_n \end{cases}$$

一阶线性齐次ODE, 分离变量法

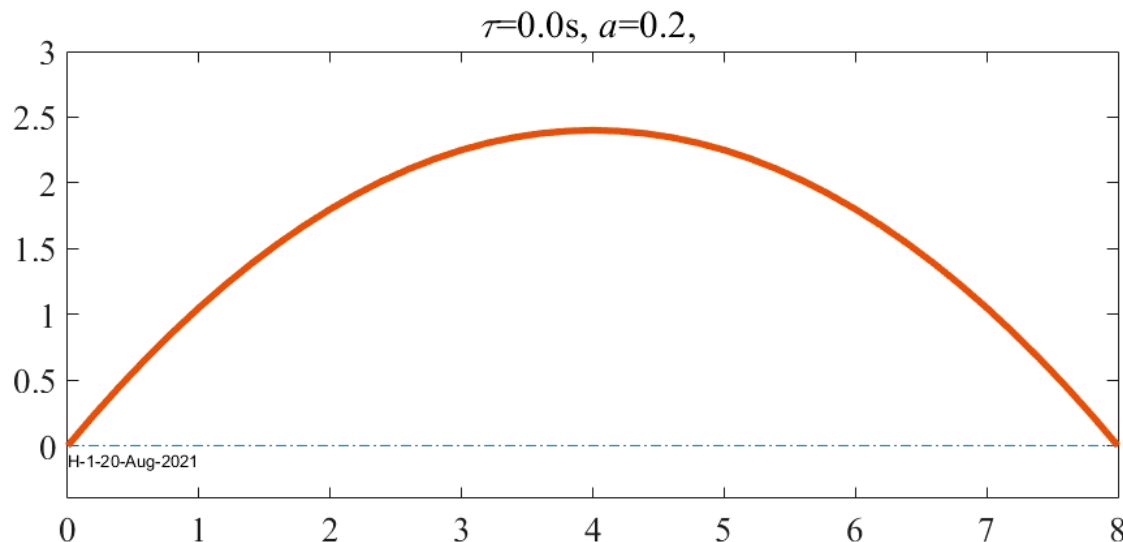
$$T_n(t) = C_{1n} \exp(-a^2 \lambda_n t)$$

根据初始条件 $T_n(0) = C_{1n} = \varphi_n$

$$T_n(t) = \frac{4l^2}{\pi^3} \cdot \frac{1 - (-1)^n}{n^3} \exp(-a^2 \lambda_n t)$$

原定解问题的解

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \frac{4l^2}{\pi^3} \cdot \sum_{n=1}^{\infty} \left[\frac{1 - \cos n\pi}{n^3} \exp(-a^2 \lambda_n t) \right] \sin\left(\frac{n\pi}{l} x \right)$$



2.2 分离变量法

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x,t)$$

一维导热定解问题

例2

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0,t) = u_0, \quad u_x(l,t) = A(1 + \sin \omega t), & t \geq 0 \\ u(x,0) = u_0, & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1非齐

2非齐

$$\begin{cases} v_t - a^2 v_{xx} = -Ax\omega \cos \omega t, & 0 < x < l, t > 0 \\ v(0,t) = 0, \quad v_x(l,t) = 0, & t \geq 0 \\ v(x,0) = -Ax, & 0 \leq x \leq l \end{cases}$$

导热/一维/非齐

1齐

2齐

解 第1步 边界条件齐次化

设 $u(x,t) = v(x,t) + w(x,t)$

可取 $w(x,t) = u_0 + A(1 + \sin \omega t)x$

各阶导数

$$\begin{aligned} w_t &= Ax\omega \cos \omega t \\ w_x &= A(1 + \sin \omega t) \\ w_{xx} &= 0 \end{aligned}$$

改造PDE

$$\begin{aligned} u_t - a^2 u_{xx} &= 0 \\ v_t + w_t - a^2 (v_{xx} + w_{xx}) &= 0 \\ v_t + Ax\omega \cos \omega t - a^2 (v_{xx} + 0) &= 0 \\ v_t - a^2 v_{xx} &= -Ax\omega \cos \omega t \end{aligned}$$

改造初始条件

$$\begin{aligned} u(x,0) &= u_0 \\ v(x,0) + w(x,0) &= u_0 \\ v(x,0) + u_0 + Ax &= u_0 \\ v(x,0) &= -Ax \end{aligned}$$

边界条件齐次化常用辅助函数 $u(x,t) = v(x,t) + w(x,t)$		
边界类型	边界条件	辅助函数
[1,1]	$u(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{l}x + g_1(t)$
[2,1]	$u_x(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = g_1(t)(x-l) + g_2(t)$
[1,2]	$u(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = g_2(t)x + g_1(t)$
[2,2]	$u_x(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{2l}x^2 + g_1(t)x$

2.2 分离变量法

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维导热定解问题

例2

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = u_0, \quad u_x(l, t) = A(1 + \sin \omega t), & t \geq 0 \\ u(x, 0) = u_0, & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1非齐

2非齐

$$\begin{cases} v_t - a^2 v_{xx} = -Ax\omega \cos \omega t, & 0 < x < l, t > 0 \\ v(0, t) = 0, \quad v_x(l, t) = 0, & t \geq 0 \\ v(x, 0) = -Ax, & 0 \leq x \leq l \end{cases}$$

导热/一维/非齐

1齐

2齐

第2步 特征值问题, 设 $v(x, t) = X(x)T(t)$

代入对应的齐次方程

$$X(x)T'(t) - a^2 X''(x)T(t) = 0$$

$$\frac{X''(x)}{X(x)} = \frac{T'(t)}{a^2 T(t)} = -\lambda$$

$$T'(t) + \lambda a^2 T(t) = 0, \quad X''(x) + \lambda X(x) = 0$$

齐次边界 $T(t)$ 不恒为0 $X(0) = X'(l) = 0$

特征值/特征函数

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, \quad X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right], \quad (n = 0, 1, 2, 3 \dots)$$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin \left(\frac{n\pi}{l} x \right)$ ($n = 1, 2, 3 \dots$)
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right]$ ($n = 0, 1, 2, 3 \dots$)
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right]$ ($n = 0, 1, 2, 3 \dots$)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \cos \left(\frac{n\pi}{l} x \right)$ ($n = 0, 1, 2, 3 \dots$)

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维导热定解问题

例2 $\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = u_0, & u_x(l, t) = A(1 + \sin \omega t), & t \geq 0 \\ u(x, 0) = u_0, & 0 \leq x \leq l \end{cases}$

导热/一维/齐次

1非齐 2非齐

$$\begin{cases} v_t - a^2 v_{xx} = -Ax\omega \cos \omega t, & 0 < x < l, t > 0 \\ v(0, t) = 0, & v_x(l, t) = 0, & t \geq 0 \\ v(x, 0) = -Ax, & 0 \leq x \leq l \end{cases}$$

导热/一维/非齐

1齐 2齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3, \dots)$$

第3步 正交分解 设定解问题形式解

$$v(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \sin \frac{(2n+1)\pi}{2l} x$$

定义在 $[0, l]$ 上函数 $f(x)$ 的Fourier展开/正交分解

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x)$$

$$f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx,$$

$$f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

自由项/初始条件按特征函数系展开

$$-Ax\omega \cos \omega t = \sum_{n=0}^{\infty} f_n X_n(x) = \sum_{n=0}^{\infty} f_n \sin \frac{(2n+1)\pi}{2l} x$$

$$f_n = \frac{2}{l} \int_0^l -Ax\omega \cos \omega t \cdot \sin \frac{(2n+1)\pi}{2l} x dx = -A\omega \cos \omega t \cdot \frac{2}{l} \int_0^l x \cdot \sin \frac{(2n+1)\pi}{2l} x dx$$

$$= -A\omega \cos \omega t \cdot \frac{2}{l} \left[\frac{4l^2}{(2n+1)^2 \pi^2} \sin \frac{(2n+1)\pi}{2} \right] = \frac{(-1)^{n+1}}{(2n+1)^2} \cdot \frac{8A\omega l}{\pi^2} \cdot \cos \omega t = k_n \cdot \cos \omega t$$

$$\sin \frac{(2n+1)\pi}{2} = (-1)^n$$

简化后续表达式

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维导热定解问题

导热/一维/齐次

1非齐 2非齐

例2

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = u_0, & u_x(l, t) = A(1 + \sin \omega t), t \geq 0 \\ u(x, 0) = u_0, & 0 \leq x \leq l \end{cases}$$

$$\begin{cases} v_t - a^2 v_{xx} = -Ax\omega \cos \omega t, & 0 < x < l, t > 0 \\ v(0, t) = 0, & v_x(l, t) = 0, t \geq 0 \\ v(x, 0) = -Ax, & 0 \leq x \leq l \end{cases}$$

导热/一维/非齐

1齐

2齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3, \dots)$$

第3步 正交分解

初始条件按特征函数系展开

$$-Ax = \sum_{n=0}^{\infty} \varphi_n X_n(x) = \sum_{n=0}^{\infty} \varphi_n \sin \frac{(2n+1)\pi}{2l} x$$

$$\varphi_n = \frac{2}{l} \int_0^l -Ax \cdot \sin \frac{(2n+1)\pi}{2l} x dx$$

$$= -\frac{2A}{l} \left[-\frac{2l}{(2n+1)\pi} x \cos \frac{(2n+1)\pi}{2l} x + \frac{4l^2}{(2n+1)^2 \pi^2} \sin \frac{(2n+1)\pi}{2l} x \right]_0^l$$

$$= \frac{(-1)^{n+1}}{(2n+1)^2} \cdot \frac{8Al}{\pi^2}$$

定义在 $[0, l]$ 上函数 $f(x)$ 的Fourier展开/正交分解

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x)$$

$$f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx,$$

$$f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维导热定解问题

导热/一维/齐次

1非齐 2非齐

例2

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = u_0, & u_x(l, t) = A(1 + \sin \omega t), t \geq 0 \\ u(x, 0) = u_0, & 0 \leq x \leq l \end{cases}$$

$$\begin{cases} v_t - a^2 v_{xx} = -Ax\omega \cos \omega t, & 0 < x < l, t > 0 \\ v(0, t) = 0, & v_x(l, t) = 0, t \geq 0 \\ v(x, 0) = -Ax, & 0 \leq x \leq l \end{cases}$$

导热/一维/非齐

1齐

2齐

第4步 建立初值ODE

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3 \dots)$$

$$v(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \sin \frac{(2n+1)\pi}{2l} x$$

根据初始条件

$$v(x, 0) = \sum_{n=1}^{\infty} T_n(0) X_n(x) = -Ax = \sum_{n=1}^{\infty} \varphi_n X_n(x)$$

假设可逐项求导, 代入原方程

$$\sum_{n=0}^{\infty} T'_n(t) X_n(x) - a^2 \sum_{n=0}^{\infty} T_n(t) X''_n(x) = \sum_{n=0}^{\infty} f_n X_n(x)$$

初值问题可表示为

$$\begin{cases} T'_n(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = \varphi_n \end{cases}$$

$$\sum_{n=0}^{\infty} T'_n(t) X_n(x) - a^2 \sum_{n=0}^{\infty} T_n(t) [-\lambda_n X_n(x)] = \sum_{n=0}^{\infty} f_n X_n(x)$$

$$\sum_{n=0}^{\infty} [T'_n(t) + a^2 \lambda_n T_n(t)] X_n(x) = \sum_{n=0}^{\infty} f_n X_n(x)$$

$$\text{所以 } T'_n(t) + a^2 \lambda_n T_n(t) = f_n = k_n \cdot \cos \omega t$$

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维导热定解问题

导热/一维/齐次

1非齐 2非齐

例2

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = u_0, & u_x(l, t) = A(1 + \sin \omega t), t \geq 0 \\ u(x, 0) = u_0, & 0 \leq x \leq l \end{cases}$$

$$\begin{cases} v_t - a^2 v_{xx} = -Ax\omega \cos \omega t, & 0 < x < l, t > 0 \\ v(0, t) = 0, & v_x(l, t) = 0, t \geq 0 \\ v(x, 0) = -Ax, & 0 \leq x \leq l \end{cases}$$

导热/一维/非齐

1齐

2齐

第5步 求初值问题

$$\begin{cases} T_n'(t) + a^2 \lambda_n T_n(t) = f_n = k_n \cdot \cos \omega t \\ T_n(0) = \varphi_n \end{cases}$$

一阶线性非齐次ODE 常数变易法

齐次方程解为 $T_n(t) = C_{1n} \exp(-a^2 \lambda_n t)$

设非齐次方程解为 $T_n(t) = C_{1n}(t) \exp(-a^2 \lambda_n t)$

$$T_n'(t) = C_{1n}'(t) \exp(-a^2 \lambda_n t) - a^2 \lambda_n C_{1n}(t) \exp(-a^2 \lambda_n t)$$

$$T_n'(t) + a^2 \lambda_n T_n(t) = C_{1n}'(t) \exp(-a^2 \lambda_n t) = k_n \cos \omega t$$

系数需满足 $C_{1n}'(t) = k_n \cos \omega t \cdot \exp(a^2 \lambda_n t)$

由积分表 $\int e^{ax} \cos bx = \frac{e^{ax} (a \cos bx + b \sin bx)}{a^2 + b^2} + C$

$$C_{1n}(t) = k_n \frac{\exp(a^2 \lambda_n t) (a^2 \lambda_n \cos \omega t + \omega \sin \omega t)}{a^4 \lambda_n^2 + \omega^2} + C$$

$$T_n(t) = C \exp(-a^2 \lambda_n t) + k_n \frac{a^2 \lambda_n \cos \omega t + \omega \sin \omega t}{a^4 \lambda_n^2 + \omega^2}$$

$$T_n(0) = C + k_n \frac{a^2 \lambda_n}{a^4 \lambda_n^2 + \omega^2} = \varphi_n, \quad C = \varphi_n - k_n \frac{a^2 \lambda_n}{a^4 \lambda_n^2 + \omega^2}$$

$$T_n(t) = \left(\varphi_n - k_n \frac{a^2 \lambda_n}{a^4 \lambda_n^2 + \omega^2} \right) \exp(-a^2 \lambda_n t) + k_n \frac{a^2 \lambda_n \cos \omega t + \omega \sin \omega t}{a^4 \lambda_n^2 + \omega^2}$$

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维导热定解问题

例2

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = u_0, & u_x(l, t) = A(1 + \sin \omega t), & t \geq 0 \\ u(x, 0) = u_0, & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1非齐

2非齐

$$\begin{cases} v_t - a^2 v_{xx} = -Ax\omega \cos \omega t, & 0 < x < l, t > 0 \\ v(0, t) = 0, & v_x(l, t) = 0, & t \geq 0 \\ v(x, 0) = -Ax, & 0 \leq x \leq l \end{cases}$$

导热/一维/非齐

1齐

2齐

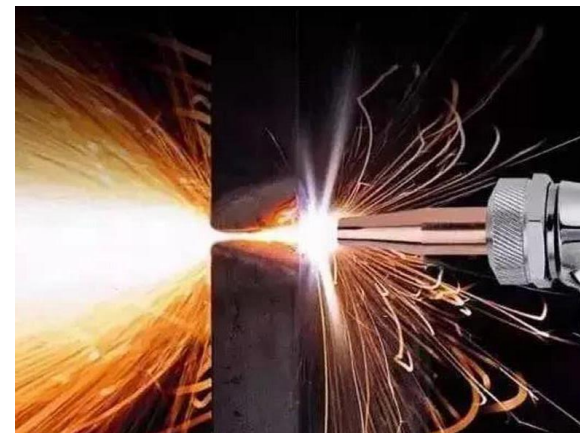
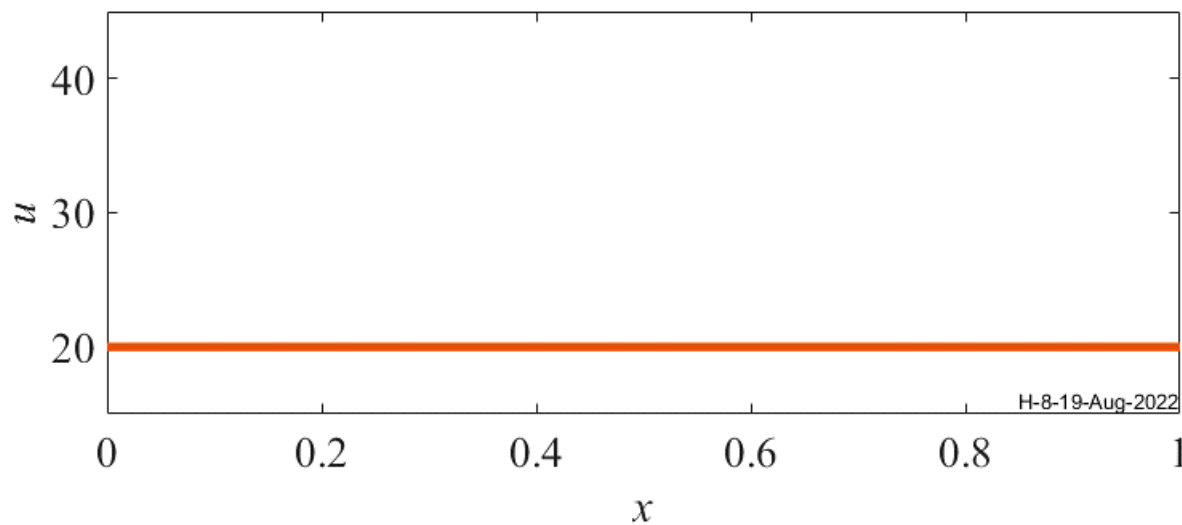
第6步 回代

$$u(x, t) = w(x, t) + v(x, t) = u_0 + A(1 + \sin \omega t)x + \sum_{n=0}^{\infty} \left[\left(\varphi_n - k_n \frac{a^2 \lambda_n}{a^4 \lambda_n^2 + \omega^2} \right) \exp(-a^2 \lambda_n^2 t) + k_n \frac{a^2 \lambda_n \cos \omega t + \omega \sin \omega t}{a^4 \lambda_n^2 + \omega^2} \right] \sin \frac{(2n+1)\pi}{2l} x$$

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \sin \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3 \dots)$$

$$\varphi_n = \frac{(-1)^{n+1}}{(2n+1)^2} \cdot \frac{8Al}{\pi^2}, k_n = \frac{(-1)^{n+1}}{(2n+1)^2} \cdot \frac{8A\omega l}{\pi^2}$$

$$\tau = 0.00s, a = 0.1, u_0 = 20, A = 100, \omega = 10,$$



2.2 分离变量法

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x,t)$$

一维导热定解问题

例3

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0,t) = f(0), u(l,t) = f(l), & t \geq 0 \\ u(x,0) = f(x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1非齐

1非齐

解 第1步 边界条件齐次化 $u(x,t) = v(x,t) + w(x,t)$

[1 1] 型边界, 辅助函数可取

$$w(x,t) = \frac{f(l) - f(0)}{l} x + f(0)$$
$$w(x) = kx + f(0)$$

原定解问题可转化为

$$\begin{cases} v_t - a^2 v_{xx} = 0, & 0 < x < l, t > 0 \\ v(0,t) = 0, v(l,t) = 0, & t \geq 0 \\ v(x,0) = f(x) - kx - f(0), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1齐

1齐

各阶导数

$w_t = 0, w_x = k, w_{xx} = 0$

改造
PDE

$$\begin{aligned} u_t - a^2 u_{xx} &= v_t + w_t - a^2 (v_{xx} + w_{xx}) \\ &= v_t + 0 - a^2 (v_{xx} + 0) \\ &= v_t - a^2 v_{xx} = 0 \end{aligned}$$

改造
初始
条件

$$\begin{aligned} u(x,0) &= f(x) \\ v(x,0) + w(x) &= f(x) \\ v(x,0) &= f(x) - kx - f(0) \end{aligned}$$

边界条件齐次化常用辅助函数 $u(x,t) = v(x,t) + w(x,t)$

边界类型	边界条件	辅助函数
[1,1]	$u(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{l} x + g_1(t)$
[2,1]	$u_x(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = g_1(t)(x-l) + g_2(t)$
[1,2]	$u(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = g_2(t)x + g_1(t)$
[2,2]	$u_x(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{2l} x^2 + g_1(t)x$

2.2 分离变量法

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维导热定解问题

例3

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = f(0), u(l, t) = f(l), & t \geq 0 \\ u(x, 0) = f(x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1非齐

1非齐

$$\begin{cases} v_t - a^2 v_{xx} = 0, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x) - kx - f(0), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1齐

1齐

第2步 特征值问题, 设 $v(x, t) = X(x)T(t)$

代入PDE $X(x)T'(t) - a^2 X''(x)T(t) = 0$

$$\frac{X''(x)}{X(x)} = \frac{T'(t)}{a^2 T(t)} = -\lambda$$

$$T'(t) + \lambda a^2 T(t) = 0, X''(x) + \lambda X(x) = 0$$

齐次边界 $T(t)$ 不恒为0 $X(0) = X(l) = 0$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$ ($n=1, 2, 3, \dots$)
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3, \dots$)
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3, \dots$)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right)$ ($n=0, 1, 2, 3, \dots$)

特征值/特征函数

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维导热定解问题

导热/一维/齐次

1非齐 1非齐

例3

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = f(0), u(l, t) = f(l), & t \geq 0 \\ u(x, 0) = f(x), & 0 \leq x \leq l \end{cases}$$

$$\begin{cases} v_t - a^2 v_{xx} = 0, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x) - kx - f(0), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1齐

1齐

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right), (n=1, 2, 3, \dots)$$

第3步 正交分解, 设定解问题的解为

$$v(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l} x \right)$$

自由项/初始条件按特征函数系展开

$$\textcircled{1} f(x) - \textcircled{2} kx - \textcircled{3} f(0) = \sum_{n=1}^{\infty} f_n X_n(x) = \sum_{n=1}^{\infty} f_n \sin\left(\frac{n\pi}{l} x \right)$$

$$\textcircled{1} f_{n1} = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx$$

$$\textcircled{2} f_{n2} = \frac{2}{l} \int_0^l -kx \sin \frac{n\pi x}{l} \cdot dx$$

$$\textcircled{3} f_{n3} = \frac{2}{l} \int_0^l -f(0) \sin \frac{n\pi x}{l} \cdot dx = -\frac{2f(0)}{l} \frac{-l}{n\pi} \cos \frac{n\pi x}{l} \Big|_0^l = \frac{2f(0)}{n\pi} [(-1)^n - 1]$$

$$= -\frac{2k}{l} \int_0^l x \sin \frac{n\pi x}{l} \cdot dx = -\frac{2k}{l} \left[-\frac{l}{n\pi} x \cos \frac{n\pi}{l} x + \frac{l^2}{n^2 \pi^2} \sin \frac{n\pi}{l} x \right] \Big|_0^l = \frac{2kl}{n\pi} \cos n\pi = (-1)^n \frac{2kl}{n\pi}$$

定义在 $[0, l]$ 上函数 $f(x)$ 的Fourier展开/正交分解

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x)$$

$$f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx,$$

$$f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维导热定解问题

导热/一维/齐次

1非齐 1非齐

例3

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = f(0), u(l, t) = f(l), & t \geq 0 \\ u(x, 0) = f(x), & 0 \leq x \leq l \end{cases}$$

$$\begin{cases} v_t - a^2 v_{xx} = 0, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x) - kx - f(0), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1齐

1齐

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right), (n=1, 2, 3, \dots)$$

初始条件按特征函数系展开为

$$\textcircled{1} f(x) - kx - f(0) = \sum_{n=1}^{\infty} f_n X_n(x) = \sum_{n=1}^{\infty} f_n \sin\left(\frac{n\pi}{l} x \right)$$

$$f_{n1} = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx \quad \left\{ \begin{aligned} f_n &= \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx + (-1)^n \frac{2kl}{n\pi} + \frac{2f(0)}{n\pi} [(-1)^n - 1] \\ f_{n2} &= \frac{2kl}{n\pi} \cos n\pi = (-1)^n \frac{2kl}{n\pi} \\ f_{n3} &= \frac{2f(0)}{n\pi} [(-1)^n - 1] \end{aligned} \right.$$

$$\begin{aligned} f_n &= \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx + \frac{2(-1)^n}{n\pi} [kl + f(0)] - \frac{2f(0)}{n\pi} \\ &= \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx + \frac{2(-1)^n}{n\pi} f(l) - \frac{2f(0)}{n\pi} \end{aligned}$$

$$k = \frac{f(l) - f(0)}{l}$$

$$f_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx + \frac{2}{n\pi} [(-1)^n f(l) - f(0)]$$

2.2 分离变量法

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维导热定解问题

例3

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = f(0), u(l, t) = f(l), & t \geq 0 \\ u(x, 0) = f(x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1非齐

1非齐

$$\begin{cases} v_t - a^2 v_{xx} = 0, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x) - kx - f(0), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1齐

1齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3\cdots)$$

第4步 建立初值ODE

$$v(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l}x\right)$$

根据PDE初始条件确定ODE初始条件

$$v(x, 0) = \sum_{n=1}^{\infty} T_n(0) X_n(x) = \sum_{n=1}^{\infty} f_n X_n(x)$$

假设可逐项求导, 代入原方程

初值问题可表示为

$$\sum_{n=1}^{\infty} T'_n(t) X_n(x) - a^2 \sum_{n=1}^{\infty} T_n(t) X''_n(x) = 0$$

$$\sum_{n=1}^{\infty} T'_n(t) X_n(x) - a^2 \sum_{n=1}^{\infty} T_n(t) [-\lambda_n X_n(x)] = 0$$

$$\sum_{n=1}^{\infty} [T'_n(t) + a^2 \lambda_n T_n(t)] X_n(x) = 0$$

$$X''_n(x) + \lambda X_n(x) = 0$$

$$\begin{cases} T'_n(t) + a^2 \lambda_n T_n(t) = 0 \\ T_n(0) = f_n \end{cases}$$

所以 $T'_n(t) + a^2 \lambda_n T_n(t) = 0$

2.2 分离变量法

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维导热定解问题

例3

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = f(0), u(l, t) = f(l), & t \geq 0 \\ u(x, 0) = f(x), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1非齐

1非齐

$$\begin{cases} v_t - a^2 v_{xx} = 0, & 0 < x < l, t > 0 \\ v(0, t) = 0, v(l, t) = 0, & t \geq 0 \\ v(x, 0) = f(x) - kx - f(0), & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

1齐

1齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

第5步 求初值问题

$$\begin{cases} T_n'(t) + a^2 \lambda_n T_n(t) = 0 \\ T_n(0) = f_n \end{cases}$$

关于 $v(x, t)$ 定解问题的解为

$$v(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} f_n \cdot \exp\left(-\frac{a^2 n^2 \pi^2}{l^2} t\right) \sin\left(\frac{n\pi}{l} x\right)$$

一阶线性齐次ODE 分离变量法

齐次方程解为

$$T_n(t) = C_{1n} \exp(-a^2 \lambda_n t)$$

根据初始条件确定系数

$$T_n(0) = C_{1n} \cdot 1 = f_n$$

初值问题解为 $T_n(t) = f_n \cdot \exp(-a^2 \lambda_n t)$

回代齐次化函数, 原定解问题的解为

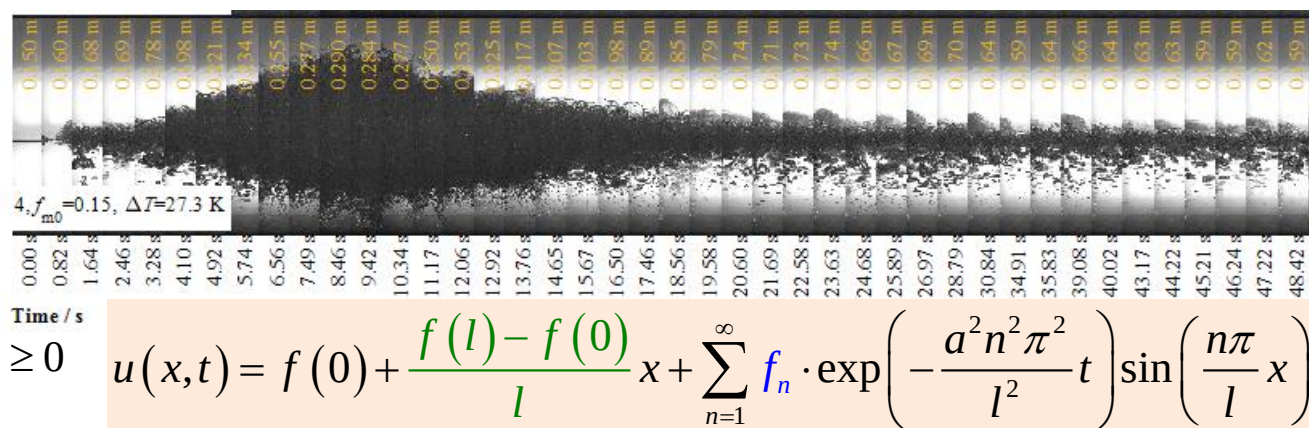
$$\begin{aligned} u(x, t) &= w(x, t) + v(x, t) \\ &= \frac{f(l) - f(0)}{l} x + f(0) + \sum_{n=1}^{\infty} f_n \cdot \exp\left(-\frac{a^2 n^2 \pi^2}{l^2} t\right) \sin\left(\frac{n\pi}{l} x\right) \end{aligned}$$

$$f_n = \frac{2}{n\pi} \left[(-1)^n f(l) - f(0) \right] + \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx$$

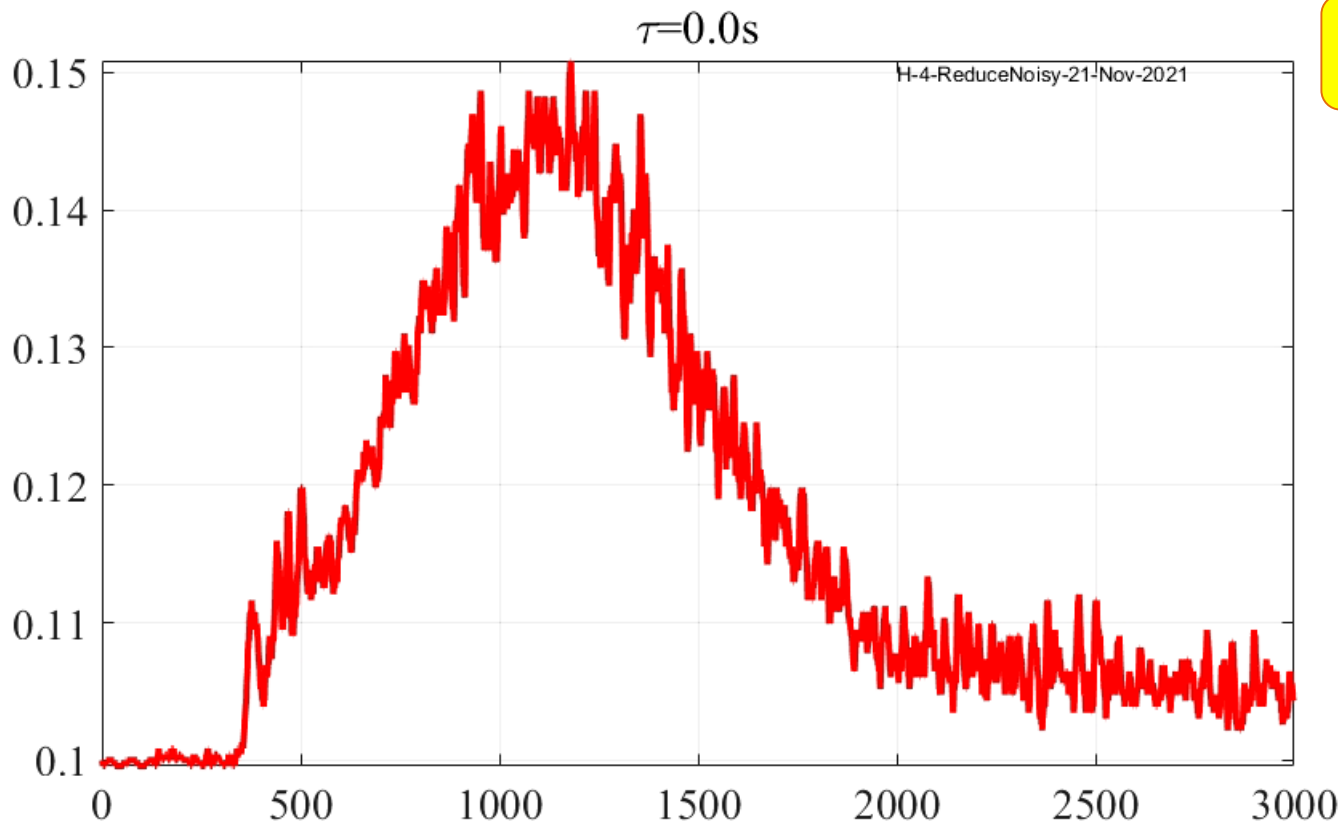
2.2 分离变量法

导热方程降噪

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = f(0), u(l, t) = f(l), & t \geq 0 \\ u(x, 0) = f(x), & 0 \leq x \leq l \end{cases}$$



$$f_n = \frac{2}{n\pi} \left[(-1)^n f(l) - f(0) \right] + \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx$$

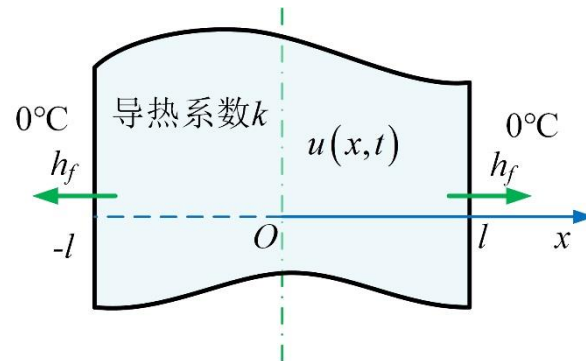


$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维导热定解问题

例4 温度均匀为 A , 厚为 $2l$ 的平板在 0 时刻竖直插入 0°C 的冰水中冷却, 平板表面尺寸远大于厚度. 若平板的导热系数为 k , 平板与水的对流传热系数为 h_f , 求平板内沿厚度方向温度场的演化



解 建立定解问题

仅考虑平板沿厚度方向的温度演化 $\rightarrow$ 一维导热问题

传热状况对称 $\rightarrow$ 仅考虑中轴线右侧半平板 $\rightarrow$ 对称轴绝热/右侧对流

对流边界 $\rightarrow$ 第三类边界

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, & t > 0 \\ u_x(0, t) = 0, & -ku_x(l, t) = h_f[u(l, t) - 0], & t \geq 0 \\ u(x, 0) = A, & 0 \leq x \leq l \end{cases} \rightarrow \begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, & t > 0 \\ u_x(0, t) = 0, & u_x(l, t) + \frac{h_f}{k} u(l, t) = 0, & t \geq 0 \\ u(x, 0) = A, & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

2齐

3齐

第三类齐次边界

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维导热定解问题

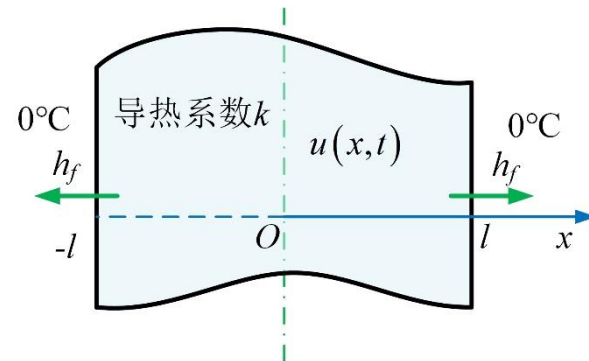
例4

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, & u_x(l, t) + \frac{h_f}{k} u(l, t) = 0, & t \geq 0 \\ u(x, 0) = A, & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

2齐

3齐



第1步 特征值问题 设 $u(x, t) = X(x)T(t)$

代入原PDE $X(x)T'(t) - a^2 X''(x)T(t) = 0$

$$\frac{X''(x)}{X(x)} = \frac{T'(t)}{a^2 T(t)} = -\lambda$$

1阶导数

$$T'(t) + \lambda a^2 T(t) = 0, \quad X''(x) + \lambda X(x) = 0$$

初值问题

特征值问题

边界条件 左端点

$$u_x(0, t) = 0$$

右端点

$$u_x(l, t) + \frac{h_f}{k} u(l, t) = 0$$

$$X'(0)T(t) = 0 \quad X'(l)T(t) + \frac{h_f}{k} X(l)T(t) = 0$$

齐次边界 $T(t)$ 不恒为0

$$X'(0) = 0 \quad X'(l) + \frac{h_f}{k} X(l) = 0$$

$$\begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X'(0) = 0, & X'(l) + \frac{h_f}{k} X(l) = 0, & t \geq 0 \end{cases}$$

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

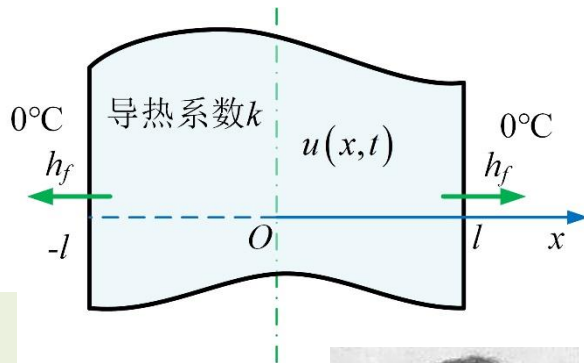
2.2 分离变量法

一维导热定解问题

例4

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u_x(l, t) + \frac{h_f}{k} u(l, t) = 0, & t \geq 0 \\ u(x, 0) = A, & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次	
2齐	3齐



J.W. Biot
1774-1862
French Phys.

第1步 特征值问题

$$\begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X'(0) = 0, X'(l) + \frac{h_f}{k} X(l) = 0, & t \geq 0 \end{cases}$$

当 $\lambda = 0$ 时, 方程的通解为 $X(x) = C_1 x + C_2$, $X'(x) = C_1$

根据边界条件得 $X'(0) = 0, C_1 = 0$

$$X'(l) + \frac{h_f}{k} X(l) = 0 + \frac{h_f}{k} C_2 = 0 \quad \text{则 } C_2 = 0$$

方程为**零解**, 所以 $\lambda = 0$ **不是特征值**

当 $\lambda > 0$ 时, 方程的通解为

$$X(x) = C_1 \cos(\sqrt{\lambda} x) + C_2 \sin(\sqrt{\lambda} x)$$

$$X'(x) = -C_1 \sqrt{\lambda} \sin(\sqrt{\lambda} x) + C_2 \sqrt{\lambda} \cos(\sqrt{\lambda} x)$$

左端点 $X'(0) = 0, C_2 = 0$

右端点

$$\begin{aligned} X'(l) + \frac{h_f}{k} X(l) &= -C_1 \sqrt{\lambda} \sin(\sqrt{\lambda} l) + \frac{h_f}{k} C_1 \cos(\sqrt{\lambda} l) = 0 \\ &= -C_1 \sqrt{\lambda} \sin(\sqrt{\lambda} l) + \frac{h_f}{k} C_1 \cos(\sqrt{\lambda} l) = 0 \end{aligned}$$

分离常数 λ 需满足超越方程

$$\frac{\sin(\sqrt{\lambda} \cdot l)}{\cos(\sqrt{\lambda} \cdot l)} = \tan(\sqrt{\lambda} \cdot l) = \frac{h_f}{k \sqrt{\lambda}} = \frac{h_f l}{k} \frac{1}{\sqrt{\lambda} \cdot l} = \frac{Bi}{\sqrt{\lambda} \cdot l}$$

**Bi数: 边界
对流/导热
相对强弱**

2.2 分离变量法

一维导热定解问题

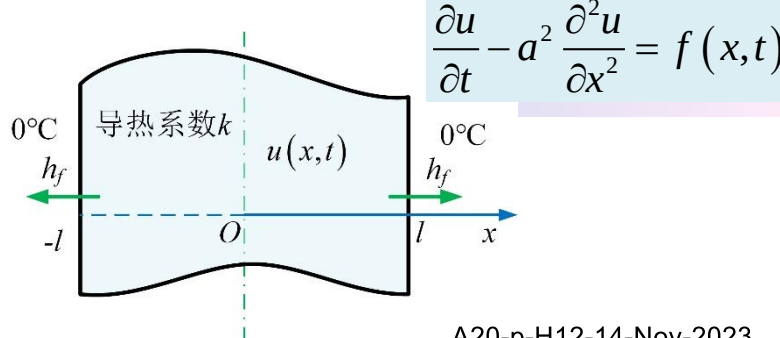
例4

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u_x(l, t) + \frac{h_f}{k} u(l, t) = 0, & t \geq 0 \\ u(x, 0) = A, & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

2齐

3齐



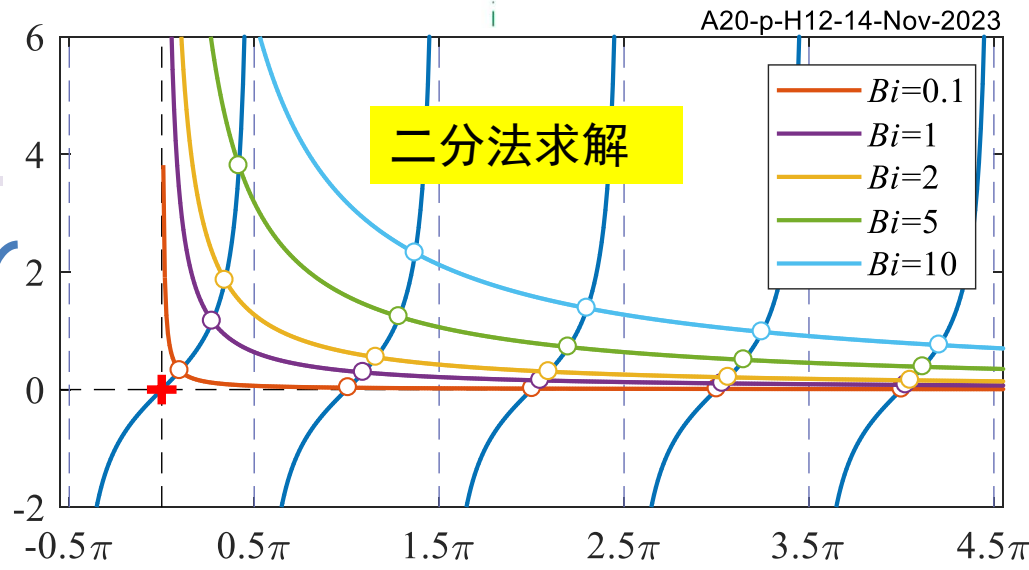
第1步 特征值问题

$$\begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X'(0) = 0, X'(l) + \frac{h_f}{k} X(l) = 0, & t \geq 0 \end{cases}$$

λ 需满足超越方程

$$\tan(\sqrt{\lambda} l) = \frac{Bi}{\sqrt{\lambda} \cdot l} \rightarrow \tan(\mu) = \frac{Bi}{\mu} \text{ 设根为 } \mu_n$$

$$\lambda_n = \left(\frac{\mu_n}{l} \right)^2, \quad \text{特征值/特征函数系}$$
$$X_n(x) = \cos\left(\sqrt{\lambda_n} x\right) = \cos\left(\frac{\mu_n}{l} x\right), \quad (n=1, 2, 3 \dots)$$



μ_n	$Bi=0.1$	1.0	2.0	5	10	...
1	0.3110	0.8603	1.0768	1.3138	1.4288	...
2	3.1730	3.4256	3.6435	4.0335	4.3058	...
3	6.2990	6.4372	6.5783	6.9095	7.2281	...
4	9.4353	9.5293	9.6295	9.8927	10.200	...
5	12.574	12.645	12.722	12.935	13.214	...
...	...	...	...	...	...	...

μ_n 不是整数/半整数

2.2 分离变量法

一维导热定解问题

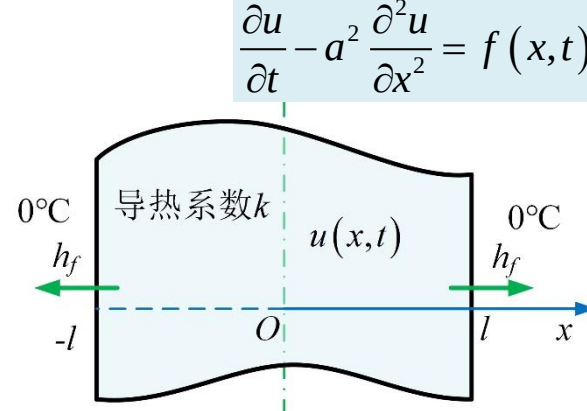
例4

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, \quad u_x(l, t) + \frac{h_f}{k} u(l, t) = 0, & t \geq 0 \\ u(x, 0) = A, & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

2齐

3齐



第2步 正交分解 设定解问题的解为

$$\lambda_n = \left(\frac{\mu_n}{l}\right)^2, \quad X_n(x) = \cos(\sqrt{\lambda_n} x) = \cos\left(\frac{\mu_n}{l} x\right), \quad (n = 1, 2, 3 \dots)$$

正交/完备 → 但 $\frac{\mu_n}{l}$ 不是整数 → 平方模需重新计算

定积分表

$$\int \cos ax \cos bxdx = \begin{cases} \frac{\sin(a+b)x}{2(a+b)} + \frac{\sin(a-b)x}{2(a-b)} + C, & (a \neq b) \\ \frac{x}{2} + \frac{1}{4a} \sin 2ax + C, & (a = b) \end{cases}$$

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x)$$

初始条件用特征函数表示

$$u(x, 0) = A = \sum_{n=1}^{\infty} \varphi_n \cdot \cos\left(\frac{\mu_n}{l} x\right)$$

$$X^2 = 2 \cdot \int_0^l \cos^2\left(\frac{\mu_m}{l} x\right) \cdot dx = 2 \cdot \frac{\mu_n + \sin \mu_n \cos \mu_n}{2\mu_n} \cdot l$$

$$\begin{aligned} \varphi_n &= \frac{2}{X^2} \int_0^l A \cdot \cos \frac{\mu_n x}{l} \cdot dx = \frac{2 \cdot A}{X^2} \int_0^l \cos \frac{\mu_n x}{l} \cdot dx \\ &= \frac{2 \cdot A}{X^2} \left[\frac{l}{\mu_n} \sin \frac{\mu_n x}{l} \right]_0^l = \frac{2 \cdot A}{2 \cdot \frac{\mu_n + \sin \mu_n \cos \mu_n}{2\mu_n} \cdot l} \cdot \frac{l}{\mu_n} \sin \mu_n = \frac{2A \sin \mu_n}{\mu_n + \sin \mu_n \cos \mu_n} \end{aligned}$$

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维导热定解问题

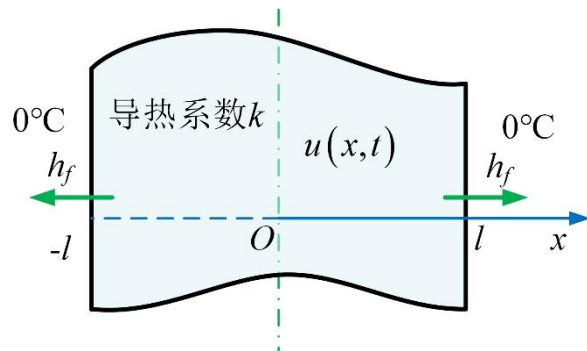
例4

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u_x(l, t) + \frac{h_f}{k} u(l, t) = 0, & t \geq 0 \\ u(x, 0) = A, & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

2齐

3齐



第3步 建立初值问题

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \cos\left(\frac{\mu_n}{l} x\right)$$

假设可逐项求导, 代入原方程

$$\sum_{n=1}^{\infty} T_n'(t) X_n(x) - a^2 \sum_{n=1}^{\infty} T_n(t) X_n''(x) = 0$$

$$\sum_{n=1}^{\infty} T_n'(t) X_n(x) - a^2 \sum_{n=1}^{\infty} T_n(t) [-\lambda_n X_n(x)] = 0$$

$$\sum_{n=1}^{\infty} [T_n'(t) + a^2 \lambda_n T_n(t)] X_n(x) = 0$$

$$\text{所以 } T_n'(t) + a^2 \lambda_n T_n(t) = 0$$

$$\lambda_n = \left(\frac{\mu_n}{l}\right)^2, \quad X_n(x) = \cos(\sqrt{\lambda_n} x) = \cos\left(\frac{\mu_n}{l} x\right), \quad (n=1, 2, 3, \dots)$$

根据初始条件

$$u(x, 0) = A = \sum_{n=1}^{\infty} T_n(0) X_n(x) = \sum_{n=1}^{\infty} \varphi_n X_n(x)$$

初值问题可表示为

$$\begin{cases} T_n'(t) + a^2 \lambda_n T_n(t) = 0 \\ T_n(0) = \varphi_n \end{cases}$$

$$X_n''(x) = -\lambda X_n(x)$$

2.2 分离变量法

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维导热定解问题

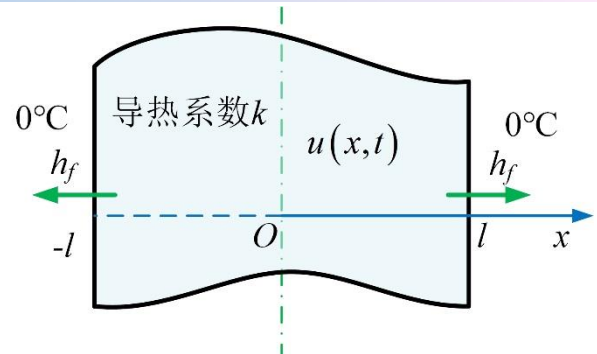
例4

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, \quad u_x(l, t) + \frac{h_f}{k} u(l, t) = 0, & t \geq 0 \\ u(x, 0) = A, & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

2齐

3齐



第4步 求解初值问题

$$\begin{cases} T'_n(t) + a^2 \lambda_n T_n(t) = 0 \\ T_n(0) = \varphi_n \end{cases}$$

一阶线性齐次ODE, 分离变量法

$$T_n(t) = C_{1n} \exp(-a^2 \lambda_n t)$$

根据初始条件 $T_n(0) = C_{1n} = \varphi_n$

故初值ODE的解为

$$T_n(t) = \varphi_n \cdot \exp(-a^2 \lambda_n t)$$

$$\lambda_n = \left(\frac{\mu_n}{l}\right)^2, \quad X_n(x) = \cos(\sqrt{\lambda_n} x) = \cos\left(\frac{\mu_n}{l} x\right), \quad (n = 1, 2, 3 \dots)$$

第6步 回代 原方程的解为

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} \varphi_n \cdot \exp\left(-a^2 \left(\frac{\mu_n}{l}\right)^2 t\right) \cos\left(\frac{\mu_n}{l} x\right) \\ &= \sum_{n=1}^{\infty} \varphi_n \cdot \exp\left(-\mu_n^2 \frac{a^2 t}{l^2}\right) \cos\left(\mu_n \frac{x}{l}\right) \\ &= \sum_{n=1}^{\infty} \varphi_n \cdot \exp(-\mu_n^2 \cdot Fo) \cos\left(\mu_n \frac{x}{l}\right) \end{aligned}$$

$Fo = \frac{a^2 t}{l^2}$

Fo数: 导热进行程度

$$\varphi_n = \frac{2A \sin \mu_n}{\mu_n + \sin \mu_n \cos \mu_n} \quad Bi = \frac{h_f l}{k}$$

其中 μ_n 是方程 $\tan(\mu) = \frac{Bi}{\mu}$ 的解



J. Fourier
1768-1830
French Math.
Phys.

2.2 分离变量法

$$\frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x,t)$$

一维导热定解问题

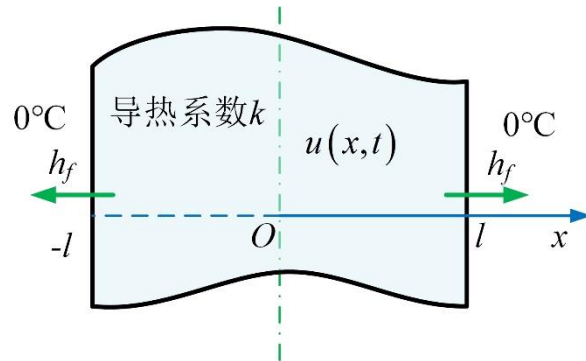
例4

$$\begin{cases} u_t - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u_x(0,t) = 0, \quad u_x(l,t) + \frac{h_f}{k} u(l,t) = 0, & t \geq 0 \\ u(x,0) = A, & 0 \leq x \leq l \end{cases}$$

导热/一维/齐次

2齐

3齐



$$Fo = \frac{a^2 t}{l^2}$$

$$Bi = \frac{h_f l}{k}$$

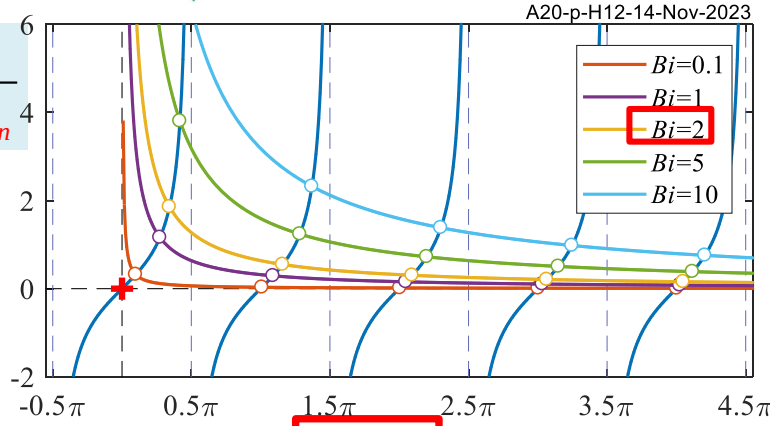
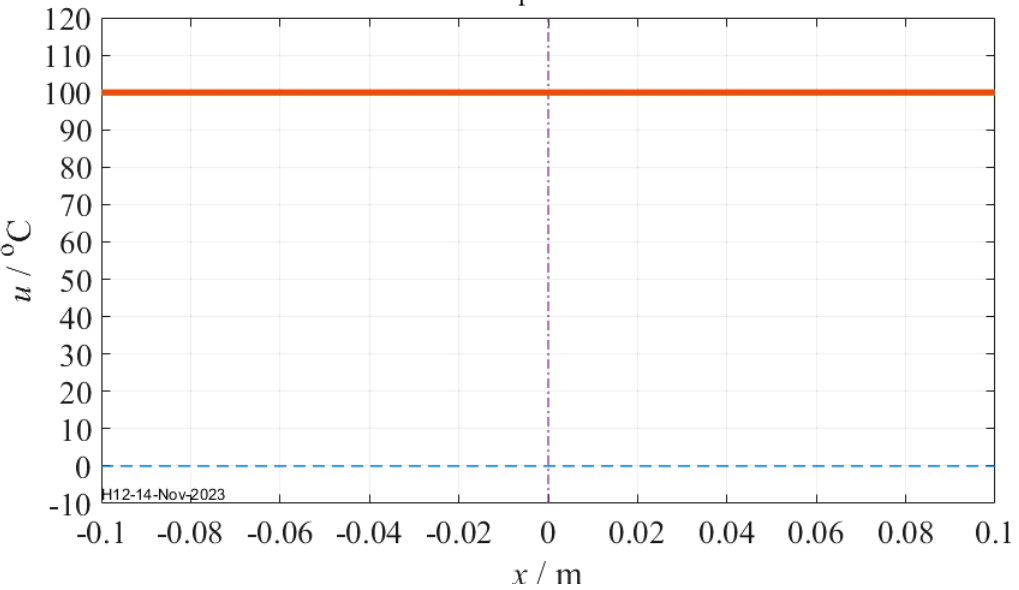
$$u(x,t) = \sum_{n=1}^{\infty} \varphi_n \cdot \exp(-\mu_n^2 \cdot Fo) \cos\left(\mu_n \frac{x}{l}\right)$$

$$\varphi_n = \frac{2A \sin \mu_n}{\mu_n + \sin \mu_n \cos \mu_n}$$

$$\lambda_n = \left(\frac{\mu_n}{l}\right)^2, \quad X_n(x) = \cos\left(\frac{\mu_n}{l} x\right), \quad (n=1, 2, 3, \dots)$$

$$\tan(\mu_n) = \frac{Bi}{\mu_n}$$

$\tau=0.000s, a=0.5, h_f=100W \cdot m^{-2} \cdot K^{-1}, Bi=2$



μ_n	$Bi=0.1$	1.0	2.0	5	10	...
1	0.3110	0.8603	1.0768	1.3138	1.4288	...
2	3.1730	3.4256	3.6435	4.0335	4.3058	...
3	6.2990	6.4372	6.5783	6.9095	7.2281	...
4	9.4353	9.5293	9.6295	9.8927	10.200	...
5	12.574	12.645	12.722	12.935	13.214	...
...	...	...	...	...	...	...

1. 数学建模和基本原理介绍

2. 分离变量法

3. Bessel函数

4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

2.1 特征值问题

2.2 分离变量法

2.3 基于MATLAB的定解问题求解方法

弦振动方程定解问题

热传导方程定解问题

平面位势方程边值问题

2.2 分离变量法

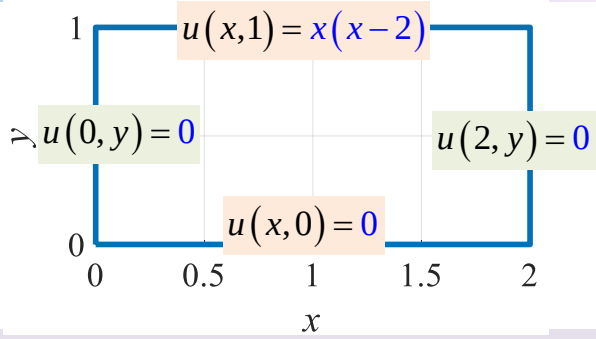
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x,y)$$

平面Laplace方程

例1

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < 2, & 0 < y < 1 \\ u(0,y) = 0, & u(2,y) = 0, & 0 \leq y \leq 1 \\ u(x,0) = 0, & u(x,1) = x(x-2), & 0 \leq x \leq 2 \end{cases}$$

Laplace/二维/齐次		
x边界	1齐	1齐
y边界	1齐	1非齐



解 方程齐次/x边界齐次→可直接分离变量
第1步 特征值问题

设 $u(x,y) = X(x)Y(y)$ 代入到原方程

$$X''(x)Y(y) + X(x)Y''(y) = 0$$

$$\frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)} = -\lambda$$

负号由非齐次维度承担

$$Y''(y) - \lambda Y(y) = 0 \quad \begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(2) = 0 \end{cases}$$

齐次维度找特征值

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$ ($n=1,2,3,\dots$)
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0,1,2,3,\dots$)
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0,1,2,3,\dots$)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right)$ ($n=0,1,2,3,\dots$)

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1,2,3,\dots)$$

x方向长度

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

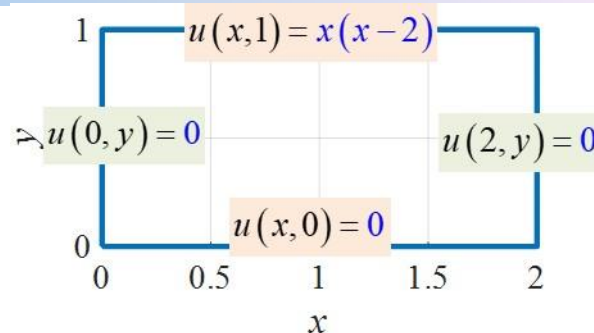
2.2 分离变量法

平面Laplace方程

例1

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < 2, & 0 < y < 1 \\ u(0, y) = 0, & u(2, y) = 0, & 0 \leq y \leq 1 \\ u(x, 0) = 0, & u(x, 1) = x(x-2), & 0 \leq x \leq 2 \end{cases}$$

La./二维/齐次		
x边界	1齐	1齐
y边界	1齐	1非齐



本例中 x 方向长度 $l=2$

双曲正/余弦函数

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2 = \left(\frac{n\pi}{2}\right)^2, \quad X_n(x) = \sin\left(\frac{n\pi}{l}x\right) = \sin\left(\frac{n\pi}{2}x\right), \quad (n=1, 2, 3, \dots)$$

$$\sinh(x) = \frac{e^x - e^{-x}}{2},$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\sinh(0) = 0, \quad \cosh(0) = 1$$

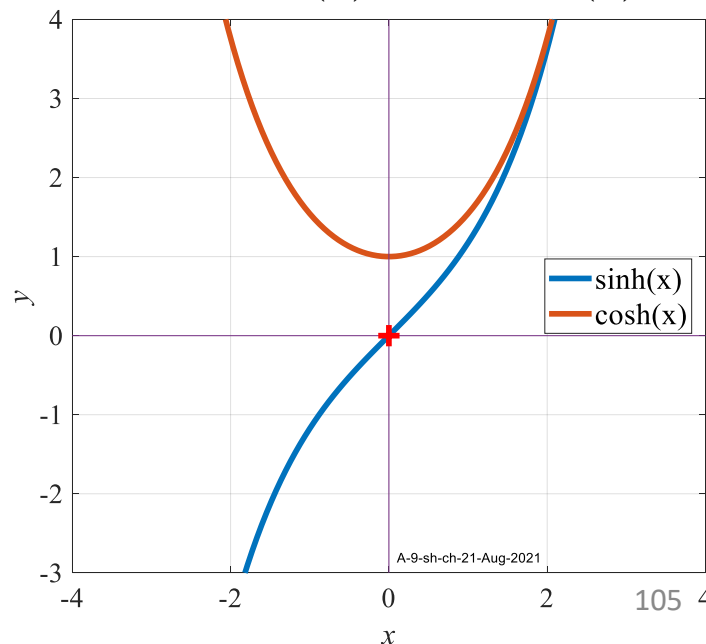
代入 y 方向方程 $Y''(y) - \lambda Y(y) = 0$

$$Y''(y) - \left(\frac{n\pi}{2}\right)^2 Y(y) = 0$$

$$Y(y) = C_{1n} \exp\left(\frac{n\pi}{2}y\right) + C_{2n} \exp\left(-\frac{n\pi}{2}y\right)$$

正/负指数函数的线性组合 $\rightarrow$ 双曲正/余弦

$$Y(y) = C_{1n} \sinh\left(\frac{n\pi}{2}y\right) + C_{2n} \cosh\left(\frac{n\pi}{2}y\right)$$



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2.2 分离变量法

平面Laplace方程

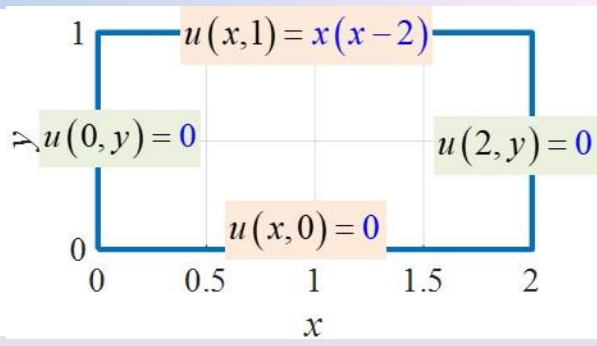
例1

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < 2, \ 0 < y < 1 \\ u(0, y) = 0, \ u(2, y) = 0, & 0 \leq y \leq 1 \\ u(x, 0) = 0, \ u(x, 1) = x(x-2), & 0 \leq x \leq 2 \end{cases}$$

La./二维/齐次

x边界	1齐	1齐
y边界	1齐	1非齐

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$



原方程的解可表示为

$$u(x, y) = \sum_{n=1}^{\infty} \left[C_{1n} \sinh\left(\frac{n\pi}{2} y\right) + C_{2n} \cosh\left(\frac{n\pi}{2} y\right) \right] \sin\left(\frac{n\pi}{2} x\right)$$

根据 y 方向边界条件

$$u(x, 0) = \sum_{n=1}^{\infty} C_{2n} \sin\left(\frac{n\pi}{2} x\right) = 0, \quad C_{2n} = 0$$

$$u(x, 1) = \sum_{n=1}^{\infty} \left[C_{1n} \sinh\left(\frac{n\pi}{2}\right) + C_{2n} \cosh\left(\frac{n\pi}{2}\right) \right] \sin\left(\frac{n\pi}{2} x\right) = x(x-2)$$

$$C_{1n} = \frac{1}{\sinh \frac{n\pi}{2}} \frac{2}{2} \int_0^2 x(x-2) \sin \frac{n\pi}{2} x dx = \frac{16 \left[(-1)^n - 1 \right]}{n^3 \pi^3 \sinh \frac{n\pi}{2}}$$

定义在[0, l]上函数 f(x) 的Fourier展开/正交分解

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x)$$
$$f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx,$$
$$f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

y 方向 ODE的解

$$Y_n(y) = \frac{16}{\pi^3} \frac{(-1)^n - 1}{n^3 \sinh \frac{n\pi}{2}} \sinh \frac{n\pi}{2} y$$

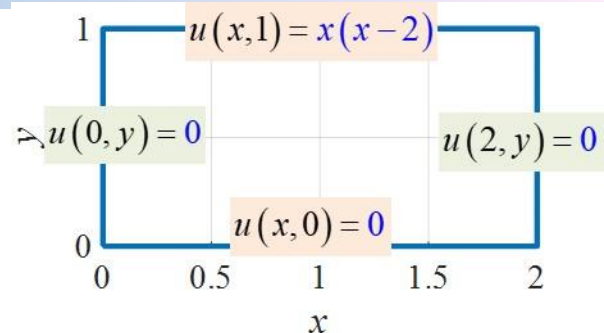
2.2 分离变量法

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

平面Laplace方程

例1
$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < 2, & 0 < y < 1 \\ u(0, y) = 0, & u(2, y) = 0, & 0 \leq y \leq 1 \\ u(x, 0) = 0, & u(x, 1) = x(x-2), & 0 \leq x \leq 2 \end{cases}$$

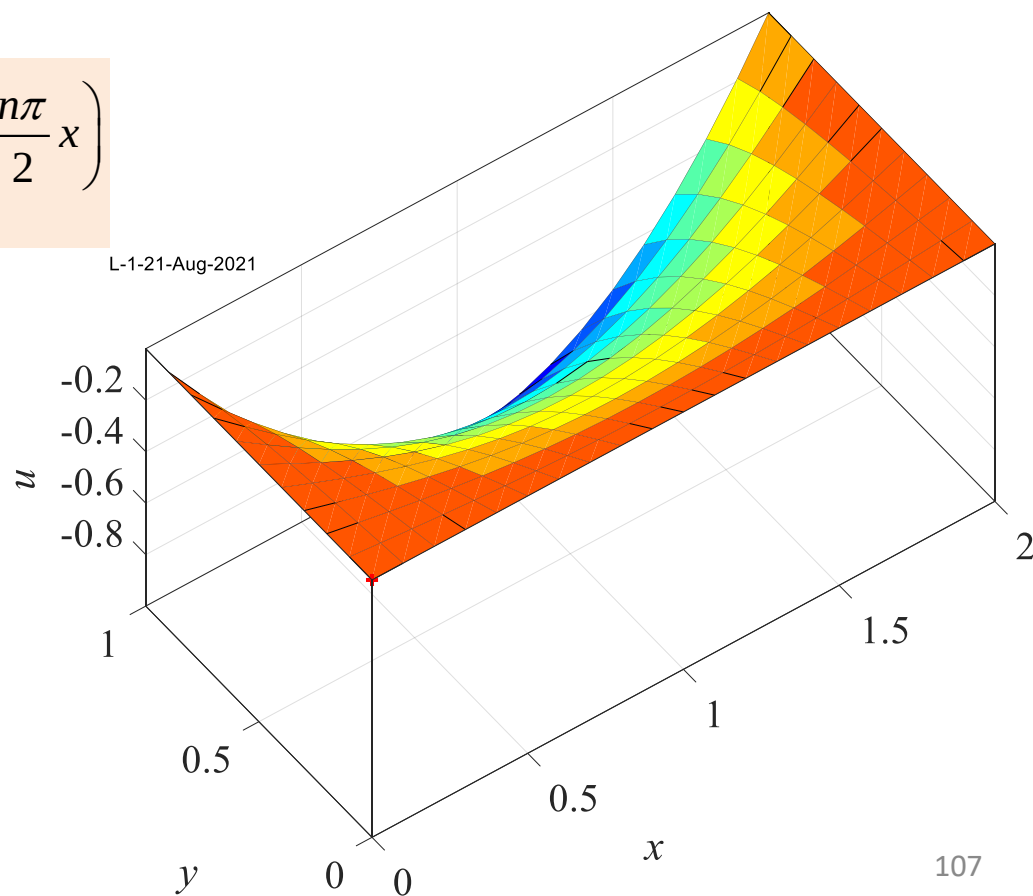
La./二维/齐次		
x边界	1齐	1齐
y边界	1齐	1非齐



原方程的解可表示为

$$u(x, y) = \frac{16}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^3 \sinh \frac{n\pi}{2}} \sinh\left(\frac{n\pi}{2} y\right) \sin\left(\frac{n\pi}{2} x\right)$$

1. 齐次边界→特征值/特征函数
2. PDE齐次→直接分离变量→并未正交分解
3. 维度地位相同→能否从 y 方向找特征值/特征函数？



2.2 分离变量法

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x,y)$$

平面Laplace方程

例1

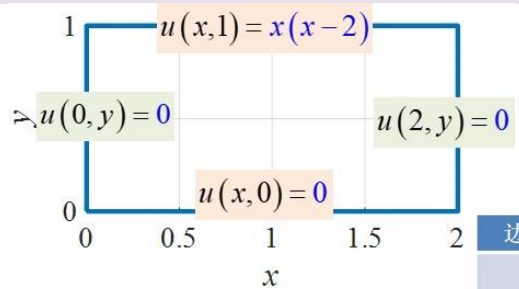
$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < 2, & 0 < y < 1 \\ u(0,y) = 0, & u(2,y) = 0, & 0 \leq y \leq 1 \\ u(x,0) = 0, & u(x,1) = x(x-2), & 0 \leq x \leq 2 \end{cases}$$

Laplace/二维/齐次		
x边界	1齐	1齐
y边界	1齐	1非齐

Poisson/二维/非齐		
x边界	1齐	1齐
y边界	1齐	1齐

$$\begin{cases} v_{xx} + v_{yy} = -2y, & 0 < x < 2, & 0 < y < 1 \\ v(0,y) = 0, & v(2,y) = 0, & 0 \leq y \leq 1 \\ v(x,0) = 0, & v(x,1) = 0, & 0 \leq x \leq 2 \end{cases}$$

解法2 从维度 y 计算



第一步 边界齐次化

设 $u(x,y) = v(x,y) + w(x,y)$

查表 $w(x,y) = x(x-2)y = x^2y - 2xy$

各阶
导数

$$\begin{aligned} w_x &= 2xy - 2y & w_y &= x^2 - 2x \\ w_{xx} &= 2y & w_{yy} &= 0 \end{aligned}$$

边界条件齐次化常用辅助函数 $u(x,t) = v(x,t) + w(x,t)$

边界类型	边界条件	辅助函数
[1,1]	$u(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{l}x + g_1(t)$
[2,1]	$u_x(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = g_1(t)(x-l) + g_2(t)$
[1,2]	$u(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = g_2(t)x + g_1(t)$
[2,2]	$u_x(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{2l}x^2 + g_1(t)x$

改造PDE

$$u_{xx} + u_{yy} = 0$$

$$v_{xx} + v_{yy} + w_{xx} + w_{yy} = 0$$

$$v_{xx} + v_{yy} = -2y$$

改造 x 边界

$$\begin{aligned} u(2,y) &= 0 \\ v(2,y) + w(2,y) &= 0 \\ v(2,y) &= 0 \end{aligned}$$

$$\begin{aligned} u(0,y) &= 0 \\ v(0,y) + w(0,y) &= 0 \\ v(0,y) &= 0 \end{aligned}$$

验证 y 边界

$$\begin{aligned} u(x,1) &= x(x-2) \\ v(x,1) + w(x,1) &= x(x-2) \\ v(x,1) &= 0 \end{aligned}$$

$$\begin{aligned} u(x,0) &= 0 \\ v(x,0) + w(x,0) &= 0 \\ v(x,0) &= 0 \end{aligned}$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x,y)$$

平面Laplace方程

例1

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < 2, & 0 < y < 1 \\ u(0,y) = 0, & u(2,y) = 0, & 0 \leq y \leq 1 \\ u(x,0) = 0, & u(x,1) = x(x-2), & 0 \leq x \leq 2 \end{cases}$$

Laplace/二维/齐次		
x边界	1齐	1齐
y边界	1齐	1非齐

Poisson/二维/非齐		
x边界	1齐	1齐
y边界	1齐	1齐

$$\begin{cases} v_{xx} + v_{yy} = -2y, & 0 < x < 2, & 0 < y < 1 \\ v(0,y) = 0, & v(2,y) = 0, & 0 \leq y \leq 1 \\ v(x,0) = 0, & v(x,1) = 0, & 0 \leq x \leq 2 \end{cases}$$

第2步 特征值问题 设 $v(x,y) = X(x)Y(y)$

代入对应的齐次PDE

$$X''(x)Y(y) + X(x)Y''(y) = 0$$

$$-\frac{X''(x)}{X(x)} = \frac{Y''(y)}{Y(y)} = -\lambda$$

x承担负号

$$X''(x) - \lambda X(x) = 0$$

$$\begin{cases} Y''(y) + \lambda Y(y) = 0 \\ Y(0) = Y(1) = 0 \end{cases}$$

特征值/特征函数

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$ ($n=1,2,3,\dots$)
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0,1,2,3,\dots$)
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0,1,2,3,\dots$)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right)$ ($n=0,1,2,3,\dots$)

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, Y_n(y) = \sin\left(\frac{n\pi}{l}y\right), (n=1,2,3,\dots)$$

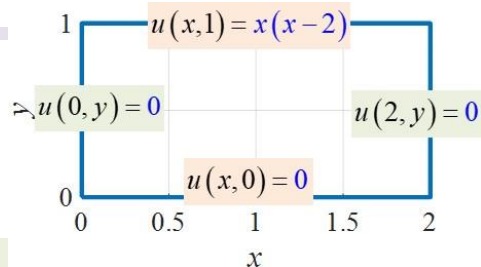
特征函数自变量用y表示

2.2 分离变量法

平面Laplace方程 例1

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < 2, & 0 < y < 1 \\ u(0, y) = 0, & u(2, y) = 0, & 0 \leq y \leq 1 \\ u(x, 0) = 0, & u(x, 1) = x(x-2), & 0 \leq x \leq 2 \end{cases}$$

Laplace/二维/齐次		
x边界	1齐	1齐
y边界	1齐	1非齐



第3步 正交分解
本例 y 方向上 $l=1$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2 = (n\pi)^2, \quad Y_n(y) = \sin(n\pi y), \quad (n=1, 2, 3, \dots)$$

$$v(x, y) = \sum_{n=1}^{\infty} X_n(x) Y_n(y) = \sum_{n=1}^{\infty} X_n(x) \sin(n\pi y)$$

从1开始

自由项按特征函数系展开

$$-2y = \sum_{n=1}^{\infty} f_n Y_n(y) = \sum_{n=1}^{\infty} f_n \sin(n\pi y)$$

$$f_n = \frac{2}{l} \int_0^l (-2y) \sin(n\pi y) dy = -4 \int_0^1 y \sin(n\pi y) dy = -4 \left[-\frac{1}{n\pi} y \cos n\pi y + \frac{1}{(n\pi)^2} \sin n\pi y \right]_0^1 = (-1)^n \frac{4}{n\pi}$$

Poisson/二维/非齐		
x边界	1齐	1齐
y边界	1齐	1齐

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

$$\begin{cases} v_{xx} + v_{yy} = -2y, & 0 < x < 2, & 0 < y < 1 \\ v(0, y) = 0, & v(2, y) = 0, & 0 \leq y \leq 1 \\ v(x, 0) = 0, & v(x, 1) = 0, & 0 \leq x \leq 2 \end{cases}$$

定义在[0, l]上函数 f(x) 的Fourier展开/正交分解

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x)$$

$$f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx,$$

$$f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

2.2 分离变量法

平面Laplace方程 例1

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < 2, \ 0 < y < 1 \\ u(0, y) = 0, \ u(2, y) = 0, & 0 \leq y \leq 1 \\ u(x, 0) = 0, \ u(x, 1) = x(x-2), & 0 \leq x \leq 2 \end{cases}$$

Laplace/二维/齐次		
x边界	1齐	1齐
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Poisson/二维/非齐		
x边界	1齐	1齐
y边界	1齐	1齐

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

$$\begin{cases} v_{xx} + v_{yy} = -2y, & 0 < x < 2, \ 0 < y < 1 \\ v(0, y) = 0, \ v(2, y) = 0, & 0 \leq y \leq 1 \\ v(x, 0) = 0, \ v(x, 1) = 0, & 0 \leq x \leq 2 \end{cases}$$

第4步 建立x方向ODE

$$v(x, y) = \sum_{n=1}^{\infty} X_n(x) Y_n(y) = \sum_{n=1}^{\infty} X_n(x) \sin(n\pi y)$$

展开式代入PDE

$$\begin{aligned} \sum_{n=1}^{\infty} X_n''(x) Y_n(y) + \sum_{n=1}^{\infty} X_n(x) Y_n''(y) &= \sum_{n=1}^{\infty} f_n Y_n(y) \\ \sum_{n=1}^{\infty} X_n''(x) Y_n(y) + \sum_{n=1}^{\infty} X_n(x) [-\lambda_n Y_n(y)] &= \sum_{n=1}^{\infty} f_n Y_n(y) \\ \sum_{n=1}^{\infty} [X_n''(x) - \lambda_n X_n(x)] Y_n(y) &= \sum_{n=1}^{\infty} f_n Y_n(y) \end{aligned}$$

得出初值ODE $X_n''(x) - \lambda_n X_n(x) = f_n$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2 = (n\pi)^2, \ Y_n(y) = \sin(n\pi y), \ (n=1, 2, 3 \dots)$$

PDE的x方向边界条件→ODE边界条件

$$\begin{aligned} v(0, y) &= \sum_{n=1}^{\infty} X_n(0) Y_n(y) = 0 \\ v(2, y) &= \sum_{n=1}^{\infty} X_n(2) Y_n(y) = 0 \end{aligned}$$

建立 x 方向定解问题

$$\begin{cases} X_n''(x) - \lambda_n X_n(x) = f_n \\ X_n(0) = X_n(2) = 0 \end{cases}$$

2.2 分离变量法

平面Laplace方程 例1

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < 2, \ 0 < y < 1 \\ u(0, y) = 0, \ u(2, y) = 0, & 0 \leq y \leq 1 \\ u(x, 0) = 0, \ u(x, 1) = x(x-2), & 0 \leq x \leq 2 \end{cases}$$

Laplace/二维/齐次		
x边界	1齐	1齐
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Poisson/二维/非齐		
x边界	1齐	1齐
y边界	1齐	1齐

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

$$\begin{cases} v_{xx} + v_{yy} = -2y, & 0 < x < 2, \ 0 < y < 1 \\ v(0, y) = 0, \ v(2, y) = 0, & 0 \leq y \leq 1 \\ v(x, 0) = 0, \ v(x, 1) = 0, & 0 \leq x \leq 2 \end{cases}$$

第5步 求解 x 方向ODE

$$\begin{cases} X_n''(x) - \lambda_n X_n(x) = f_n \\ X_n(0) = X_n(2) = 0 \end{cases}$$

$$\begin{aligned} X_n(x) &= C_{1n} \exp(n\pi x) + C_{2n} \exp(-n\pi x) + \bar{T}_n \\ &= C_{1n} \sinh(n\pi x) + C_{2n} \cosh(n\pi x) + \bar{T}_n \end{aligned}$$

自由项为常数→特解为常数 $\bar{T}_n = C$

$$\text{各阶导数 } \bar{T}_n' = \bar{T}_n'' = 0$$

代入ODE $0 - \lambda_n \bar{T}_n = f_n$ 得特解

$$\bar{T}_n = \frac{f_n}{-\lambda_n} = \frac{(-1)^n \frac{4}{n\pi}}{(n\pi)^2} = -(-1)^n \frac{4}{(n\pi)^3}$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2 = (n\pi)^2, \ Y_n(y) = \sin(n\pi y), \ (n=1, 2, 3\cdots)$$

$$X_n(x) = C_{1n} \sinh(n\pi x) + C_{2n} \cosh(n\pi x) - (-1)^n \frac{4}{(n\pi)^3}$$

利用ODE边界条件确定系数

$$X_n(0) = C_{2n} - (-1)^n \frac{4}{(n\pi)^3} = 0 \quad C_{2n} = (-1)^n \frac{4}{(n\pi)^3}$$

$$X_n(2) = C_{1n} \sinh(2n\pi) + C_{2n} \cosh(2n\pi) - (-1)^n \frac{4}{(n\pi)^3} = 0$$

$$\begin{aligned} C_{1n} &= \frac{1}{\sinh(2n\pi)} \left[(-1)^n \frac{4}{(n\pi)^3} - C_{2n} \cosh(2n\pi) \right] \\ &= (-1)^n \frac{4}{(n\pi)^3} \frac{1 - \cosh(2n\pi)}{\sinh(2n\pi)} \end{aligned}$$

2.2 分离变量法

平面Laplace方程 例1

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < 2, \ 0 < y < 1 \\ u(0, y) = 0, \ u(2, y) = 0, & 0 \leq y \leq 1 \\ u(x, 0) = 0, \ u(x, 1) = x(x-2), & 0 \leq x \leq 2 \end{cases}$$

Laplace/二维/齐次		
x边界	1齐	1齐
y边界	1齐	1非齐

Poisson/二维/非齐		
x边界	1齐	1齐
y边界	1齐	1齐

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

第6步 回代

$$C_{1n} = (-1)^n \frac{4}{(n\pi)^3} \frac{1 - \cosh(2n\pi)}{\sinh(2n\pi)} \quad C_{2n} = (-1)^n \frac{4}{(n\pi)^3}$$

$$\begin{aligned} X_n(x) &= C_{1n} \sinh(n\pi x) + C_{2n} \cosh(n\pi x) - (-1)^n \frac{4}{(n\pi)^3} \\ &= (-1)^n \frac{4}{(n\pi)^3} \left[\frac{1 - \cosh(2n\pi)}{\sinh(2n\pi)} \sinh(n\pi x) + \cosh(n\pi x) - 1 \right] \end{aligned}$$

得原定解问题的另一种等价解

$$\begin{aligned} u(x, y) &= w(x, y) + v(x, y) \\ &= x(x-2)y + \frac{4}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \left[\frac{1 - \cosh(2n\pi)}{\sinh(2n\pi)} \sinh(n\pi x) + \cosh(n\pi x) - 1 \right] \sin(n\pi y) \end{aligned}$$

$$\begin{cases} v_{xx} + v_{yy} = -2y, & 0 < x < 2, \ 0 < y < 1 \\ v(0, y) = 0, \ v(2, y) = 0, & 0 \leq y \leq 1 \\ v(x, 0) = 0, \ v(x, 1) = 0, & 0 \leq x \leq 2 \end{cases}$$

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, \ X_n(x) = \sin\left(\frac{n\pi}{l} x \right), \ (n=1, 2, 3 \dots)$$

$$w(x, y) = x(x-2)y = x^2y - 2xy$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x,y)$$

平面Laplace方程 例1

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < 2, & 0 < y < 1 \\ u(0,y) = 0, & u(2,y) = 0, & 0 \leq y \leq 1 \\ u(x,0) = 0, & u(x,1) = x(x-2), & 0 \leq x \leq 2 \end{cases}$$

Laplace/二维/齐次		
x边界	1齐	1齐
y边界	1齐	1非齐

Poisson/二维/非齐		
x边界	1齐	1齐
y边界	1齐	1齐

$$\begin{cases} v_{xx} + v_{yy} = -2y, & 0 < x < 2, & 0 < y < 1 \\ v(0,y) = 0, & v(2,y) = 0, & 0 \leq y \leq 1 \\ v(x,0) = 0, & v(x,1) = 0, & 0 \leq x \leq 2 \end{cases}$$

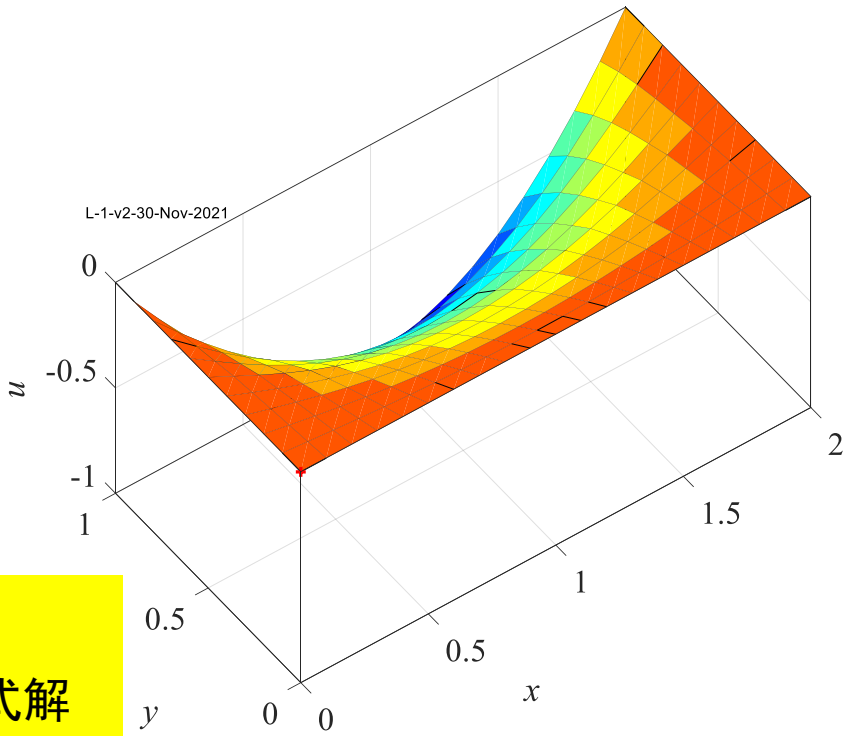
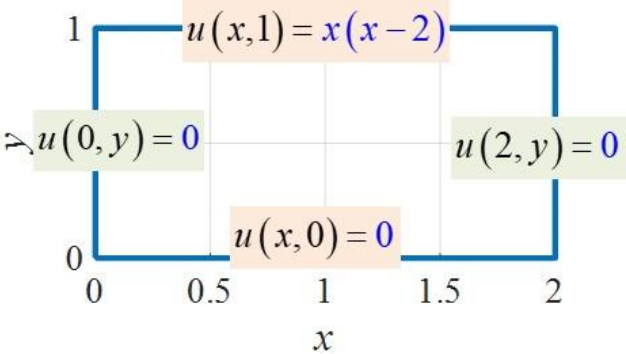
讨论

解法2

$$u(x,y) = x(x-2)y + \frac{4}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \left[\frac{1 - \cosh(2n\pi)}{\sinh(2n\pi)} \sinh(n\pi x) + \cosh(n\pi x) - 1 \right] \sin(n\pi y)$$

解法1

$$u(x,y) = \frac{16}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^3 \sinh \frac{n\pi}{2}} \sinh\left(\frac{n\pi}{2} y\right) \sin\left(\frac{n\pi}{2} x\right)$$



- 1.空间维度地位相等→任一维度都可求解

2.同一个定解问题可用不同特征函数系表达形式解

2.2 分离变量法

圆域/扇域/环域 Poisson方程→极坐标更方便

预备知识1 极坐标下 Δu 展开式

直角系下 $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$

极坐标定义 $\rho = \sqrt{x^2 + y^2}, \theta = \tan^{-1} \frac{y}{x}$



$$\frac{\partial \rho}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}}, \frac{\partial \rho}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$$



$$\left(\frac{\partial \rho}{\partial x}\right)^2 + \left(\frac{\partial \rho}{\partial y}\right)^2 = 1$$



$$\left[\left(\frac{\partial \rho}{\partial x}\right)^2 + \left(\frac{\partial \rho}{\partial y}\right)^2\right] \frac{\partial^2 u}{\partial \rho^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

$$\frac{\partial^2 \rho}{\partial x^2} = \frac{y^2}{(x^2 + y^2)^{\frac{3}{2}}}, \frac{\partial^2 \rho}{\partial y^2} = \frac{x^2}{(x^2 + y^2)^{\frac{3}{2}}}$$



$$\frac{\partial^2 \rho}{\partial x^2} + \frac{\partial^2 \rho}{\partial y^2} = \frac{1}{\rho}$$



$$+ \left(\frac{\partial^2 \rho}{\partial x^2} + \frac{\partial^2 \rho}{\partial y^2}\right) \frac{\partial u}{\partial \rho}$$

$$\frac{\partial \theta}{\partial x} = -\frac{y}{x^2 + y^2}, \frac{\partial \theta}{\partial y} = \frac{x}{x^2 + y^2}$$



$$\left(\frac{\partial \theta}{\partial x}\right)^2 + \left(\frac{\partial \theta}{\partial y}\right)^2 = \frac{1}{\rho^2}$$



$$+ \left[\left(\frac{\partial \theta}{\partial x}\right)^2 + \left(\frac{\partial \theta}{\partial y}\right)^2\right] \frac{\partial^2 u}{\partial \theta^2}$$

$$\frac{\partial^2 \theta}{\partial x^2} = \frac{2xy}{(x^2 + y^2)^2}, \frac{\partial^2 \theta}{\partial y^2} = -\frac{2xy}{(x^2 + y^2)^2}$$



$$\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} = 0$$



$$+ \left(\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2}\right) \frac{\partial u}{\partial \theta}$$



$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \Delta u = \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2}$$

一阶导数

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \rho} \frac{\partial \rho}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \rho} \frac{\partial \rho}{\partial y} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial y}$$

二阶导数

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial^2 u}{\partial \rho^2} \left(\frac{\partial \rho}{\partial x} \right)^2 + \frac{\partial u}{\partial \rho} \frac{\partial^2 \rho}{\partial x^2} + \frac{\partial^2 u}{\partial \theta^2} \left(\frac{\partial \theta}{\partial x} \right)^2 + \frac{\partial u}{\partial \theta} \frac{\partial^2 \theta}{\partial x^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial^2 u}{\partial \rho^2} \left(\frac{\partial \rho}{\partial y} \right)^2 + \frac{\partial u}{\partial \rho} \frac{\partial^2 \rho}{\partial y^2} + \frac{\partial^2 u}{\partial \theta^2} \left(\frac{\partial \theta}{\partial y} \right)^2 + \frac{\partial u}{\partial \theta} \frac{\partial^2 \theta}{\partial y^2}$$

两式相加



2.2 分离变量法

极坐标Poisson方程→分离变量→产生Euler方程

预备知识2

Euler方程解法

Euler方程 $t^n \frac{d^n x}{dt^n} + a_1 t^{n-1} \frac{d^{n-1} x}{dt^{n-1}} + \cdots + a_{n-1} t \frac{dx}{dt} + a_n x = F(t)$



$a_1 = 1, F(t) = 0$

二阶齐次

$$t^2 \frac{d^2 x}{dt^2} + t \frac{dx}{dt} + a_n x = 0$$

1739

L.Euler
1707-1783
Swiss Math.
Physical Scientist



解 令 $t = \exp s$, 则 $s = \ln t$

$$\frac{dx}{dt} = \frac{dx}{ds} \frac{ds}{dt} = \frac{1}{t} \frac{dx}{ds}$$

$$\begin{aligned} \frac{d^2 x}{dt^2} &= \frac{d}{dt} \left[\frac{1}{t} \frac{dx}{ds} \right] = \frac{dx}{ds} \frac{d}{dt} \left[\frac{1}{t} \right] + \frac{1}{t} \frac{d^2 x}{ds^2} \frac{ds}{dt} \\ &= -\frac{1}{t^2} \frac{dx}{ds} + \frac{1}{t^2} \frac{d^2 x}{ds^2} \end{aligned}$$

代入原方程 $t^2 \frac{d^2 x}{dt^2} + t \frac{dx}{dt} + a_n x = 0$

$$t^2 \left(-\frac{1}{t^2} \frac{dx}{ds} + \frac{1}{t^2} \frac{d^2 x}{ds^2} \right) + t \frac{1}{t} \frac{dx}{ds} + a_n x = 0$$

$$\frac{d^2 x}{ds^2} + a_n x = 0$$

$\frac{d^2 x}{ds^2} + a_n x = 0$ Euler方程化为常系数ODE

特征方程

$m^2 + a_n = 0$

复数m

$$\begin{cases} a_n < 0, & x = C_1 \exp(\sqrt{-a_n} s) + C_2 \exp(-\sqrt{-a_n} s) \\ a_n = 0, & x = C_1 s + C_2 \\ a_n > 0, & x = C_1 \cos(\sqrt{a_n} s) + C_2 \sin(\sqrt{a_n} s) \end{cases}$$

还原变量

$s = \ln t$

$$t^2 \frac{d^2 x}{dt^2} + t \frac{dx}{dt} + a_n x = 0 \begin{cases} a_n < 0, & x = C_1 t^{\sqrt{-a_n}} + C_2 t^{-\sqrt{-a_n}} \\ a_n = 0, & x = C_1 \ln t + C_2 \\ a_n > 0, & x = C_1 \cos(\sqrt{a_n} \ln t) + C_2 \sin(\sqrt{a_n} \ln t) \end{cases}$$

变量代换 $t = \exp s$ 实现了变系数ODE(Euler)方程的求解

2.2 分离变量法

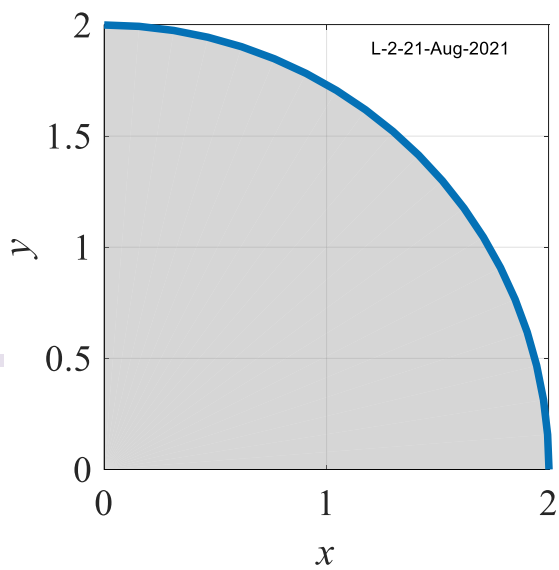
$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

平面Laplace方程

例2

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x, \ 0 < y, \ x^2 + y^2 < 4 \\ u(x, 0) = 0, & 0 \leq x \leq 2 \\ u(0, y) = 0, & 0 \leq y \leq 2 \\ u(x, y) = xy, & x^2 + y^2 = 4 \end{cases}$$

Laplace/二维/齐次		
x边界		
y边界		



解 第1步 坐标转换

扇形区域→直角系无齐次边界→向极坐标转换

$$x = \rho \cos \theta, \ y = \rho \sin \theta$$

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < 2, \ 0 < \theta < \frac{\pi}{2} \\ u(\rho, 0) = 0, \ u\left(\rho, \frac{\pi}{2}\right) = 0, & 0 \leq \rho \leq 2 \\ u(2, \theta) = 2 \sin \theta \cdot 2 \cos \theta = 2 \sin 2\theta, & 0 \leq \theta \leq \frac{\pi}{2} \end{cases}$$

Laplace/二维/齐次		
周向θ	1齐	1齐
径向ρ	1齐	1非齐

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

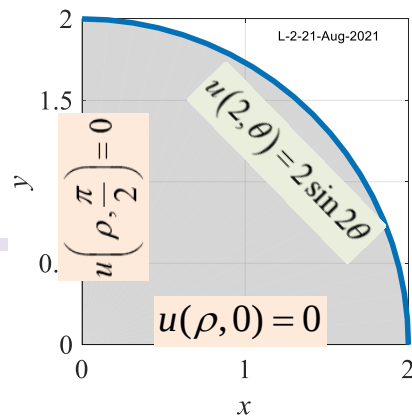
2.2 分离变量法

平面Laplace方程

例2

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x, & 0 < y, & x^2 + y^2 < 4 \\ u(x, 0) = 0, & 0 \leq x \leq 2 \\ u(0, y) = 0, & 0 \leq y \leq 2 \\ u(x, y) = xy, & x^2 + y^2 = 4 \end{cases} \rightarrow \begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < 2, & 0 < \theta < \frac{\pi}{2} \\ u(\rho, 0) = 0, & u\left(\rho, \frac{\pi}{2}\right) = 0, & 0 \leq \rho \leq 2 \\ u(2, \theta) = 2 \sin \theta \cdot 2 \cos \theta = 2 \sin 2\theta, & 0 \leq \theta \leq \frac{\pi}{2} \end{cases}$$

Laplace/二维/齐次		
周向 θ	1齐	1齐
径向 ρ	1齐	1非齐



第2步 特征值问题 令 $u(\rho, \theta) = R(\rho)\Phi(\theta)$

$$R''(\rho)\Phi(\theta) + \frac{1}{\rho} R'(\rho)\Phi(\theta) + \frac{1}{\rho^2} R(\rho)\Phi''(\theta) = 0$$

$$\frac{1}{\rho^2} R(\rho)\Phi''(\theta) = -\left[R''(\rho) + \frac{1}{\rho} R'(\rho)\right]\Phi(\theta)$$

$$\frac{\Phi''(\theta)}{\Phi(\theta)} = -\frac{R''(\rho) + \frac{1}{\rho} R'(\rho)}{\frac{1}{\rho^2} R(\rho)} = -\lambda$$

$$\underbrace{\rho^2 R''(\rho) + \rho R'(\rho) - \lambda R(\rho)}_{\text{径向}} = 0 \quad \underbrace{\Phi''(\theta) + \lambda \Phi(\theta)}_{\text{周向}} = 0$$

极坐标下→周向边
界齐次→确定特征
值 / 特征函数

2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

平面Laplace方程

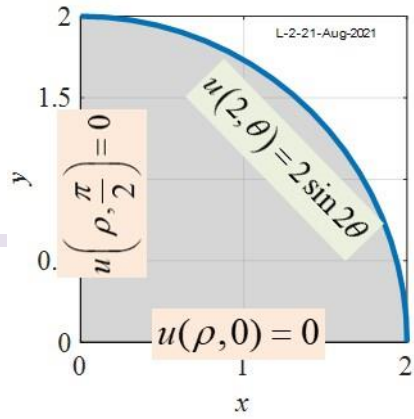
例2

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x, 0 < y, x^2 + y^2 < 4 \\ u(x, 0) = 0, & 0 \leq x \leq 2 \\ u(0, y) = 0, & 0 \leq y \leq 2 \\ u(x, y) = xy, & x^2 + y^2 = 4 \end{cases}$$



$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < 2, 0 < \theta < \frac{\pi}{2} \\ u(\rho, 0) = 0, & u\left(\rho, \frac{\pi}{2}\right) = 0, & 0 \leq \rho \leq 2 \\ u(2, \theta) = 2 \sin \theta \cdot 2 \cos \theta = 2 \sin 2\theta, & 0 \leq \theta \leq \frac{\pi}{2} \end{cases}$$

Laplace/二维/齐次		
周向θ	1齐	1齐
径向ρ	1齐	1非齐



第2步 特征值问题

不恒为0

扇域非周期边界

周向边界

$$\begin{aligned} u(\rho, 0) &= R(\rho) \Phi(0) = 0 \\ u\left(\rho, \frac{\pi}{2}\right) &= R(\rho) \Phi\left(\frac{\pi}{2}\right) = 0 \end{aligned}$$



$$\begin{cases} \Phi(0) = 0, & \Phi\left(\frac{\pi}{2}\right) = 0 \\ \Phi''(\theta) + \lambda \Phi(\theta) = 0 \end{cases}$$



周向[1,1]型齐次边界

本例中

$$l = \frac{\pi}{2}$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2 = \left(\frac{n\pi}{\pi/2}\right)^2 = 4n^2, \quad (n=1, 2, 3, \dots)$$

$$\Phi_n(\theta) = \sin\left(\frac{n\pi}{l}\theta\right) = \sin\left(\frac{n\pi}{\pi/2}\theta\right) = \sin(2n\theta)$$

周向解毕/分离常数确定→径向可解

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$ (n=1, 2, 3...)
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right]$ (n=0, 1, 2, 3...)
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$ (n=0, 1, 2, 3...)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right)$ (n=0, 1, 2, 3...)

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

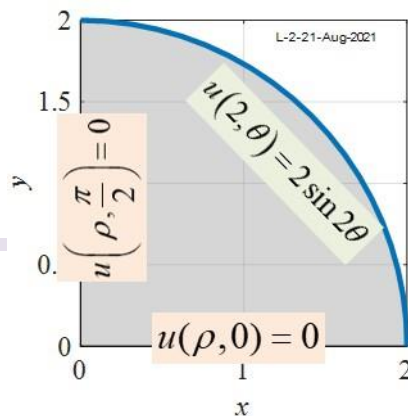
2.2 分离变量法

平面Laplace方程

例2

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x, & 0 < y, & x^2 + y^2 < 4 \\ u(x, 0) = 0, & 0 \leq x \leq 2 \\ u(0, y) = 0, & 0 \leq y \leq 2 \\ u(x, y) = xy, & x^2 + y^2 = 4 \end{cases} \rightarrow \begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < 2, & 0 < \theta < \frac{\pi}{2} \\ u(\rho, 0) = 0, & u\left(\rho, \frac{\pi}{2}\right) = 0, & 0 \leq \rho \leq 2 \\ u(2, \theta) = 2 \sin \theta \cdot 2 \cos \theta = 2 \sin 2\theta, & 0 \leq \theta \leq \frac{\pi}{2} \end{cases}$$

Laplace/二维/齐次		
周向 θ	1齐	1齐
径向 ρ	1齐	1非齐



第3步 解径向ODE

$$\lambda_n = 4n^2, \quad \Phi_n(x) = \sin(2n\theta), \quad (n=1, 2, 3, \dots)$$

特征值代入径向ODE

$$\rho^2 R''(\rho) + \rho R'(\rho) - \lambda R(\rho) = 0$$

$$\rho^2 R''(\rho) + \rho R'(\rho) - 4n^2 R(\rho) = 0$$

Euler方程令 $\rho = \exp s$ 各阶导数

$$s = \ln \rho$$

$$\left\{ \begin{aligned} R'_n(\rho) &= R'_n(s) \frac{ds}{d\rho} = \frac{1}{\rho} R'_n(s) \\ R''_n(\rho) &= \frac{d\left[\frac{1}{\rho} R'_n(s)\right]}{d\rho} = R'_n(s) \frac{d\left[\frac{1}{\rho}\right]}{d\rho} + \frac{1}{\rho} \frac{d[R'_n(s)]}{ds} \frac{ds}{d\rho} \\ &= -\frac{1}{\rho^2} R'_n(s) + \frac{1}{\rho^2} R''_n(s) \end{aligned} \right.$$

$$\text{代入径向ODE} \quad \rho^2 \left[-\frac{1}{\rho^2} R'_n(s) + \frac{1}{\rho^2} R''_n(s) \right] + \rho \frac{1}{\rho} R'_n(s) - 4n^2 R_n = 0$$

$$R''_n(s) - 4n^2 R_n(s) = 0$$

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

2.2 分离变量法

平面Laplace方程

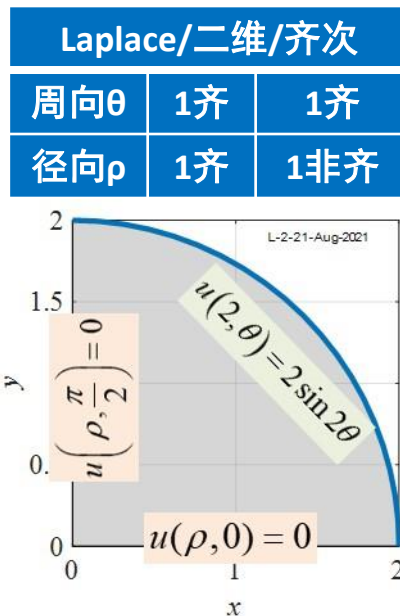
例2

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x, & 0 < y, & x^2 + y^2 < 4 \\ u(x, 0) = 0, & 0 \leq x \leq 2 \\ u(0, y) = 0, & 0 \leq y \leq 2 \\ u(x, y) = xy, & x^2 + y^2 = 4 \end{cases}$$



$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < 2, & 0 < \theta < \frac{\pi}{2} \\ u(\rho, 0) = 0, & u\left(\rho, \frac{\pi}{2}\right) = 0, & 0 \leq \rho \leq 2 \\ u(2, \theta) = 2 \sin \theta \cdot 2 \cos \theta = 2 \sin 2\theta, & 0 \leq \theta \leq \frac{\pi}{2} \end{cases}$$

$$\lambda_n = 4n^2, \quad \Phi_n(x) = \sin(2n\theta), \quad (n = 1, 2, 3, \dots)$$



第3步 解径向ODE

$$R_n''(s) - 4n^2 R_n = 0 \quad n \text{ 从 } 1 \text{ 开始, 无需考虑 } n=0$$

由 $n > 0$ 可知通解为

$$R_n(s) = C_{1n} \exp(2ns) + C_{2n} \exp(-2ns)$$

还原自变量 $s = \ln \rho$, $R_n(\rho) = C_{1n} \rho^{2n} + C_{2n} \rho^{-2n}$

考虑自然边界条件 $|R_n(0)| < \infty$, $C_{2n} = 0$

实际物理量
不可能为 ∞

径向ODE通解

$$R_n(\rho) = C_{1n} \rho^{2n}$$

Euler 方程的解

$$t^2 \frac{d^2 x}{dt^2} + t \frac{dx}{dt} + a_n x = 0$$

$$\begin{cases} a_n < 0, & x = C_1 t^{\sqrt{-a_n}} + C_2 t^{-\sqrt{-a_n}} \\ a_n = 0, & x = C_1 \ln t + C_2 \\ a_n > 0, & x = C_1 \cos(\sqrt{a_n} \ln t) + C_2 \sin(\sqrt{a_n} \ln t) \end{cases}$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

平面Laplace方程

例2

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x, 0 < y, x^2 + y^2 < 4 \\ u(x, 0) = 0, & 0 \leq x \leq 2 \\ u(0, y) = 0, & 0 \leq y \leq 2 \\ u(x, y) = xy, & x^2 + y^2 = 4 \end{cases}$$



$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < 2, 0 < \theta < \frac{\pi}{2} \\ u(\rho, 0) = 0, & u\left(\rho, \frac{\pi}{2}\right) = 0, & 0 \leq \rho \leq 2 \\ u(2, \theta) = 2 \sin \theta \cdot 2 \cos \theta = 2 \sin 2\theta, & 0 \leq \theta \leq \frac{\pi}{2} \end{cases}$$

Laplace/二维/齐次		
周向 θ	1齐	1齐
径向 ρ	1齐	1非齐

第3步 解径向ODE 原PDE形式解

$$u(\rho, \theta) = \sum_{n=1}^{\infty} R_n(\rho) \Phi_n(\theta) = \sum_{n=1}^{\infty} C_{1n} \rho^{2n} \sin(2n\theta)$$

利用径向非齐次边界确定系数

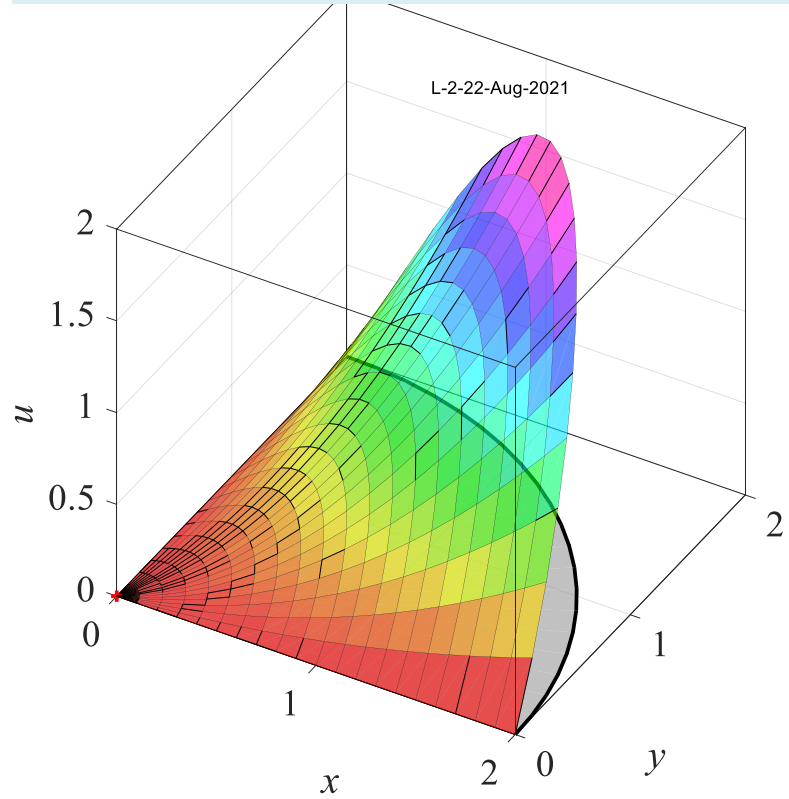
$$u(2, \theta) = \sum_{n=1}^{\infty} C_{1n} 2^{2n} \sin(2n\theta) = 2 \sin 2\theta$$

比较系数, 只可能 $n=1, C_{1n} = \frac{1}{2}$ 可去掉 Σ

原PDE的解 $u(\rho, \theta) = \frac{1}{2} \rho^2 \sin 2\theta$

还原为直角系 $u(x, y) = xy$

$$\lambda_n = 4n^2, \Phi_n(x) = \sin(2n\theta), (n=1, 2, 3 \dots)$$



2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

平面Laplace方程

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$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < 2, 0 < \theta < \frac{\pi}{2} \\ u(\rho, 0) = 0, \quad u\left(\rho, \frac{\pi}{2}\right) = 0, & 0 \leq \rho \leq 2 \\ u(2, \theta) = 2 \sin \theta \cdot 2 \cos \theta = 2 \sin 2\theta, & 0 \leq \theta \leq \frac{\pi}{2} \end{cases}$$

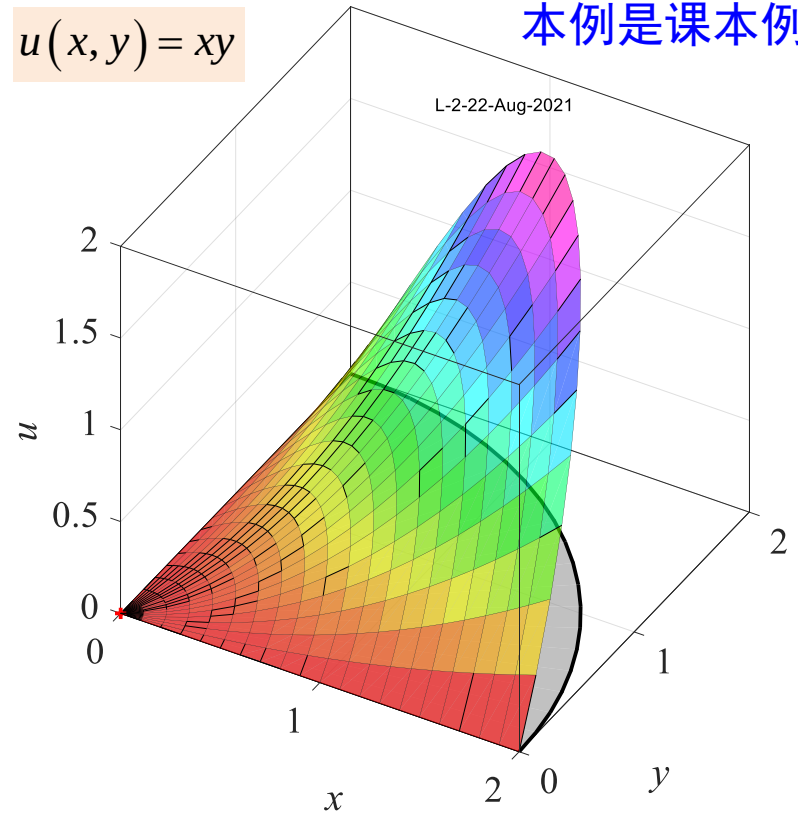
Laplace/二维/齐次		
周向 θ	1齐	1齐
径向 ρ	1齐	1非齐

原方程的解 $u(\rho, \theta) = \frac{1}{2} \rho^2 \sin 2\theta$ 还原为直角系 $u(x, y) = xy$

本例是课本例2.8

点评 分离变量法诞生200年来求解 Poisson方程最经典的例题

- 1. 直角系下非齐→化为极坐标→周向齐次
- 2. 利用齐次周向求特征值→未使用周期边界
- 3. 径向为Euler方程→经典变量代换法
- 4. 自然边界条件→简化了Euler方程通解
- 5. 比较系数→确定形式解→竟然去掉了 Σ
- 6. 最终解还原为直角系→简单明了



2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

平面Laplace方程

例3
$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < a, \quad 0 \leq \theta < 2\pi \\ u(a, \theta) = \varphi(\theta), & 0 \leq \theta < 2\pi \end{cases}$$

Laplace/二维/齐次		
周向θ	周期	
径向ρ	自然	1齐

解 周向定义域延拓至 **R**, 满足周期性边界

令 $u = R(\rho)\Phi(\theta)$ 代入极坐标方程

$$R''(\rho)\Phi(\theta) + \frac{1}{\rho} R'(\rho)\Phi(\theta) + \frac{1}{\rho^2} R(\rho)\Phi''(\theta) = 0$$

$$\frac{1}{\rho^2} R(\rho)\Phi''(\theta) = - \left[R''(\rho) + \frac{1}{\rho} R'(\rho) \right] \Phi(\theta)$$

$$\frac{\Phi''(\theta)}{\Phi(\theta)} = - \frac{R''(\rho) + \frac{1}{\rho} R'(\rho)}{\frac{1}{\rho^2} R(\rho)} = -\lambda$$

$$\rho^2 R''(\rho) + \rho R'(\rho) - \lambda R(\rho) = 0$$

$$\Phi''(\theta) + \lambda \Phi(\theta) = 0$$

$$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(\theta) = \Phi(\theta + 2\pi), \quad \Phi'(\theta) = \Phi'(\theta + 2\pi) \end{cases}$$
 特征值
特征函数

$$\lambda_n = n^2, \quad \Phi_n = \{1, \cos n\theta, \sin n\theta\}, \quad (n = 0, 1, 2, \dots)$$

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数		
齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3, \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3, \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0, 1, 2, 3, \dots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1, 2, 3, \dots)$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

平面Laplace方程

例3
$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < a, \ 0 \leq \theta < 2\pi \\ u(a, \theta) = \varphi(\theta), & 0 \leq \theta < 2\pi \end{cases}$$

Laplace/二维/齐次		
周向 θ	周期	
径向 ρ	自然	1齐

$$\lambda_n = n^2, \ \Phi_n = C_{1n} \cos n\theta + C_{2n} \sin n\theta, \ (n = 0, 1, 2, \dots)$$

将特征值代入径向方程

$$\rho^2 R''(\rho) + \rho R'(\rho) - n^2 R(\rho) = 0$$

Euler方程令 $\rho = \exp s$ $s = \ln \rho$

各阶导数
$$\begin{aligned} R'_n(\rho) &= R'_n(s) \frac{ds}{d\rho} = \frac{1}{\rho} R'_n(s) \\ R''_n(\rho) &= -\frac{1}{\rho^2} R'_n(s) + \frac{1}{\rho^2} R''_n(s) \end{aligned}$$

$$\rho^2 \left[-\frac{1}{\rho^2} R'_n(s) + \frac{1}{\rho^2} R''_n(s) \right] + \rho \frac{1}{\rho} R'_n(s) - n^2 R_n = 0$$

代入原方程
$$R''_n(s) - n^2 R_n = 0$$

根据 n 分类讨论

① 当 $n=0$ 时, $R''_n(s) - n^2 R_n = 0$ 的通解为

$$R_0(s) = C_{10} + C_{20}s, \ R_0(\rho) = C_{10} + C_{20} \ln \rho$$

圆心自然边界 $|u(0, \theta)| < \infty, \ |R(0)| < \infty$

故 $C_{20} = 0$ $R_0(\rho) = C_{10}$

② 当 $n>0$ 时, $R''_n(s) - n^2 R_n = 0$ 的通解为

$$R_n(s) = C_{1n} \exp(ns) + C_{2n} \exp(-ns)$$

$$R_n(\rho) = C_{1n} \rho^n + C_{2n} \rho^{-n}$$

根据自然边界 $C_{2n} = 0$ $R_n(\rho) = C_{1n} \rho^n$

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

2.2 分离变量法

Laplace/二维/齐次

周向 θ

周期

径向 ρ

自然

1齐

平面Laplace方程

例3

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < a, \quad 0 \leq \theta < 2\pi \\ u(a, \theta) = \varphi(\theta), & 0 \leq \theta < 2\pi \end{cases}$$

定义在 $[0, l]$ 上函数 $f(x)$ 的Fourier展开/正交分解

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x)$$

$$f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx,$$

$$f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

原方程的解可表示为

$n=0$ 单列

$$\begin{aligned} u(\rho, \theta) &= R_0(\rho) \Phi_0(\theta) + \sum_{n=1}^{\infty} R_n(\rho) \Phi_n(\theta) \\ &= C_{10} + \sum_{n=1}^{\infty} \rho^n (C_{1n} \cos n\theta + C_{2n} \sin n\theta) \end{aligned}$$

根据边界条件

$$u(a, \theta) = C_{10} + \sum_{n=1}^{\infty} a^n (C_{1n} \cos n\theta + C_{2n} \sin n\theta) = \varphi(\theta)$$

$$C_{10} = \frac{1}{2\pi} \int_0^{2\pi} \varphi(\theta) \cdot 1 \cdot d\theta = \frac{1}{2\pi} \int_0^{2\pi} \varphi(\theta) d\theta$$

$$C_{1n} = \frac{1}{a^n} \frac{1}{\pi} \int_0^{2\pi} \varphi(\theta) \cos n\theta d\theta = \frac{1}{a^n \pi} \int_0^{2\pi} \varphi(\theta) \cos n\theta d\theta$$

$$C_{2n} = \frac{1}{a^n} \frac{1}{\pi} \int_0^{2\pi} \varphi(\theta) \sin n\theta d\theta = \frac{1}{a^n \pi} \int_0^{2\pi} \varphi(\theta) \sin n\theta d\theta$$

原定解问题的解

$$\begin{aligned} u(\rho, \theta) &= C_{10} + \sum_{n=1}^{\infty} \rho^n (C_{1n} \cos n\theta + C_{2n} \sin n\theta) \\ &= \frac{1}{2\pi} \int_0^{2\pi} \varphi(\theta) d\theta \\ &\quad + \frac{1}{\pi} \sum_{n=1}^{\infty} \left(\frac{\rho}{a} \right)^n \left[\int_0^{2\pi} \varphi(\theta) \cos n\theta d\theta \cdot \cos n\theta \right. \\ &\quad \left. + \int_0^{2\pi} \varphi(\theta) \sin n\theta d\theta \cdot \sin n\theta \right] \end{aligned}$$

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

2.2 分离变量法

平面Laplace方程

$$\lambda_n = n^2, \Phi_n = C_{1n} \cos n\theta + C_{2n} \sin n\theta, (n = 0, 1, 2, \dots)$$

例3

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < a, 0 \leq \theta < 2\pi \\ u(a, \theta) = \varphi(\theta), & 0 \leq \theta < 2\pi \end{cases}$$

Laplace/二维/齐次

周向 θ

周期

径向 ρ

自然

1齐

讨论

举例, 若取 $\varphi(\theta) = \sin^2 \theta$, $a = 1$

$$u(\rho, \theta) = C_{10} + \sum_{n=1}^{\infty} \rho^n (C_{1n} \cos n\theta + C_{2n} \sin n\theta)$$

$$C_{10} = \frac{1}{2\pi} \int_0^{2\pi} \varphi(\theta) d\theta = \frac{1}{2\pi} \int_0^{2\pi} \sin^2 \theta d\theta = \frac{1}{2}$$

$$C_{1n} = \frac{1}{a^n \pi} \int_0^{2\pi} \varphi(\theta) \cos n\theta d\theta$$

$$= \frac{1}{\pi} \int_0^{2\pi} \sin^2 \theta \cos n\theta d\theta$$

$$= \frac{1}{\pi} \int_0^{2\pi} \frac{1 - \cos 2\theta}{2} \cos n\theta d\theta$$

$$= \frac{1}{\pi} \left[\frac{1}{2} \int_0^{2\pi} 1 \cdot \cos n\theta d\theta - \frac{1}{2} \int_0^{2\pi} \cos 2\theta \cos n\theta d\theta \right]$$

$$= \frac{1}{\pi} \left[0 - \frac{1}{2} \cdot \pi \right] = -\frac{1}{2} \quad (n=2)$$

仅 $n=2$ 时
不为0

可去掉 Σ

$$C_{2n} = \frac{1}{a^n \pi} \int_0^{2\pi} \varphi(\theta) \sin n\theta d\theta$$

$$= \frac{1}{\pi} \int_0^{2\pi} \sin^2 \theta \sin n\theta d\theta = 0$$

$$= \frac{1}{\pi} \int_0^{2\pi} \frac{1 - \cos 2\theta}{2} \sin n\theta d\theta$$

$$= \frac{1}{\pi} \left[\frac{1}{2} \int_0^{2\pi} \sin n\theta d\theta - \frac{1}{2} \int_0^{2\pi} \cos 2\theta \sin n\theta d\theta \right]$$

$$= 0$$

=0

=0

2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

平面Laplace方程

$$\lambda_n = n^2, \Phi_n = C_{1n} \cos n\theta + C_{2n} \sin n\theta, \quad (n = 0, 1, 2, \dots)$$

例3

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < a, \quad 0 \leq \theta < 2\pi \\ u(a, \theta) = \varphi(\theta), & 0 \leq \theta < 2\pi \end{cases}$$

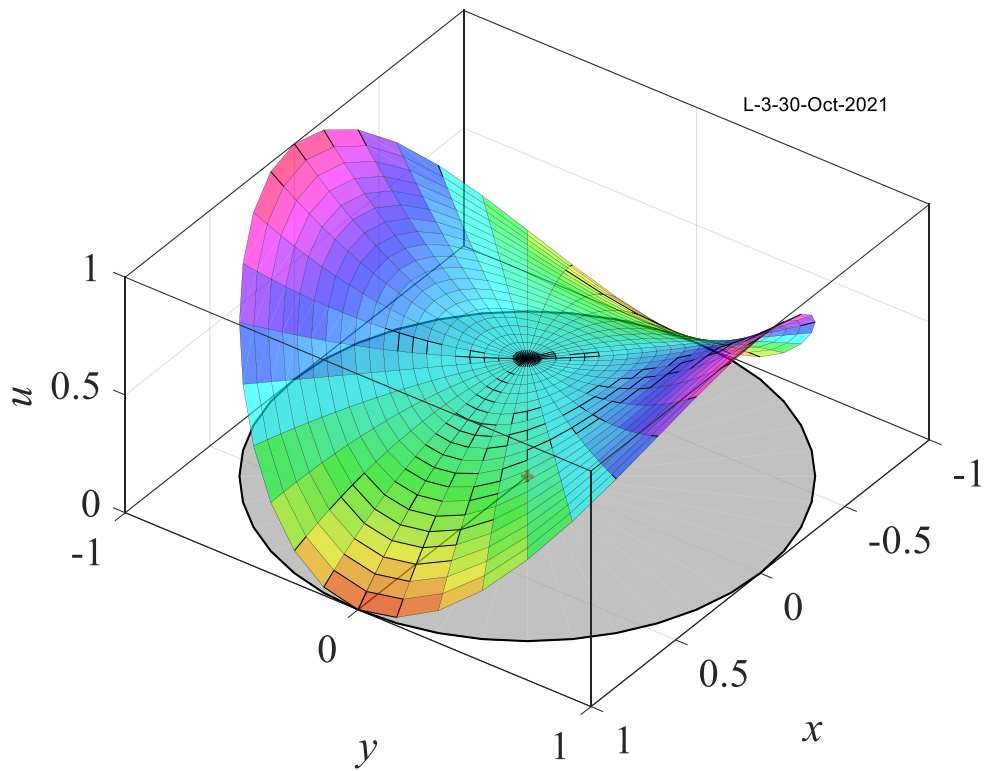
Laplace/二维/齐次		
周向 θ	周期	
径向 ρ	自然	1齐

讨论 举例, 若取 $\varphi(\theta) = \sin^2 \theta, \quad a = 1$

$$C_{10} = \frac{1}{2}, \quad C_{1n} = -\frac{1}{2}, \quad (n = 2), \quad C_{2n} = 0$$

$$u(\rho, \theta) = C_{10} + \sum_{n=1}^{\infty} \rho^n (C_{1n} \cos n\theta + C_{2n} \sin n\theta)$$

原方程的解为
$$u(\rho, \theta) = \frac{1}{2} (1 - \rho^2 \cos 2\theta)$$



2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

平面Laplace方程

$$\lambda_n = n^2, \quad \Phi_n = C_{1n} \cos n\theta + C_{2n} \sin n\theta, \quad (n = 0, 1, 2, \dots)$$

例3
$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < a, \quad 0 \leq \theta < 2\pi \\ u(a, \theta) = \varphi(\theta), & 0 \leq \theta < 2\pi \end{cases}$$

Laplace/二维/齐次

周向 θ	周期	
径向 ρ	自然	1齐

讨论 举例, 若取 $\varphi(\theta) = \sin \theta$, $a = 1$

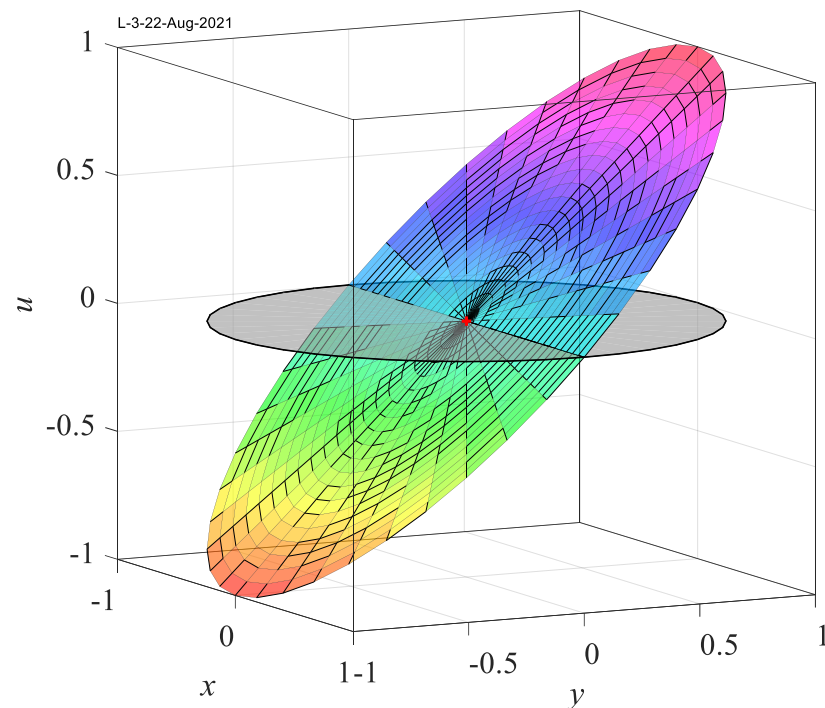
$$C_{10} = \frac{1}{2\pi} \int_0^{2\pi} \varphi(\theta) d\theta = \frac{1}{2\pi} \int_0^{2\pi} \sin \theta d\theta = 0$$

$$\begin{aligned} C_{1n} &= \frac{1}{a^n \pi} \int_0^{2\pi} \varphi(\theta) \cos n\theta d\theta \\ &= \frac{1}{\pi} \int_0^{2\pi} \sin \theta \cos n\theta d\theta = 0 \end{aligned}$$

$$\begin{aligned} C_{2n} &= \frac{1}{a^n \pi} \int_0^{2\pi} \varphi(\theta) \sin n\theta d\theta \\ &= \frac{1}{\pi} \int_0^{2\pi} \sin \theta \sin n\theta d\theta = \begin{cases} = 1, & (n = 1) \\ = 0, & (n > 1) \end{cases} \end{aligned}$$

$$u(\rho, \theta) = C_{10} + \sum_{n=1}^{\infty} \rho^n (C_{1n} \cos n\theta + C_{2n} \sin n\theta)$$

$$u(\rho, \theta) = \rho \sin \theta$$



2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

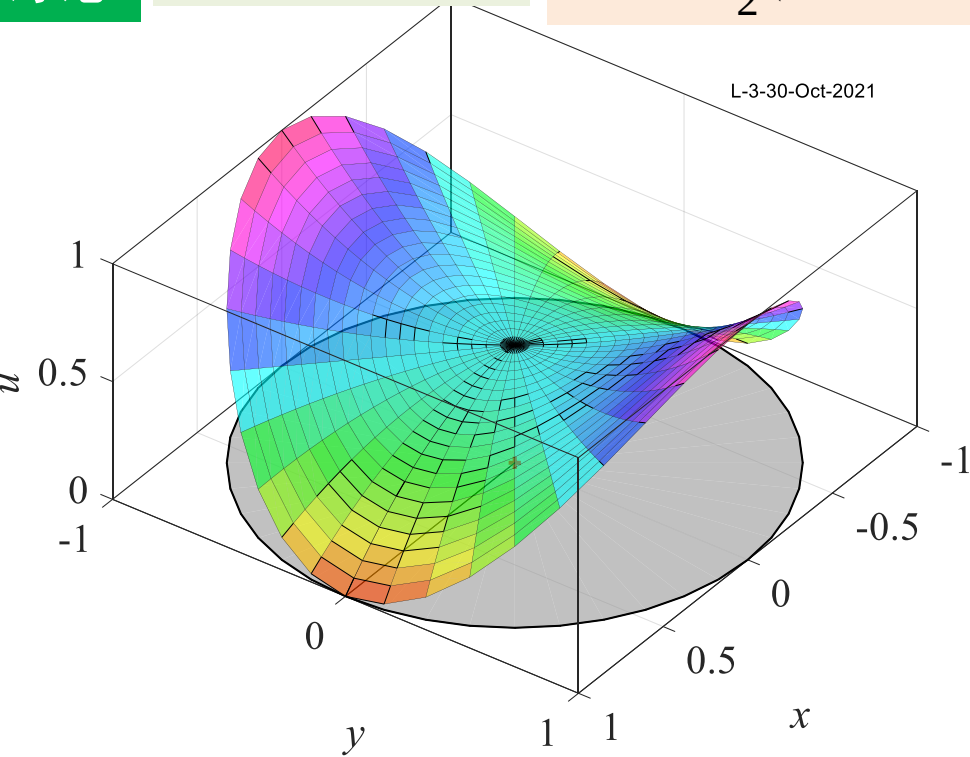
平面Laplace方程

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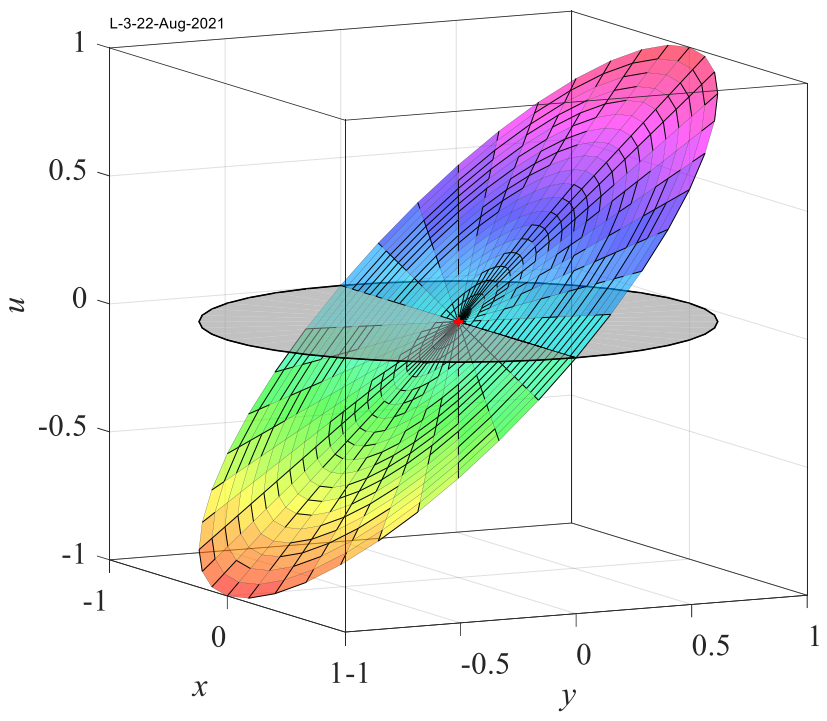
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Laplace/二维/齐次		
周向 θ	周期	
径向 ρ	自然	1齐

讨论 $\varphi(\theta) = \sin^2 \theta, \quad a = 1 \quad u(\rho, \theta) = \frac{1}{2}(1 - \rho^2 \cos 2\theta)$



$\varphi(\theta) = \sin \theta, \quad a = 1 \quad u(\rho, \theta) = \rho \sin \theta$



给定边界→内部唯一确定→能否用通式表达

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

2.2 分离变量法

平面Laplace方程

例3

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Laplace/二维/齐次

周向 θ

周期

径向 ρ

自然

1齐

S. Poisson
1781-1840

French Math.
Physical Scientist



$$u(\rho, \theta) = C_{10} + \sum_{n=1}^{\infty} \rho^n (C_{1n} \cos n\theta + C_{2n} \sin n\theta)$$

调整积分
变量

$$= \frac{1}{2\pi} \int_0^{2\pi} \varphi(\tau) d\tau + \frac{1}{\pi} \sum_{n=1}^{\infty} \left(\frac{\rho}{a} \right)^n \left[\int_0^{2\pi} \varphi(\tau) \cos n\tau d\tau \cdot \cos n\theta + \int_0^{2\pi} \varphi(\tau) \sin n\tau d\tau \cdot \sin n\theta \right]$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \varphi(\tau) d\tau + \frac{1}{\pi} \sum_{n=1}^{\infty} \left(\frac{\rho}{a} \right)^n \int_0^{2\pi} [\varphi(\tau) \cos n\tau \cdot \cos n\theta + \varphi(\tau) \sin n\tau \cdot \sin n\theta] d\tau$$

合并积分

$$= \frac{1}{2\pi} \int_0^{2\pi} \varphi(\tau) \left[1 + 2 \sum_{n=1}^{\infty} \left(\frac{\rho}{a} \right)^n \cos n(\theta - \tau) \right] d\tau$$

和角公式/积分
累加交换次序

$$e^{ix} = \cos x + i \sin x \quad \text{Euler公式}$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \varphi(\tau) \left[1 + 2 \operatorname{Re} \left[\sum_{n=1}^{\infty} \left(\frac{\rho}{a} \right)^n e^{in(\theta - \tau)} \right] \right] d\tau = \frac{1}{2\pi} \int_0^{2\pi} \varphi(\tau) \left[1 + 2 \operatorname{Re} \left[\frac{\left(\frac{\rho}{a} \right) e^{i(\theta - \tau)}}{1 - \left(\frac{\rho}{a} \right) e^{i(\theta - \tau)}} \right] \right] d\tau$$

等比级数求和

$$u(\rho, \theta) = \frac{1}{2\pi} \int_0^{2\pi} \frac{(a^2 - \rho^2)}{a^2 + \rho^2 - 2a\rho \cos(\theta - \tau)} \varphi(\tau) d\tau$$

圆域上调和函数的Poisson公式

通解/数值/Green公式

2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

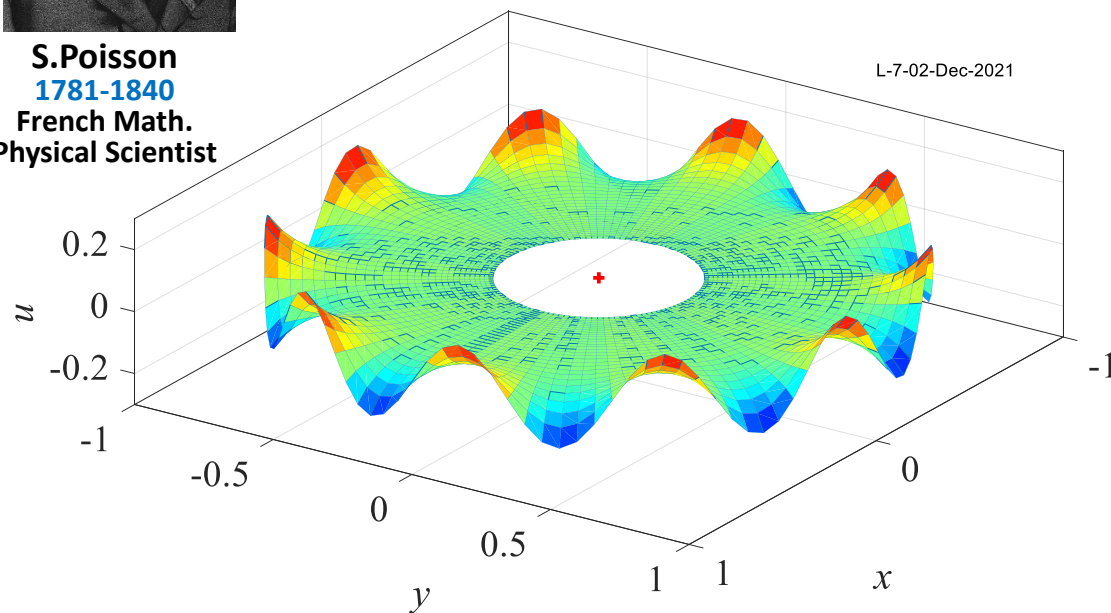
Poisson公式的应用



S. Poisson
1781-1842
French Math.
Physical Scientist

$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0, & 0 < \rho < a, \quad 0 \leq \theta < 2\pi \\ u(a, \theta) = \varphi(\theta), & 0 \leq \theta < 2\pi \end{cases} \quad \text{圆域上调和函数Poisson公式}$$
$$u(\rho, \theta) = \frac{1}{2\pi} \int_0^{2\pi} \frac{(a^2 - \rho^2)}{a^2 + \rho^2 - 2a\rho \cos(\theta - \tau)} \varphi(\tau) d\tau$$

给定边界条件 $\varphi(\theta) = 0.15 \cdot a \sin(10\theta)$ 采用数值积分可计算圆域内变量分布



1. Poisson公式相当于Laplace方程的通解, 可计算整个圆域/也可计算部分区域
2. 难点是数值积分中算法的组织
3. Poisson公式+边界条件→可作为复杂曲面的表示形式

2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

平面Laplace方程

例4
$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 12\rho^2 \cos 2\theta, & a < \rho < b, 0 \leq \theta < 2\pi \\ u(a, \theta) = 0, \quad u_{\rho}(b, \theta) = 0, & 0 \leq \theta \leq 2\pi \end{cases}$$

Poisson/二维环域/非齐		
周向θ	周期	
径向ρ	1齐	2齐

解 第1步 特征值问题

令 $u = R(\rho)\Phi(\theta)$ 代入齐次方程

$$R''(\rho)\Phi(\theta) + \frac{1}{\rho} R'(\rho)\Phi(\theta) + \frac{1}{\rho^2} R(\rho)\Phi''(\theta) = 0$$
$$\frac{1}{\rho^2} R(\rho)\Phi''(\theta) = - \left[R''(\rho) + \frac{1}{\rho} R'(\rho) \right] \Phi(\theta)$$
$$\frac{\Phi''(\theta)}{\Phi(\theta)} = - \underbrace{\frac{R''(\rho) + \frac{1}{\rho} R'(\rho)}{\frac{1}{\rho^2} R(\rho)}}_{\lambda} = -\lambda$$
$$\rho^2 R''(\rho) + \rho R'(\rho) - \lambda R(\rho) = 0$$
$$\Phi''(\theta) + \lambda \Phi(\theta) = 0$$

周期性边界

$$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(\theta) = \Phi(\theta + 2\pi), \quad \Phi'(\theta) = \Phi'(\theta + 2\pi) \end{cases}$$
$$\lambda_n = n^2, \quad \Phi_n = \{1, \cos n\theta, \sin n\theta\}, \quad (n = 0, 1, 2, \dots)$$

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数		
齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3, \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3, \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0, 1, 2, 3, \dots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1, 2, 3, \dots)$

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

2.2 分离变量法

平面Laplace方程

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$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 12\rho^2 \cos 2\theta, & a < \rho < b, 0 \leq \theta < 2\pi \\ u(a, \theta) = 0, & u(b, \theta) = 0, & 0 \leq \theta \leq 2\pi \end{cases}$$

Poisson/二维环域/非齐

周向 θ

周期

径向 ρ

1齐

2齐

$$\lambda_n = n^2, \quad \Phi_n = \{1, \cos n\theta, \sin n\theta\}, \quad (n = 0, 1, 2, \dots)$$

第2步 正交分解

$$u(\rho, \theta) = \sum_{n=0}^{\infty} R_n(\rho) \Phi_n(\theta)$$

$$12\rho^2 \cos 2\theta = \sum_{n=0}^{\infty} f_n \cdot \Phi_n(\theta)$$

$$= f_0 \cdot 1 \quad \rightarrow \quad f_0 = \frac{1}{2\pi} \int_0^{2\pi} 12\rho^2 \cos 2\theta \cdot 1 \cdot d\theta = 0$$

$$+ \sum_{n=0}^{\infty} f_{n,\cos} \cdot \cos n\theta \quad \rightarrow \quad f_{n,\cos} = \frac{1}{\pi} \int_0^{2\pi} 12\rho^2 \cos 2\theta \cdot \cos n\theta \cdot d\theta = \begin{cases} 12\rho^2, & (n=2) \\ 0, & (n \neq 2) \end{cases}$$

$$+ \sum_{n=0}^{\infty} f_{n,\sin} \cdot \sin n\theta \quad \rightarrow \quad f_{n,\sin} = \frac{1}{\pi} \int_0^{2\pi} 12\rho^2 \cos 2\theta \cdot \sin n\theta \cdot d\theta = 0$$

$$= 0 + f_{2,\cos} \cdot \cos 2\theta + 0$$

$$= 12\rho^2 \cdot \cos 2\theta$$

虽与展开前相同→
径向ODE分类讨论

自由项

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

2.2 分离变量法

平面Laplace方程

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$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 12\rho^2 \cos 2\theta, & a < \rho < b, 0 \leq \theta < 2\pi \\ u(a, \theta) = 0, & u(b, \theta) = 0, 0 \leq \theta \leq 2\pi \end{cases}$$

Poisson/二维环域/非齐

周向 θ

周期

径向 ρ

1齐

2齐

$$\lambda_n = n^2, \Phi_n = \{1, \cos n\theta, \sin n\theta\}, (n = 0, 1, 2, \dots)$$

第3步 建立径向ODE

$$\sum_{n=0}^{\infty} R_n''(\rho) \Phi_n(\theta) + \frac{1}{\rho} \sum_{n=0}^{\infty} R_n'(\rho) \Phi_n(\theta) + \frac{1}{\rho^2} \sum_{n=0}^{\infty} R_n(\rho) \Phi_n''(\theta) = \sum_{n=0}^{\infty} f_n \Phi_n(\theta)$$

$$\sum_{n=0}^{\infty} R_n''(\rho) \Phi_n(\theta) + \frac{1}{\rho} \sum_{n=0}^{\infty} R_n'(\rho) \Phi_n(\theta) - \frac{1}{\rho^2} \sum_{n=0}^{\infty} R_n(\rho) \lambda_n \Phi_n(\theta) = \sum_{n=0}^{\infty} f_n \Phi_n(\theta)$$

$$\sum_{n=0}^{\infty} \left[R_n''(\rho) + \frac{1}{\rho} R_n'(\rho) - \frac{\lambda_n}{\rho^2} R_n(\rho) \right] \Phi_n(\theta) = \sum_{n=0}^{\infty} f_n \Phi_n(\theta)$$

实现径向边界条件

$$R_n''(\rho) + \frac{1}{\rho} R_n'(\rho) - \frac{\lambda_n}{\rho^2} R_n(\rho) = f_n$$

$$\begin{aligned} u(a, \theta) = 0 &= \sum_{n=0}^{\infty} R_n(a) \Phi_n(\theta) & R_n(a) &= 0 \\ u(b, \theta) = 0 &= \sum_{n=0}^{\infty} R_n(b) \Phi_n(\theta) & R_n(b) &= 0 \end{aligned} \quad \begin{cases} R_n''(\rho) + \frac{1}{\rho} R_n'(\rho) - \frac{\lambda_n}{\rho^2} R_n(\rho) = f_n, & a < \rho < b \\ R_n(a) = 0, R_n(b) = 0, & 0 \leq \theta \leq 2\pi \end{cases}$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

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Poisson/二维环域/非齐		
周向 θ	周期	
径向 ρ	1齐	2齐

第4步 解径向ODE

$$\begin{cases} R_n''(\rho) + \frac{1}{\rho} R_n'(\rho) - \frac{\lambda_n}{\rho^2} R_n(\rho) = f_n, & a < \rho < b \\ R_n(a) = 0, \quad R_n'(b) = 0, & 0 \leq \theta \leq 2\pi \end{cases}$$

$\rho^2 R_n'' + \rho R_n' - \lambda_n R_n = \rho^2 f_n$

Euler方程令 $\rho = \exp s$

$R_n''(s) - \lambda_n R_n(s) = e^{2s} f_n$

$\left\{ \begin{array}{l} n=0, \\ \lambda_0 = 0, f_0 = 0 \\ \\ n=2, \\ \lambda_2 = 2^2, f_2 = 12\rho^2 \\ \\ n>0, n\neq 2, \\ \lambda_n = n^2, f_n = 0 \end{array} \right.$

$$\lambda_n = n^2, \quad \Phi_n = \{1, \cos n\theta, \sin n\theta\}, \quad (n = 0, 1, 2, \dots)$$

$$R_0''(s) - 0 \cdot R_0(s) = 0$$

直接积分法 $R_0(s) = C_{0,1}s + C_{0,2}$

还原变量 $R_0(\rho) = C_{0,1} \ln \rho + C_{0,2}$

$$R_0'(\rho) = C_{0,1} \frac{1}{\rho}$$

由径向内外环边界条件

$$R_0(a) = C_{0,1} \ln a + C_{0,2} = 0$$

$$R_0'(b) = C_{0,1} \frac{1}{b} = 0$$

$\rightarrow C_{0,1} = C_{0,2} = 0$

特解 $R_0(\rho) = 0$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

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周向 θ	周期	
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$$\lambda_n = n^2, \quad \Phi_n = \{1, \cos n\theta, \sin n\theta\}, \quad (n = 0, 1, 2, \dots)$$

第4步 解径向ODE

$$\begin{cases} R_n''(\rho) + \frac{1}{\rho} R_n'(\rho) - \frac{\lambda_n}{\rho^2} R_n(\rho) = f_n, & a < \rho < b \\ R_n(a) = 0, \quad R_n'(b) = 0, & 0 \leq \theta \leq 2\pi \end{cases}$$

$\rho^2 R_n'' + \rho R_n' - \lambda_n R_n = \rho^2 f_n$

Euler方程令 $\rho = \exp s$

$$R_n''(s) - \lambda_n R_n(s) = e^{2s} f_n$$

$n = 0,$
 $\lambda_0 = 0, f_0 = 0$

$n = 2,$
 $\lambda_2 = 2^2, f_2 = 12\rho^2$

$n > 0, n \neq 2,$
 $\lambda_n = n^2, f_n = 0$

$$R_2''(s) - 2^2 R_2(s) = 12e^{4s}$$

$s = 4$ 不是特征根 $\bar{T}_2(s) = C e^{4s}$ 易得 $C = 1$

通解 $R_2(s) = C_{2,1} e^{ns} + C_{2,2} e^{-ns} + e^{4s}$

还原变量 $R_2(\rho) = C_{2,1} \rho^n + C_{2,2} \rho^{-n} + \rho^4$

由边界条件 $R_2'(\rho) = 2C_{2,1} \rho - 2C_{2,2} \rho^{-3} + 4\rho^3$

$$\begin{aligned} R_2(a) &= C_{2,1} a^2 + C_{2,2} a^{-2} + a^4 = 0 & C_{2,1} &= -\frac{a^6 + 2b^6}{a^4 + b^4} \\ R_2'(b) &= 2C_{2,1} b - 2C_{2,2} b^{-3} + 4b^3 = 0 & C_{2,2} &= -\frac{a^4 b^4 (a^2 - 2b^2)}{a^4 + b^4} \end{aligned}$$

$$R_2(\rho) = -\frac{a^6 + 2b^6}{a^4 + b^4} \rho^2 - \frac{a^4 b^4 (a^2 - 2b^2)}{a^4 + b^4} \rho^{-2} + \rho^4$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

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Poisson/二维环域/非齐		
周向 θ	周期	
径向 ρ	1齐	2齐

$$\lambda_n = n^2, \quad \Phi_n = \{1, \cos n\theta, \sin n\theta\}, \quad (n = 0, 1, 2, \dots)$$

第4步 解径向ODE

$$\begin{cases} R_n''(\rho) + \frac{1}{\rho} R_n'(\rho) - \frac{\lambda_n}{\rho^2} R_n(\rho) = f_n, & a < \rho < b \\ R_n(a) = 0, \quad R_n'(b) = 0, & 0 \leq \theta \leq 2\pi \end{cases}$$

$\rho^2 R_n'' + \rho R_n' - \lambda_n R_n = \rho^2 f_n$

Euler方程令 $\rho = \exp s$

$$R_n''(s) - \lambda_n R_n(s) = e^{2s} f_n$$

$n = 0,$
 $\lambda_0 = 0, f_0 = 0$

$n = 2,$
 $\lambda_2 = 2^2, f_2 = 12\rho^2$

$n > 0, n \neq 2,$
 $\lambda_n = n^2, f_n = 0$

$R_n''(s) - n^2 R_n(s) = e^{2s} \cdot 0$

通解 $R_n(s) = C_{n,1} e^{ns} + C_{n,2} e^{-ns}$

还原变量 $R_n(\rho) = C_{n,1} \rho^n + C_{n,2} \rho^{-n}$

$$R_n'(\rho) = C_{n,1} n \rho^{n-1} + C_{n,2} (-n) \rho^{-n-1}$$

由边界条件

$$R_n'(b) = C_{n,1} b^{n-1} - C_{n,2} b^{-n-1} = 0 \quad C_{n,1} b^{2n} = C_{n,2}$$
$$R_n(a) = C_{n,1} a^n + C_{n,2} a^{-n} = C_{n,1} (a^n + b^{2n} a^{-n}) = 0$$
$$C_{n,1} = C_{n,2} = 0$$
$$R_n(\rho) = 0$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = f(\rho, \theta)$$

平面Laplace方程

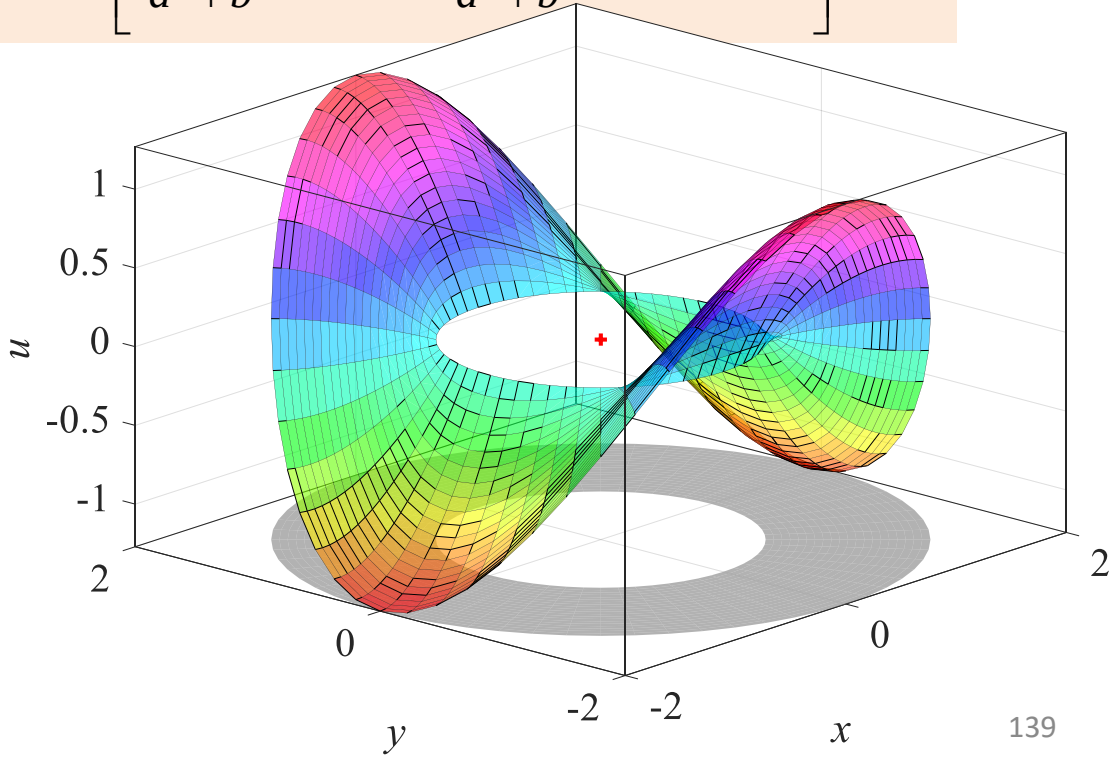
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$$\begin{cases} u_{\rho\rho} + \frac{1}{\rho} u_{\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 12\rho^2 \cos 2\theta, & a < \rho < b, 0 \leq \theta < 2\pi \\ u(a, \theta) = 0, \quad u_{\rho}(b, \theta) = 0, & 0 \leq \theta \leq 2\pi \end{cases}$$

Poisson/二维环域/非齐		
周向 θ	周期	
径向 ρ	1齐	2齐

第5步 回代

$$\lambda_n = n^2, \quad \Phi_n = \{1, \cos n\theta, \sin n\theta\}, \quad (n = 0, 1, 2, \dots)$$

$$u(\rho, \theta) = \sum_{n=0}^{\infty} R_n(\rho) \Phi_n(\theta) = R_2(\rho) \Phi_2(\theta) = - \left[\frac{a^6 + 2b^6}{a^4 + b^4} \rho^2 + \frac{a^4 b^4 (a^2 - 2b^2)}{a^4 + b^4} \rho^{-2} - \rho^4 \right] \cos 2\theta$$



本章作业

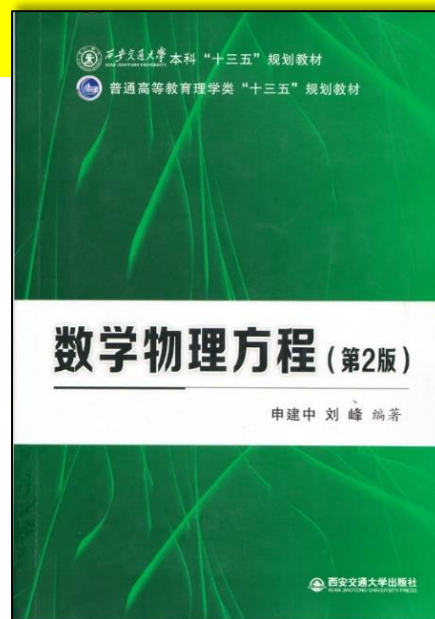
习题2

① 特征值问题 2 (1) (3)

② 波动方程 5 (2) (4)

③ 导热方程 6 (1) (3)

④ Poisson方程 7 (1) (4) 课本(第2版)例 2.8



$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x, 0 < y, x^2 + y^2 < 4 \\ u(x, 0) = 0, & 0 \leq x \leq 2 \\ u(0, y) = 0, & 0 \leq y \leq 2 \\ u(x, y) = xy, & x^2 + y^2 = 4 \end{cases}$$

大作业 在你专业领域or兴趣方向找1个PDE定解问题→建模/求解→撰写报告

内容要求

1. 物理现象背景简介

2. 建模→定解问题

3. 定解问题求解过程

4. 典型工况图示

5. 结论

6. 参考文献
- 明确对象

简化假设

数理推导

定解条件

执行要求

合作完成:自由组队, 1~5人

提交时间 **2025.12.31** 之前

电子版 Word文档

“张三-李四-王五-XXX研究报告”→思源学堂

2022年西安交通大学本科教学改革研究项目（未来人才培养专项）立项名单

序号	项目编号	项目名称	项目主持人	申报单位	项目类别
1	22WL01Z	基于产教融合的“智能制造系统中人工智能应用及实践项目”课程建设	王保建	机械学院	重点
2	22WL02Z	机械设计课程的智能化改造及产教融合路径探究	李宝童	机械学院	重点
3	22WL03Z	《航空叶片制造系统的大数据与云计算应用及实践》项目课程建设与改革	李晶	机械学院	重点
4	22WL04Z	基于采集式学习的实践教学模式探索	刘均	电信学部	重点
5	22WL05Z	产教融合、多学科贯通的《电化学基础》课程建设与实践	徐友龙	电信学部	重点
6	22WL06Z	信息技术赋能计算机科学基础与高级程序设计	夏秦	电信学部	重点
7	22WL07Z	面向人工智能专业的现代控制工程课程建设与实践	刘妹琴	人工智能学院	重点
8	22WL08Z	数字教材在数字教学中的探索—以“双师型”影像学数字教学为例	张明	医学部	重点
9	22WL09Z	医学专业临床课程产教融合课程体系构建的研究与实践	刘进军	医学部	重点
10	22WL10Z	面向理工交叉的《医用电子技术基础》产教融合式课程创新与教学模式探索	吴春生	医学部	重点
11	22WL01Y	基于产教融合的概率论与数理统计课程改革与实践	赵小艳	数学学院	一般
12	22WL02Y	面向医学专业《原子物理学》课程的产教融合探索与实践	王兴	物理学院	一般
13	22WL03Y	工程项目贯穿式材料力学教学改革与实践	李培	机械学院	一般
14	22WL04Y	面向未来技术学院的“人工智能基础”课程建设	张小林	机械学院	一般
15	22WL05Y	“产教融合”在储能专业《数学物理方程》教学中的实践与评价	张丹	能动学院	一般
16	22WL06Y	初级项目1: 心电信号检测电路	范世全	电信学部	一般
17	22WL07Y	医学《人体分子与细胞》产教融合课程体系的构建与实践	李冬民	医学部	一般
18	22WL08Y	基于知识图谱的采煤教学新模式在医学基础项目课程中的探索与实践	徐小玲	医学部	一般
19	22WL09Y	面向前沿探索的工程训练实践课程知识图谱构建及应用	栗茂林	实践教学中心	一般

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